

Interspeech 2026 Tutorial

On The Landscape of Spoken Language Models





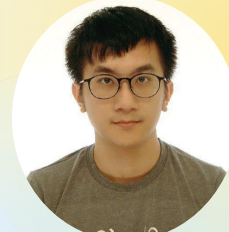
Kai-Wei Chang
(MIT)



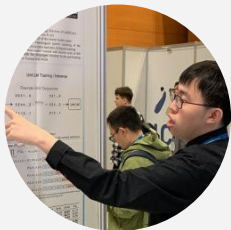
Chung-Ming Chien
(TTIC)



Shinji Watanabe
(CMU)



Ke-Han Lu
(NTU)



Chih-Kai Yang
(NTU)



Siddhant Arora
(CMU)



Hung-yi Lee
(NTU)



Karen Livescu
(TTIC)

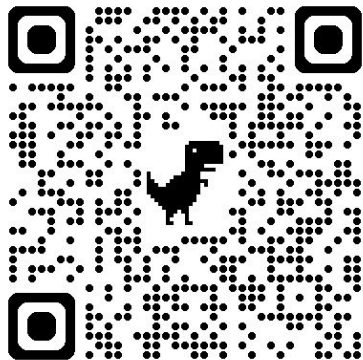




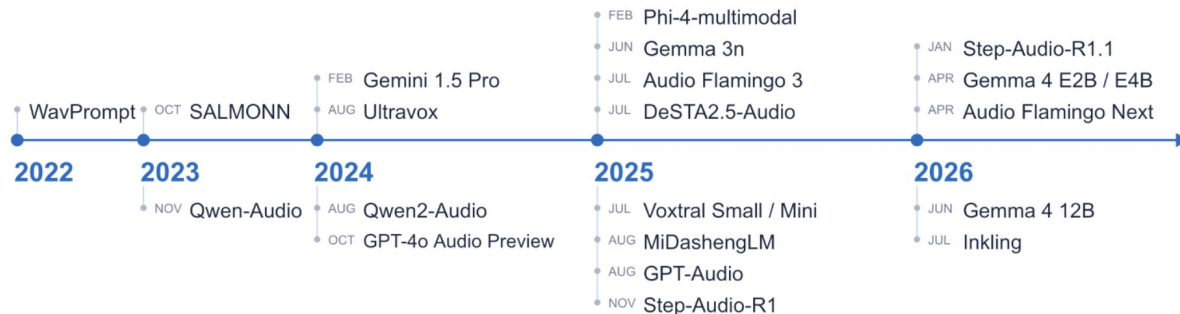
On The Landscape of Spoken Language Models

Tutorial Website

<https://interspeech2026slm.github.io/>



Why a Tutorial on SLMs



Speech version of LLM that can

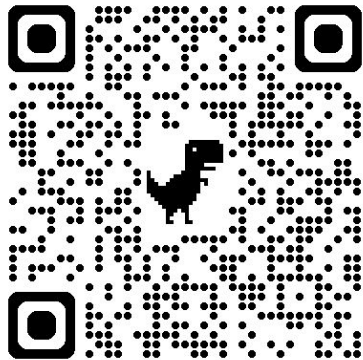
- Take meeting notes and identify action items
- Summarize lectures, interviews, and podcasts
-



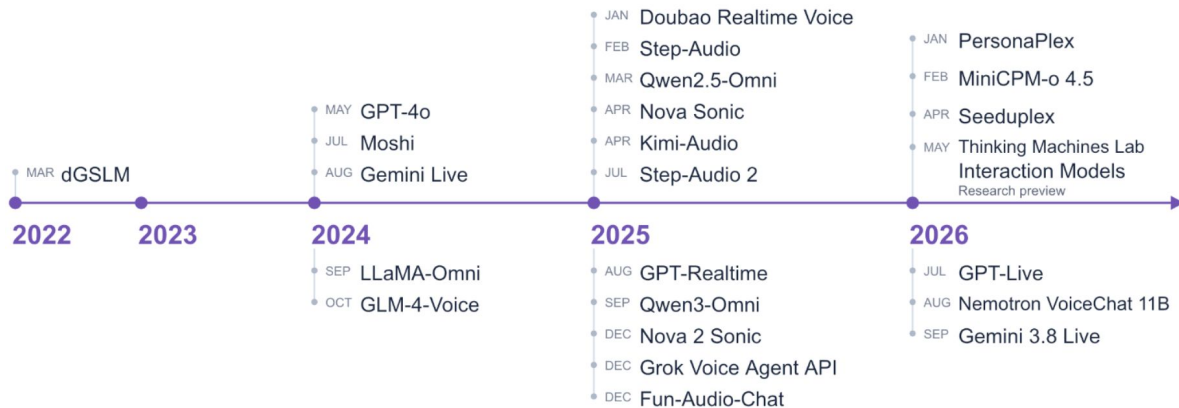
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Why a Tutorial on SLMs



Personal voice assistant.

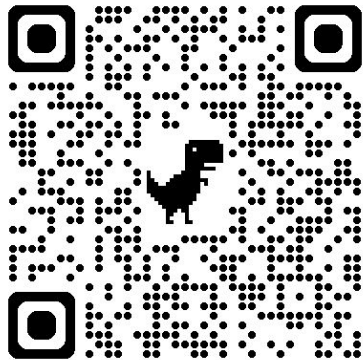
- Get live guidance while cooking or troubleshooting
- Manage schedules and everyday tasks through voice
- Practice languages with a conversational tutor



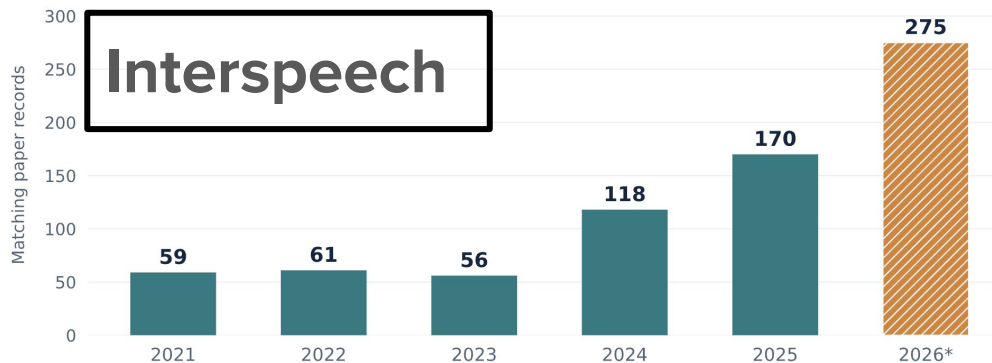
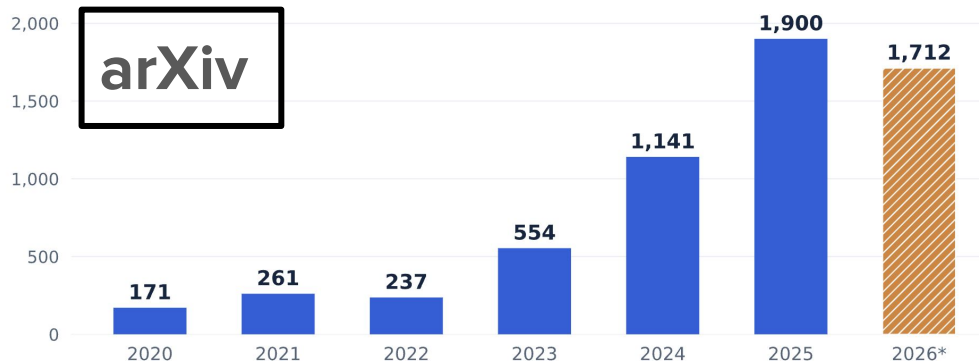
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Why a Tutorial on SLMs



We searched arXiv titles and abstracts for both speech-related and language-model-related terms

Agenda

01. Introduction

– Kai-Wei Chang

02. Speech representation

– Chung-Ming Chien

03. Typical Training

– Shinji Watanabe

04. Advanced Training

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05. Hands-on Session

– Ke-Han Lu

06. Full-duplex

– Siddhant Arora

07. Evaluation

– Hung-yi Lee

08. Emerging Topics

– Chih-Kai Yang

09. Conclusion

– Karen Livescu

The tutorial is based on the survey paper

Published in Transactions on Machine Learning Research (10/2025)

On The Landscape of Spoken Language Models: A Comprehensive Survey

Siddhant Arora^{1*} Kai-Wei Chang^{2*} Chung-Ming Chien^{1*} Yifan Peng^{1*} Haibin Wu^{2,*†}
Yossi Adi^{1*} Emmanuel Dupoux^{1*} Hung-Yi Lee^{2*} Karen Livescu^{1*} Shinji Watanabe^{1*}

¹ Carnegie Mellon University, USA

² National Taiwan University, Taiwan

³ Toyota Technological Institute at Chicago, USA

⁴ Hebrew University of Jerusalem, Israel

⁵ ENS - PSL, EHESS, CNRS, France

Reviewed on OpenReview: <https://openreview.net/forum?id=8vxaF3sVbA>

Abstract

The field of spoken language processing is undergoing a shift from training custom-built, task-specific models toward using and optimizing *spoken language models* (SLMs) which act as universal speech processing systems. This trend is similar to the progression toward universal language models that has taken place in the field of (text) natural language processing. SLMs include both “pure” language models of speech—models of the distribution of tokenized speech sequences—and models that combine speech encoders with text language models, often including both spoken and written input or output. Work in this area is very diverse, with a range of terminology and evaluation settings. This paper aims to contribute an improved understanding of SLMs via a unifying literature survey of recent work in the context of the evolution of the field. Our survey categorizes the work in this area by model architecture, training, and evaluation choices, and describes some key challenges and directions for future work.

1 Introduction

In the last few years, the field of natural language processing (NLP) has evolved from (1) training many task-specific models from scratch, to (2) combining pre-trained multi-purpose contextual representation models (such as BERT (Devlin et al., 2019)) with a small number of task-specific parameters, to (3) training generative universal large language models (LLMs) (Brown et al., 2020; OpenAI et al., 2024)¹ that perform arbitrary text tasks given natural language instructions (prompts) and can generalize to unseen domains and tasks (Wei et al., 2022a; Lin et al., 2023), and finally to (4) dialogue (chatbot) systems that function as assistants and perform tasks while directly interacting with the user.

The field of speech processing has been undergoing a similar evolution, although with some lag, and has mainly focussed on stages (1) and (2). The current state of the art (SOTA) for common specialized speech tasks—including automatic speech recognition (ASR), speech translation (ST), spoken language understanding (SLU) tasks, and speaker identification (SID)—involves combining a pre-trained self-supervised encoder (Mohamed et al., 2022) with a task-specific prediction “head”. For some very common tasks—namely, ASR and ST—in relatively high-resource languages, large supervised models (Radford et al., 2023; Peng et al., 2023b) also have consistently good (if not SOTA) performance.

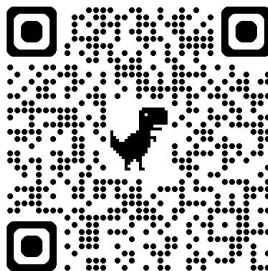
¹Equal-contribution first authors listed in alphabetical order. ²Equal-contribution last authors listed in alphabetical order. ³Work was done before Haibin Wu joined Microsoft.

⁴Throughout the paper, we use the terms LLMs and LMs interchangeably to refer to modern language models.

On The Landscape of Spoken Language Models: A Comprehensive Survey

Siddhant Arora*, Kai-Wei Chang*, Chung-Ming Chien*, Yifan Peng*, Haibin Wu*,
Yossi Adi†, Emmanuel Dupoux†, Hung-yi Lee†, Karen Livescu†, Shinji Watanabe†

*Co-first author †Co-last author – alphabetical order

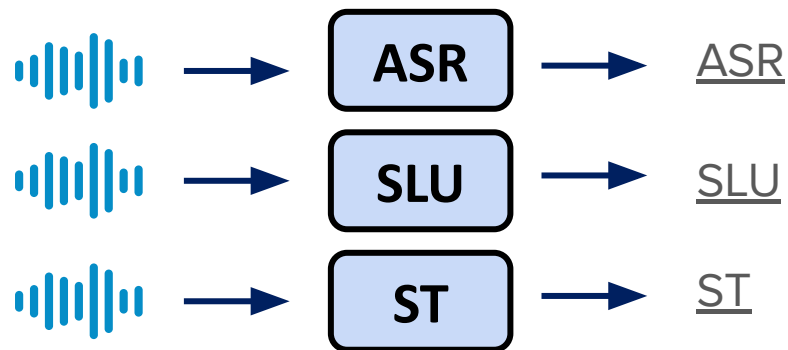


Transactions on Machine Learning Research (TMLR),
October 2025. arxiv.org/abs/2504.08528

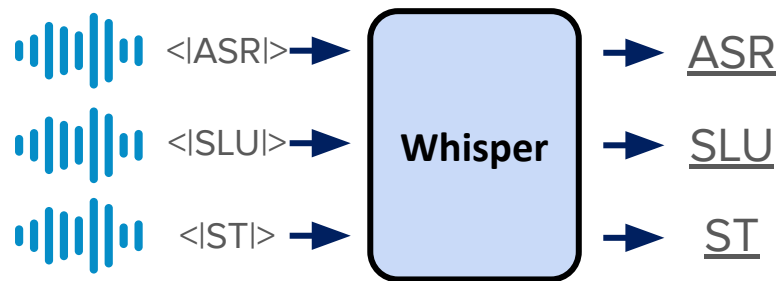
PART 01 · Kai-Wei Chang

Introduction to Spoken Language Model

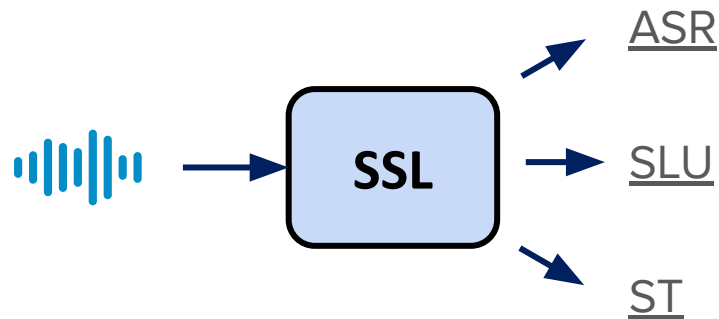
Task Expert Model



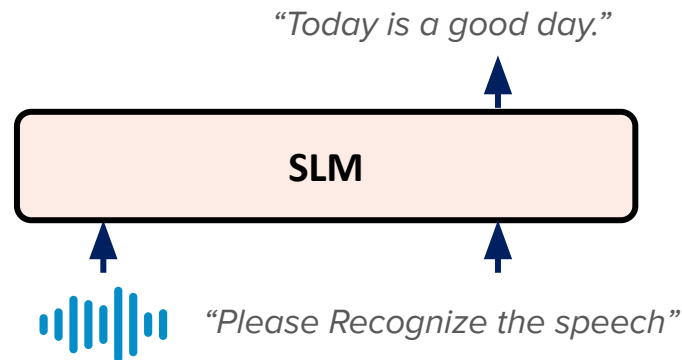
Task Specifier



Self-supervised Speech Encoder



Spoken Language Model

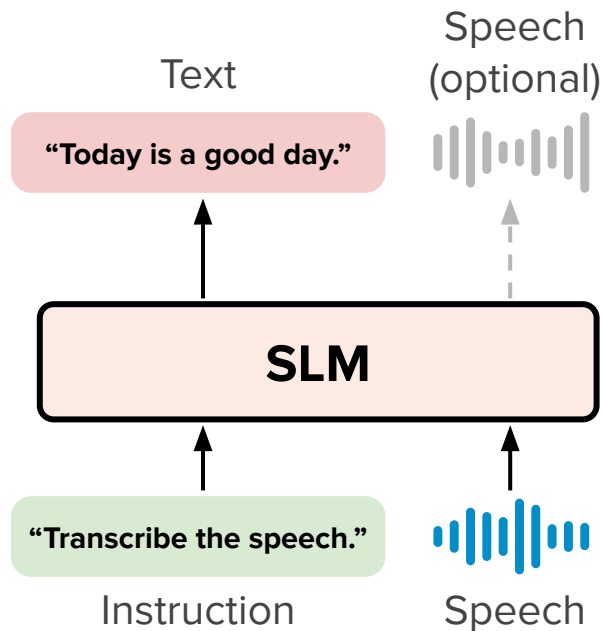


What is a Spoken Language Model?

Spoken Language Model

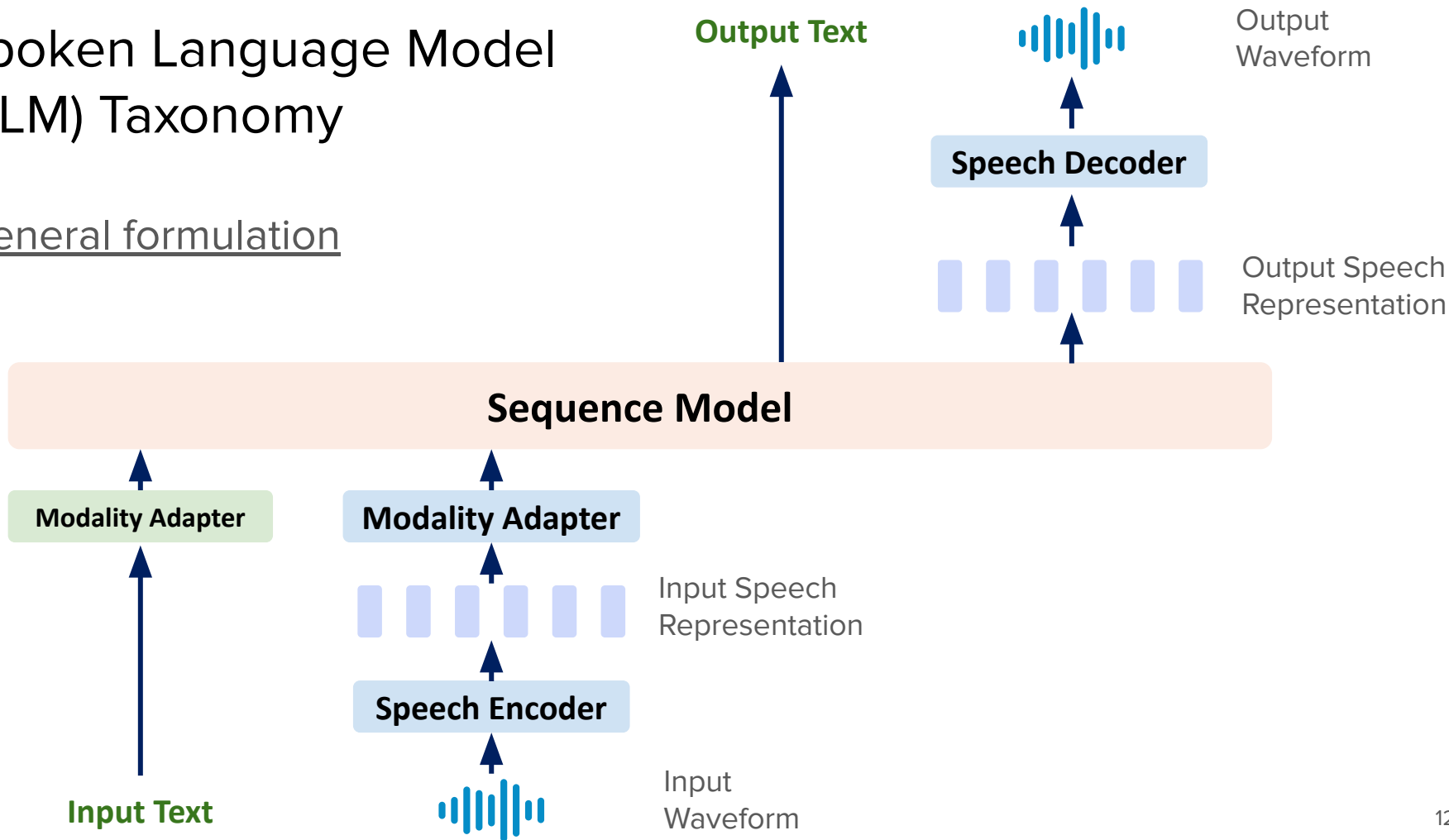
Functional definition. This tutorial scope:

- **Speech in; text or speech out.**
- **Intended to be universal.**
One model to solve diverse speech tasks; not a single one
e.g. VALL-E excluded
- **Instructed in natural language.**
The task instruction is typed or spoken; not a task tag.
e.g. Whisper and SpeechPrompt excluded

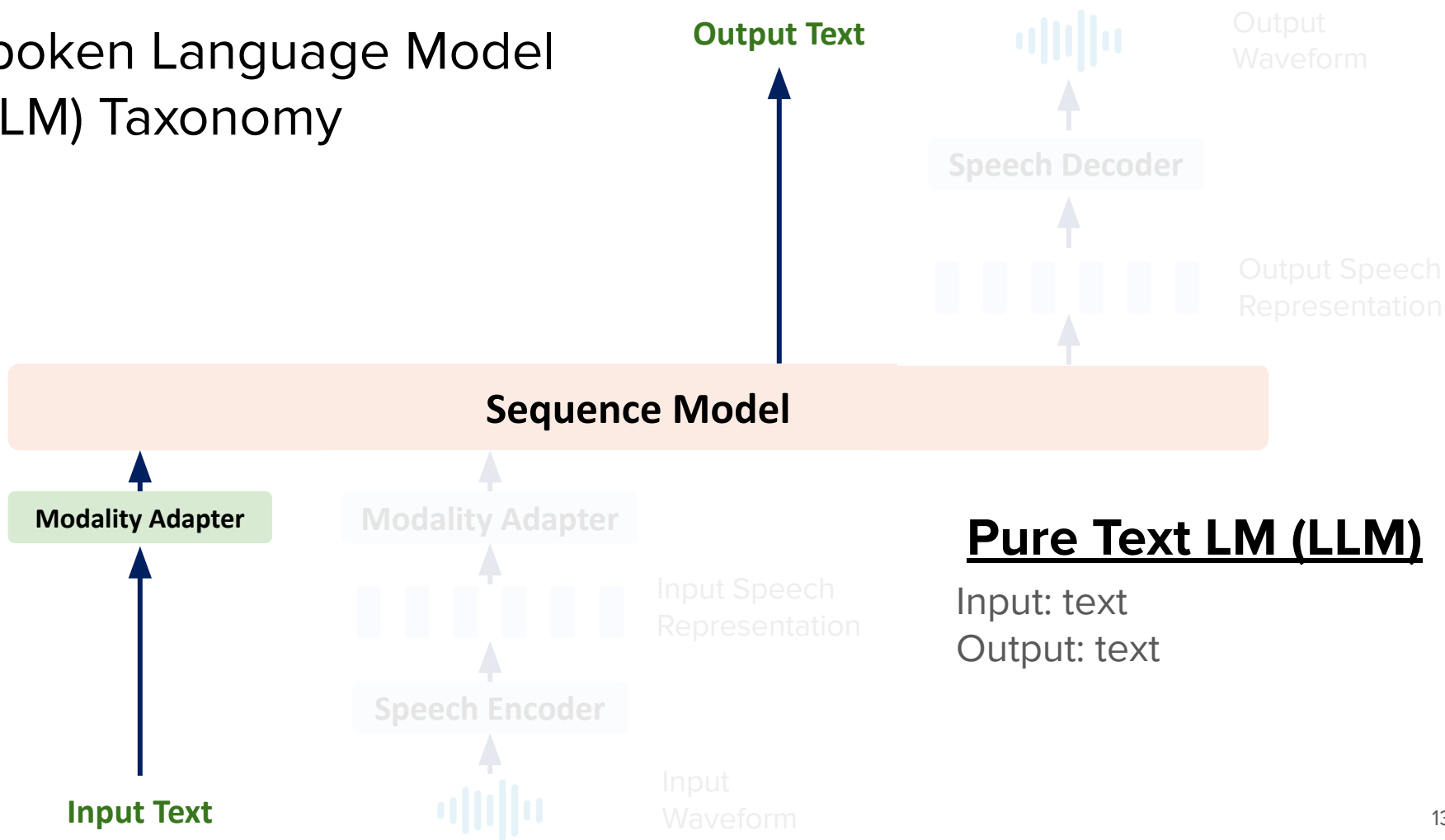


Spoken Language Model (SLM) Taxonomy

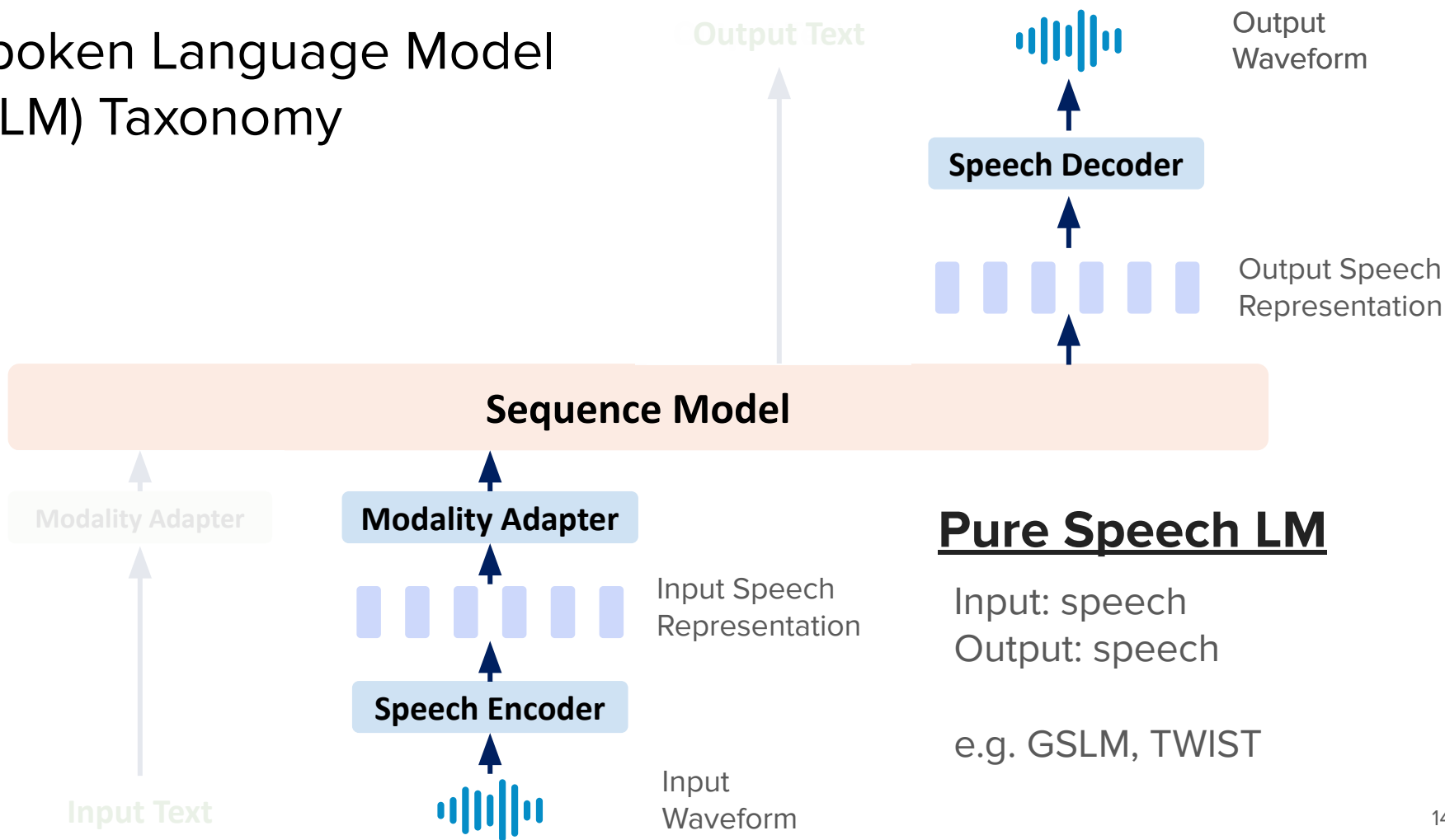
General formulation



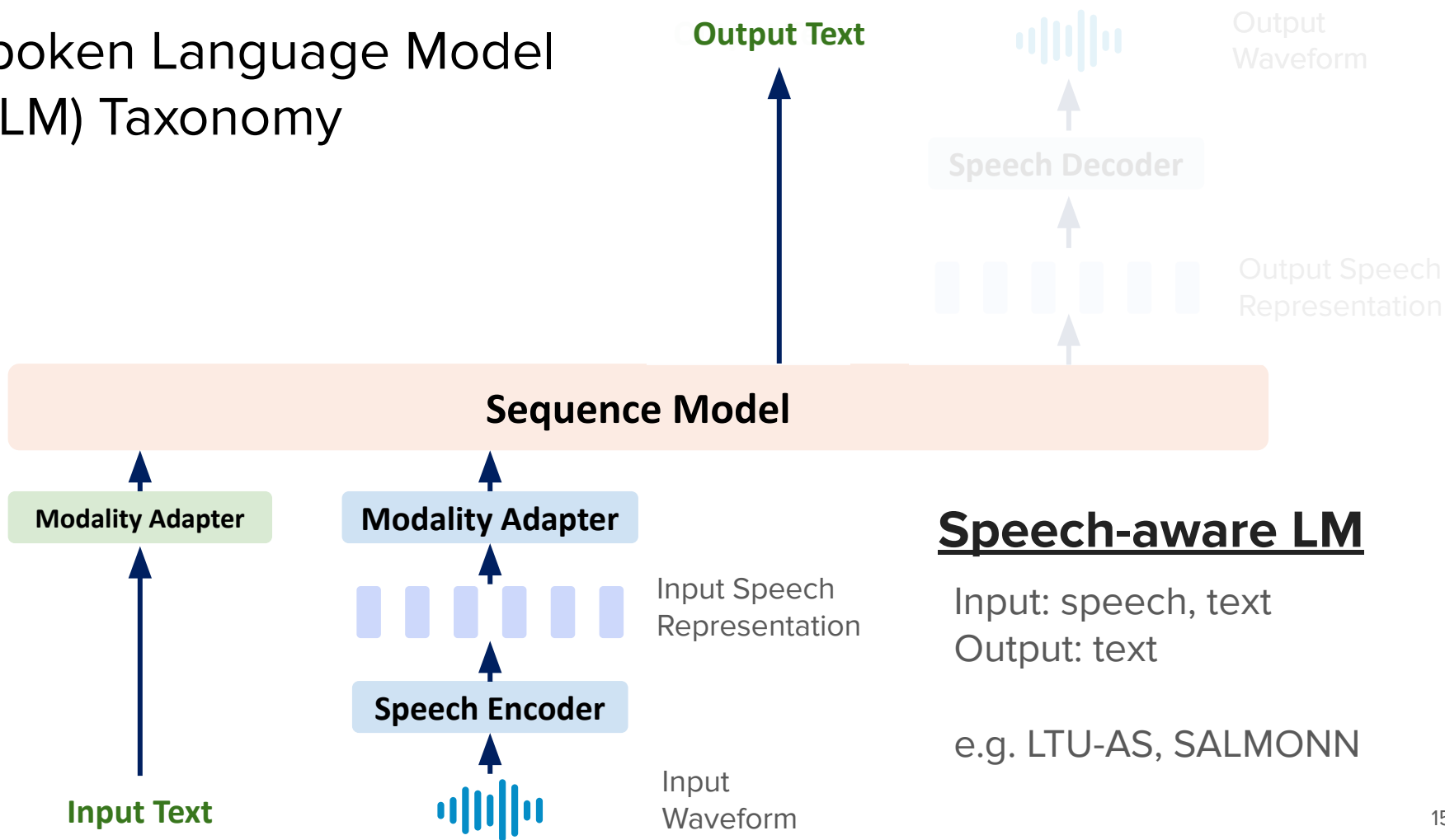
Spoken Language Model (SLM) Taxonomy



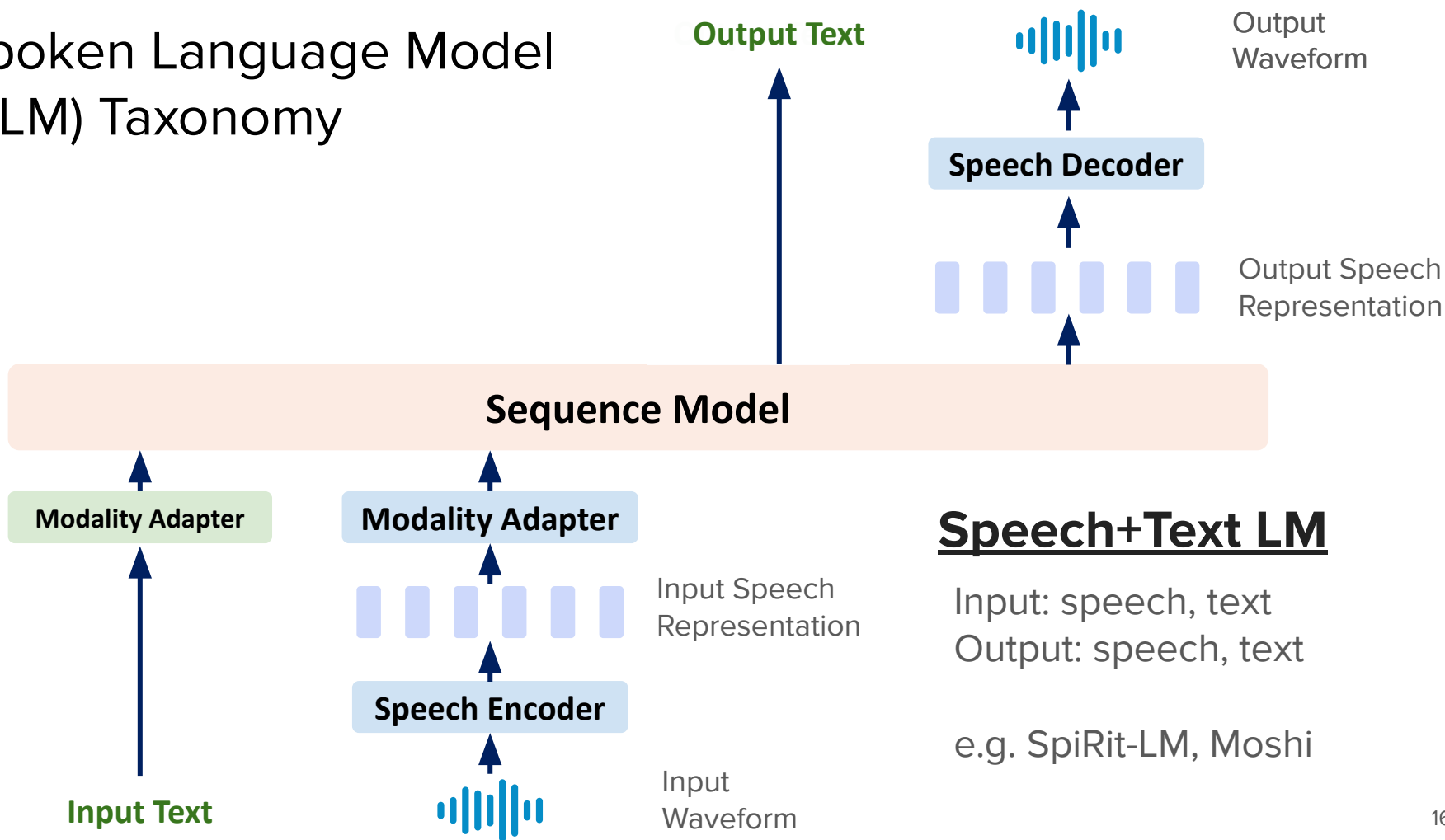
Spoken Language Model (SLM) Taxonomy



Spoken Language Model (SLM) Taxonomy



Spoken Language Model (SLM) Taxonomy



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PART 02 · Chung-Ming Chien

Speech Representation, Tokenization, and Generation

Roadmap for This Part

1 Overview of Spoken Language Model Architectures

Speech encoder · modality adapter · sequence model · speech decoder

2 Input Modules: Mapping Speech into the LM's Representation Space

Discrete vs. continuous representations · phonetic vs. codec tokens · temporal compression · modality adapter

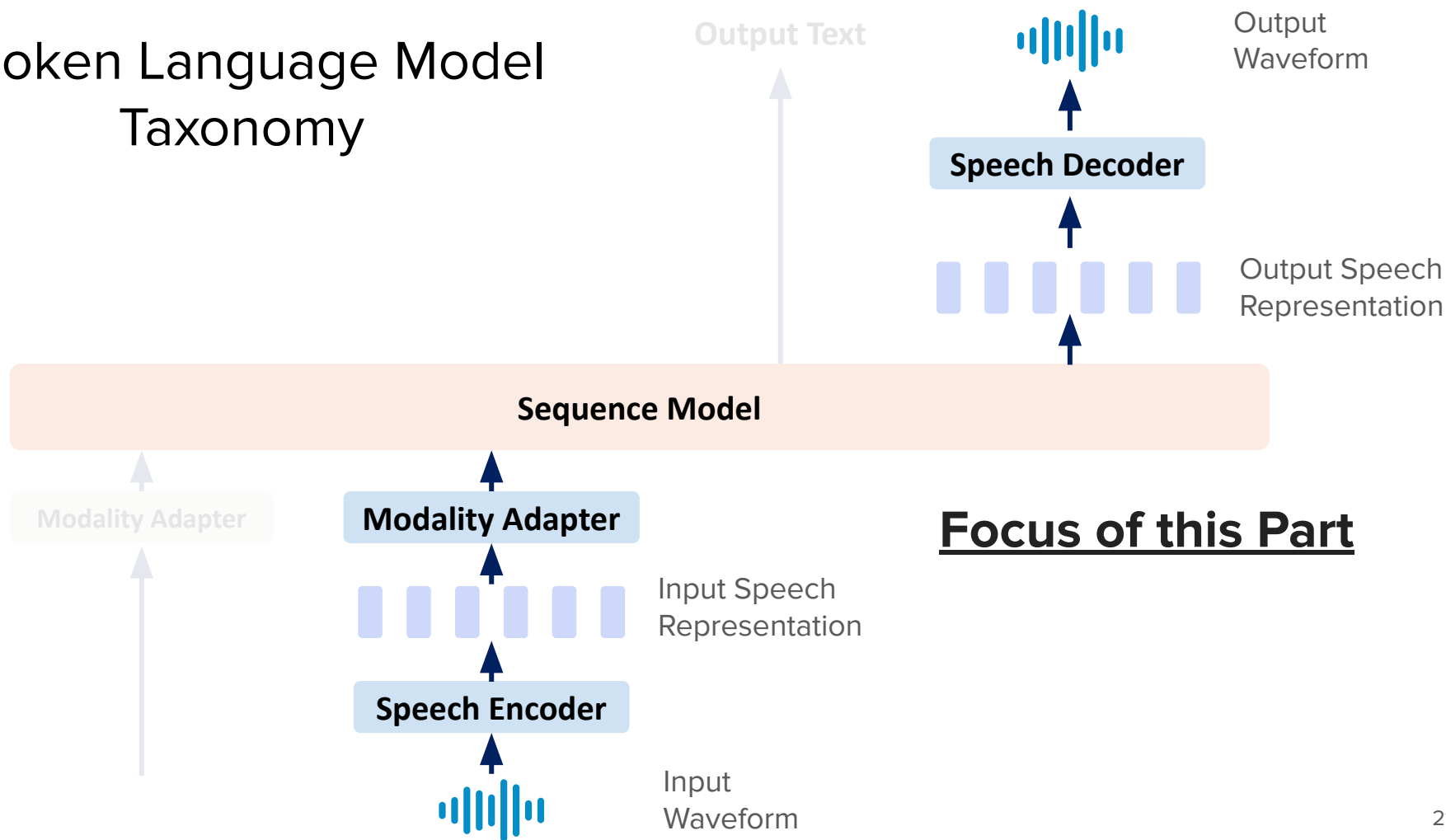
3 Sequence Model: Modeling Mixed Speech-Text Sequences

Speech-text coupling · speech-text alignment · sequence formulation · hierarchical generation

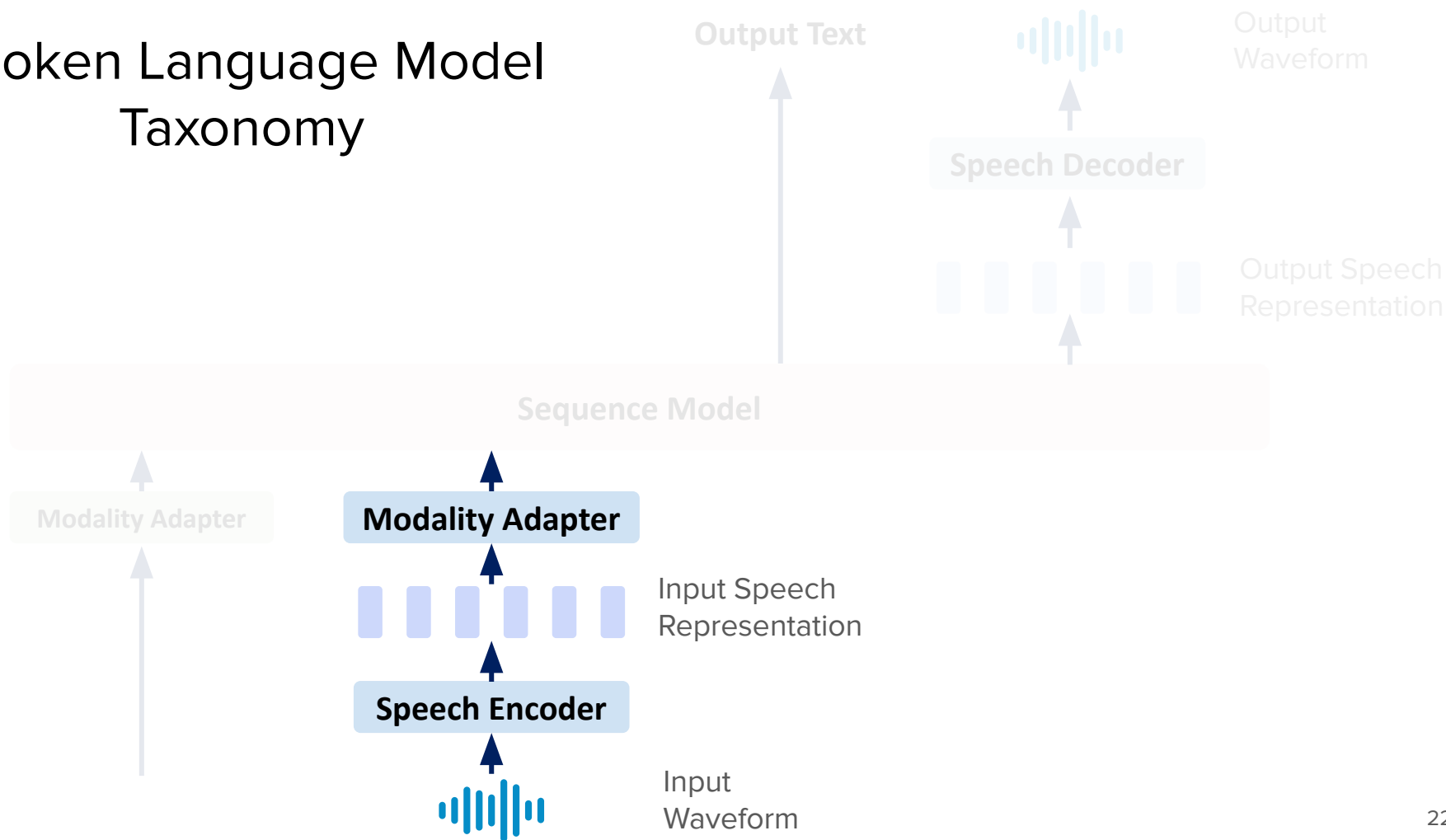
4 Output Modules: Generating Waveforms

Vocoder · codec decoder

Spoken Language Model Taxonomy

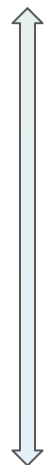


Spoken Language Model Taxonomy



Input Modules

Primary goal: convert speech into representations that LLMs can process



- Text: discrete, semantic, short sequence.
- Speech: continuous, acoustic, long sequence with speaker, intonation, etc.

Design Considerations

Information which non-textual cues to preserve: speaker, prosody, accent, ...

Sequence length how to balance context length and computation?

Causality whether speech processing can use future context?

Real-time latency constraints for online applications.

→ Critical for streaming & full-duplex systems.

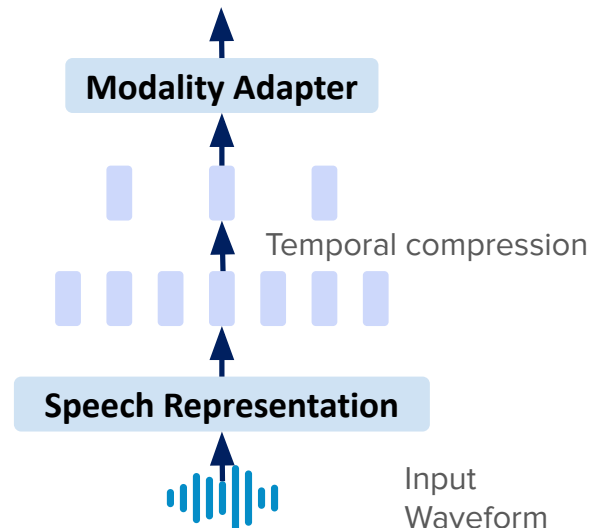
Input Modules

Primary goal: convert speech into representations that LLMs can process

Speech representation model encodes continuous signals into vector sequences

Temporal compression (optional) reduces sequence length while preserving relevant information

Modality adapter maps speech representations into the LM's representation space



Speech Representations

A pre-trained model that encodes continuous speech signals into a vector sequence

Must include textual content and may **additionally** encode non-textual information

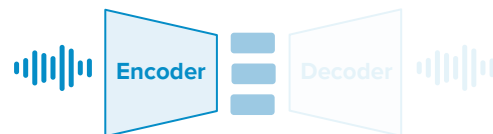
A ASR/Supervised training



- ✓ textual content

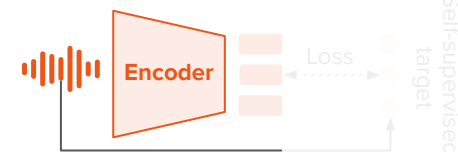
Examples Whisper (Radford et al.), USM (Zhang et al., 2023a)

B Reconstruction training



- ✓ rich textual and non-textual information
 - ✗ limited compression
- Examples** audio codec such as EnCodec (Défossez et al., 2023)

C Self-supervised training



- ✓ compact representations with strong downstream performance

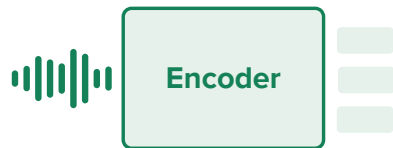
Examples HuBERT (Hsu et al.), WavLM (Chen et al.)

Other representations such as Mel filterbanks and spectrograms

Continuous Features vs. Discrete Tokens

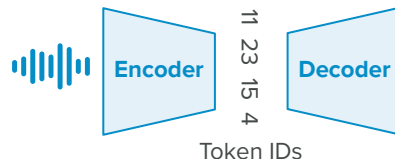
LLMs inherently operate on discrete units

What about speech?



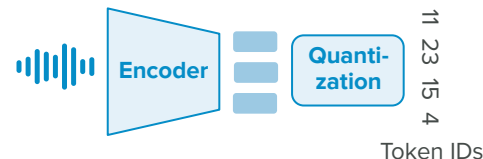
Directly using continuous features

- Avoid information loss from discretization
- Harder to predict with an LM, but still possible (e.g., CALM (Rouard et al.))
- Commonly used for speech understanding
- Example: Whisper features used in SALMONN (Tang et al.) and LTU-AS (Gong et al.)



Learned discrete representations

- Information encoded depends on the representation learning objective → may include non-content information such as speaker identity
- Codec: natural for speech generation
- Example: mimi codec used in Moshi (Défossez et al., 2024)



Post-hoc quantization

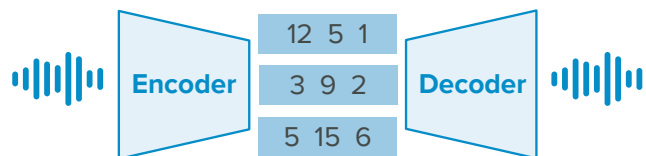
- Configurable quantization (e.g. number of clusters, layer selection) → flexibility in information extraction
- Example: quantized HuBERT features used in GSLM (Lakhotia et al.)

Discrete Tokens

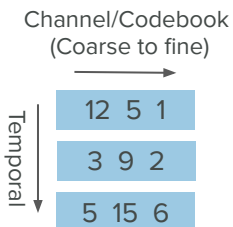
Discrete Tokens: Acoustic Codec vs. Phonetic Tokens

Acoustic codec tokens

Optimized for audio compression and reconstruction

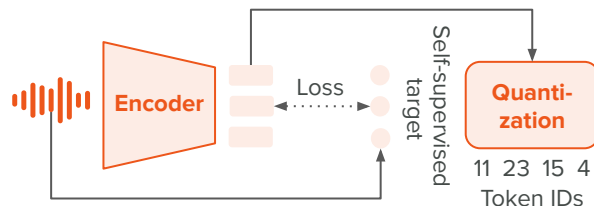


- Well-suited for audio generation
- Residual vector quantization (RVQ) is commonly used → flexibly control the number of channels modeled by the LM
- Usually longer sequences than phonetic tokens, but ultra-low-rate codec are now available (see Wu et al.)



Phonetic Tokens *also known as “semantic tokens” in literature

Quantized self-supervised speech representations

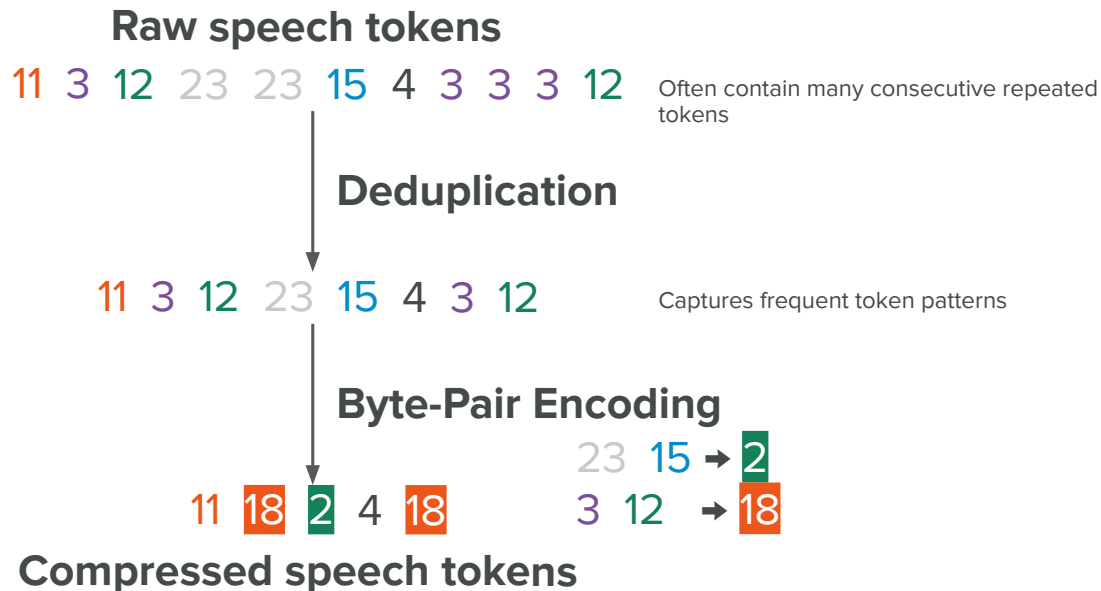


- Self-supervised models already learn compact representations
- Quantized features can be treated as pseudo-text (Lakhotia et al.)
- Little speaker information, more phonetic (textual) information (Pasad et al.)
- Well-suited for content modeling

They can be combined – e.g., WavLM (Chen et al.) phonetic tokens are distilled into the first layer of the Mimi codec (Défossez et al., 2024)

Temporal Compression for Speech Token Sequences

Discrete speech token sequences can be as long as 50 tokens/sec., while text tokens for the same content can be 10x shorter.

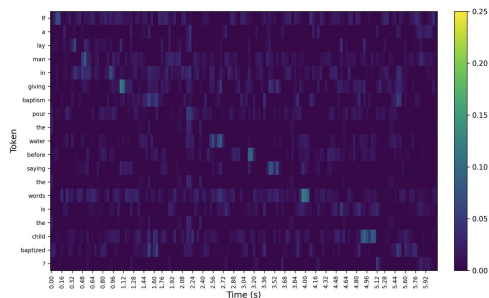


Modality Adapters

How are text embeddings and speech representations different?

Mismatched feature spaces · Different sequence lengths · Variable alignment durations

A Mismatched feature spaces

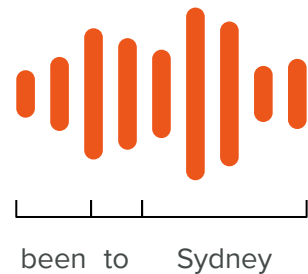


Similarity between text and speech representations of the same content in an SLM (Kim et al.)

B Different sequence lengths



C Variable alignment durations



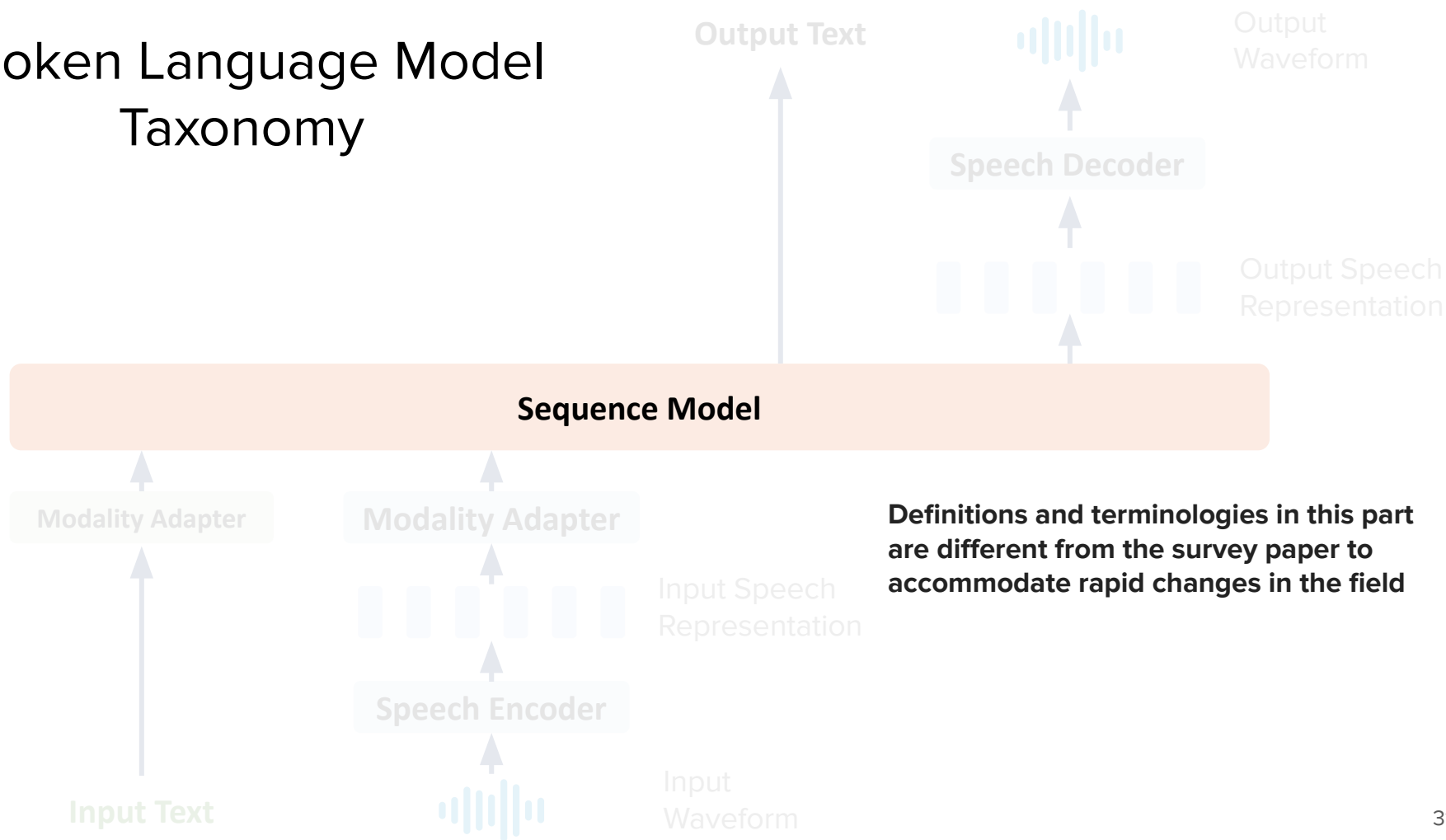
Modality Adapters

How are text embeddings and speech representations different?

Mismatched feature spaces · Different sequence lengths · Variable alignment durations

Modality adapters	Feature space	Sequence length <i>Speech can be much longer than text</i>	Alignment duration <i>Text tokens span variable durations</i>
Linear layer(s)	✓		
CNN downsampling	✓	✓	
Q-Former (Li et al.)	✓	✓	
CTC-based compression	✓	✓	✓
AlignFormer (Fan et al.)	✓	✓	✓

Spoken Language Model Taxonomy



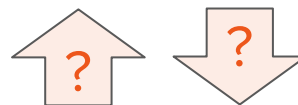
Sequence Modeling

Most (except for pure speech models) spoken language models handle both text and speech

How do LMs model mixed speech-text sequences?

Sequence Model

Text and speech tokens of the same utterance
(This example will be used in the following few slides)



Text tokens

I

wish

I

could

visit

Sydney

Speech tokens

11

13

8

2

11

18

5

4

8

24

Speech-Text Coupling

Text tokens	I	wish	I	could	visit	Sydney
Speech tokens	11	13 8 2	11	18	5 4	8 24

Different ways to couple speech and text

Replacement

Use either speech or text

Sequence I 13 8 2 11 could 5 4 Sydney

- What LLMs can do with text can potentially be done with speech (e.g., text reasoning → speech reasoning)

- Also used in speech-text LM pre-training such as SpiRit-LM (Nguyen et al.)

Complementary

Model both speech and text

Text tokens I wish I could visit Sydney

Speech tokens 11 13 8 2 11 18 5 4 8 24

- Speech and text contain substantial shared linguistic information
- Useful in speech generation as text can provide strong guidance for speech tokens, such as SpeechGPT (Zhang et al., 2023b)

Text tokens

I

wish

I

could

visit

Sydney

Speech-Text Alignment

Speech tokens

11

13

8

2

11

18

5

4

8

24

Word/token-level alignment

Speech tokens are explicitly aligned to individual words or text tokens

Text tokens	I	wish			I	could	visit		Syd- ney	
Speech tokens	11	13	8	2	11	18	5	4	8	24

- More preprocessing and alignment supervision
- Fine-grained manipulation such as time-specific information injection is possible (Chien et al.)
- Examples: Moshi (Défossez et al., 2024), MoshiRAG (Chien et al.)

Utterance-level alignment

Speech and text are aligned only at utterance boundaries

Text tokens	I	wish	I	could	visit	Syd- ney					But	I
Speech tokens	11	13	8	2	11	18	5	4	8	24	3	13

- No fine-grained alignment required
- Robust against minor alignment errors
- Requires the LM to “plan ahead” future text tokens
- Examples: Mini-Omni (Xie & Wu), LLaMA-Omni (Fang et al.)

Sequence Formulation

Text tokens

wish

could

visit

Sydney

Speech tokens

11

13

8

2

11

18

5

4

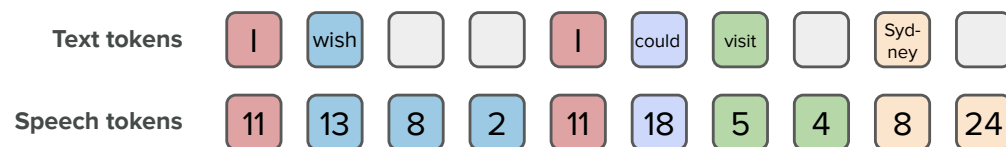
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24

How to arrange complementary speech & text into a sequence of LM input vectors?

Channel fusion

Fuse individual speech & text vectors at each position



- Speech and text information are mixed at each position
- The LM must learn to disentangle the two modalities from mixed latents when needed
- Example: Moshi

Interleaving

Keep speech and text vectors in separate positions



- Longer sequence
- Modality identity is explicit from token position/type
- Speech and text representations remain individually accessible
- Example: SALMONN-omni (Yu et al.)

Sequence Modeling

How do LMs model mixed speech-text sequences?

Speech-text coupling · Speech-text alignment · Sequence formulation

Speech-Text Coupling	Speech-Text Alignment	Sequence Formulation (for complementary inputs)	Examples
replacement	word-level	—	SpiRit-LM
replacement	utterance-level	—	SALMONN
complementary	word-level	channel fusion	Moshi
complementary	word-level	interleaving	MiniCPM-o 4.5 (Cui et al.)
complementary	utterance-level	channel fusion	Mini-Omni, SALM-Duplex (Hu et al.), Qwen3-Omni* (Xu et al.), Kimi-Audio (Kimi Team) , VoiceChat (NVIDIA)
complementary	utterance-level	interleaving	GLM-4-Voice (Zeng et al.), Baichuan-Omni-1.5 (Li et al.), SALMONN-omni, Step-Audio 2 (Wu et al.)

* Qwen3-Omni adopts a thinker-talker architecture. Speech and text tokens are combined only in the talker LM; speech tokens are not fed back to the thinker LM as autoregressive inputs.

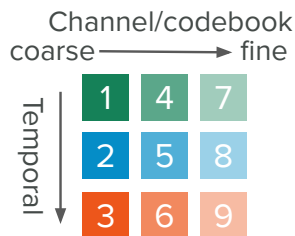
Hierarchical Generation

Speech codecs may contain multiple layers; how should we define their order of generation?

Make use of the hierarchy to model coarse- & fine-grained units

A Coarse-to-fine

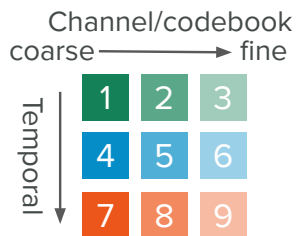
Focus on high-level information first



Example: AudioLM (Borsos et al.)

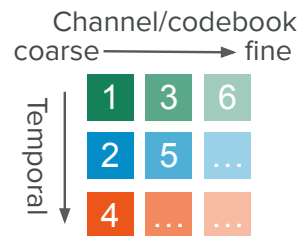
B Interleaving

Generate each timestep causally



C Delayed interleaving

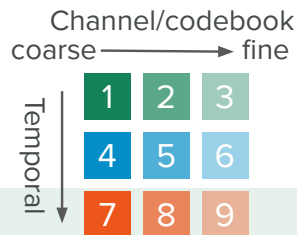
Look ahead future coarse tokens to improve acoustic details



Example: Moshi

Text can also be viewed as the coarsest level of the hierarchy

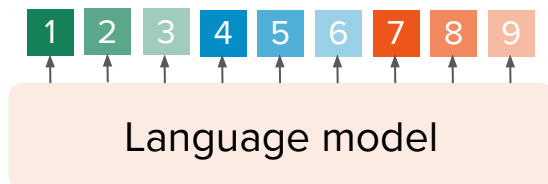
Hierarchical Generation



Model architectures for hierarchical generation

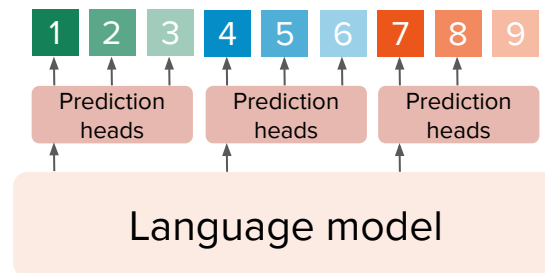
Assuming an interleaving setup

Single autoregressive LM



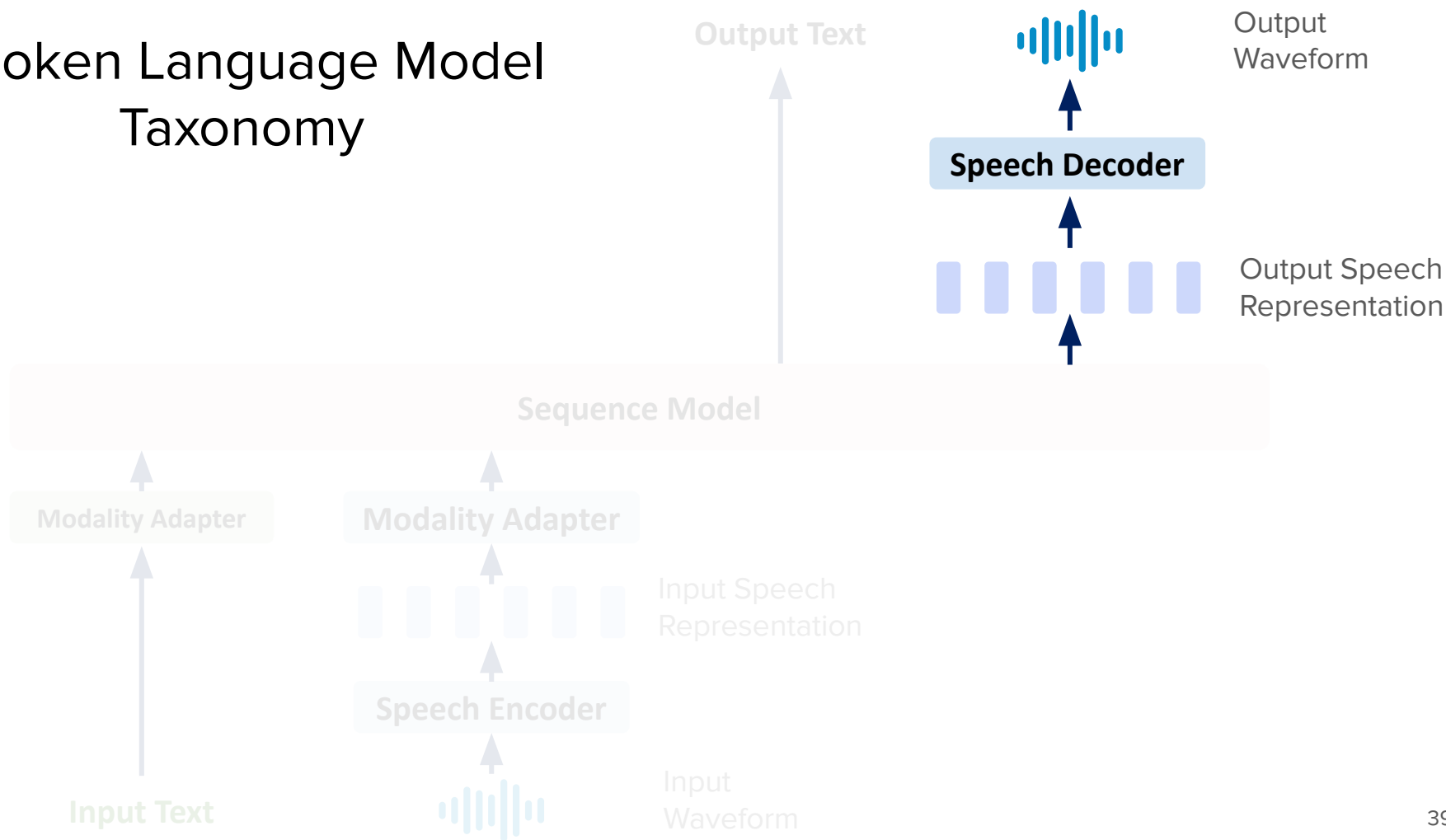
- Simple model
- Longer sequences

Codebook-wise prediction heads



- Reduced sequence length
- Prediction head designs:
 - Codebook-axis autoregressive heads, e.g., Moshi, Baichuan-Omni-1.5, Qwen3-Omni
 - Parallel heads, e.g., Mini-Omni, SALM-Duplex

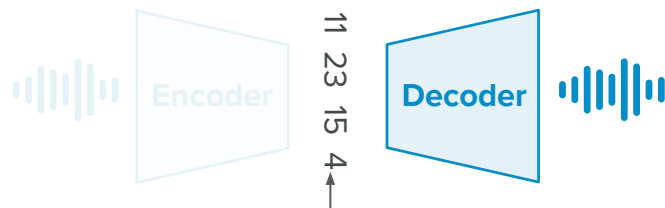
Spoken Language Model Taxonomy



Output Modules

If the encoder is trained for reconstruction

No matter continuous or discrete

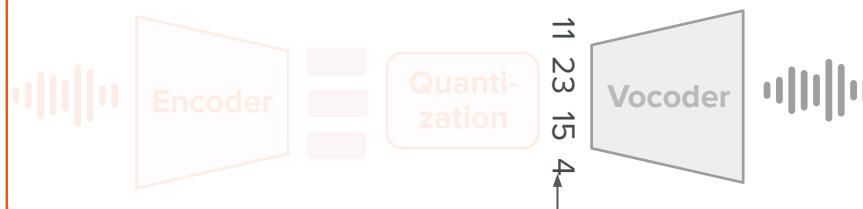


Sequence Model

- Reuse the original decoder
- Representation retains information needed for waveform generation
- VQ-VAE decoder, codec decoder, etc.

If the encoder is not trained for reconstruction

No matter continuous or discrete



Sequence Model

- Requires a separately trained decoder/vocoder to reconstruct waveforms
- Information discarded by the representation must be inferred (e.g., speaker, prosody, acoustic details, or temporal information lost through temporal compression)

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Agenda

01. Introduction

– Kai-Wei Chang

02. Speech representation

– Chung-Ming Chien

03. Typical Training

– Shinji Watanabe

04. Advanced Training

– Kai-Wei Chang

05. Hands-on Session

– Ke-Han Lu

06. Full-duplex

– Siddhant Arora

07. Evaluation

– Hung-yi Lee

08. Emerging Topics

– Chih-Kai Yang

09. Conclusion

– Karen Livescu



PART 03 · Shinji Watanabe

Typical Training Methods for Spoken Language Models

Roadmap for This Part

1

Where training enters an SLM

A one-slide recap of the anatomy, and what “pre-training” vs. “post-training” mean for SLMs

2

Three starting points

From scratch on speech · from a text LLM · from a conditional speech model (ASR/ST)

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Four pre-training objectives

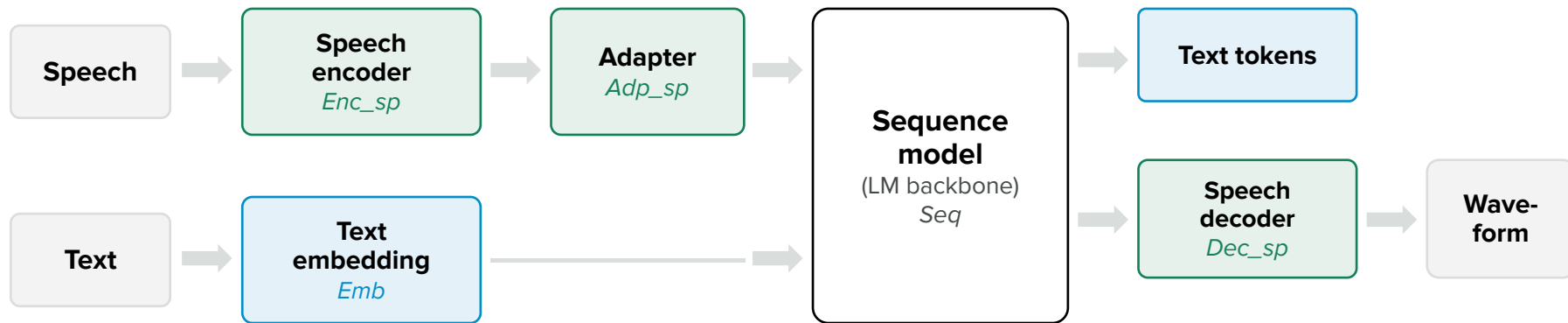
$p(\text{speech})$ · $p(\text{text}, \text{speech})$ · continual pre-training from $p(\text{text})$ · $p(\text{text} | \text{speech})$

4

Making the two modalities talk to each other

Implicit and explicit speech–text alignment, the data each objective needs, and a typical recipe

Recap: Where Training Enters an SLM



- Pre-training fixes **which distribution** the sequence model represents
- Warm starts everywhere: *Enc_sp* from SSL or ASR, *Seq* from a text LLM, *Dec_sp* from a codec
- Variations in learnable parameters: adapter-only → partial → full model

Notation follows Part 02 (§3 of the survey).

Which Distribution Are We Modeling?

Type of LM	Pre-training	Examples
pure text LM	$p(\text{text})$	GPT, LLaMA → ChatGPT
pure speech LM	$p(\text{speech})$	GSLM, AudioLM, TWIST → Align-SLM
speech + text LM	$p(\text{text}, \text{speech})$	SpiRit-LM, Moshi → Moshi, Mini-Omni
speech-aware text LM	$p(\text{text} \mid \text{speech}, \text{text})$	SALMONN, Qwen-Audio-Chat

The speech output models will be discussed in full-duplex systems

→ The taxonomy and the “universal” criteria come from **Part 01 (Kai-Wei)** · the post-training column is **Part 04 (Kai-Wei)**

Pre-training vs. Post-training

Pre-training

Any training that does not explicitly aim at making the model a universal speech processing system.

- Usually multi-stage: text next-token prediction → continual training on speech
- May train for specific tasks (e.g. ASR) to enhance modality alignment
- Includes the text LLM's own post-training

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Post-training

Training aimed at a wide range of downstream tasks, so the model becomes a more-or-less universal system.

- Task-specific multi-task training with task specifiers
- Instruction tuning — tasks named in natural language
- Preference optimization (RLHF, DPO)

The line is drawn by the goal, not the recipe: post-training a text LLM counts as pre-training of the SLM, and pre-training may still include supervised tasks.

→ Everything in the right-hand card is **Part 04 (Kai-Wei)**: instruction tuning, preference optimization, reasoning, tool calling

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Three Starting Points

A From scratch, on speech

Objective $p(\text{speech})$

You need unlabeled speech only

You get textless LM, prosody, speech output

Examples GSLM, AudioLM, SpeechSSM

B From a text LLM

Objective continual $p(\text{text}) \rightarrow p(\text{speech})$ or $p(\text{text}, \text{speech})$

You need a text LLM + speech (+ pairs)

You get the LLM's knowledge and reasoning

Examples TWIST, SpiRit-LM, Moshi, SALMONN

C From a conditional speech model

Objective $p(\text{text} | \text{speech})$

You need a trained ASR/ST model

You get speech–text alignment for free

Examples Whisper / OWSM as initialization

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Examples Whisper / OWSM as initialization

Most systems since 2024 start from B. A is the route for unwritten languages and speech-native generation; C buys an alignment stage you would otherwise have to pay for.

→ Route B is built live in **Hands-on 1 (Ke-Han)**: a frozen Whisper encoder and a trained MLP adapter on top of a text LLM

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p(speech): Pure Speech Pre-training

$$L = - \sum_t \log p(s_t \mid s_1, \dots, s_{t-1})$$

next-token prediction over discrete speech tokens s_t

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- Tokenize with an SSL encoder (HuBERT, WavLM, w2v-BERT, XEUS) plus k-means or VQ, then train a plain LM
- Non-phonetic streams can be modeled jointly: pitch and duration (pGSLM), or **codec tokens** in a coarse-to-fine hierarchy (AudioLM)
- **No text anywhere** in the pipeline — the spoken analogue of GPT pre-training

The line of work

GSLM (2021) first textless speech LM

pGSLM prosody-aware — pitch + duration

AudioLM hierarchical phonetic → codec tokens

TWIST warm start from a text LLM

SpeechSSM · Flow-SLM long-form; continuous features

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What you gain

Works for **unwritten languages**; keeps **prosody and other paralinguistics** that transcripts throw away.

The line of work

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What it costs

25–50 tokens/s means **long sequences**, **weak semantics**, and **long-form coherence** remain hard.

→ Where the tokens come from — **Part 02 (Chung-Ming)**: continuous vs. discrete, codec vs. phonetic, hierarchical generation

p(text, speech): Joint Pre-training

$$L = - \sum_t \log p(u_t | u_1, \dots, u_{t-1}) \quad u_t \in V_{\text{text}} \cup V_{\text{speech}} \text{ (interleaved)} \cdot u_t = (w_t, s_t) \text{ (parallel)}$$

Token layouts in detail: §3.3.2 of the survey.

→ **Part 02 (Chung-Ming)** gave the full layout taxonomy — coupling, alignment, formulation. Moshi's channels: **Part 06 (Siddhant)**

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(a) Interleaved — one stream, alignment decided in advance

*SpiRit-LM,
GLM-4-Voice*



(b) Parallel streams — text and speech as separate, time-aligned channels

text



Moshi

speech



- Text acts as a **semantic support** for speech generation — the two modalities share one loss
- Where does the **parallel aligned data** come from? Time-aligned ASR corpora, forced alignment, a text→speech-token model, or TTS at scale
- This is what lets a model take speech in, **“think in text”**, and speak back

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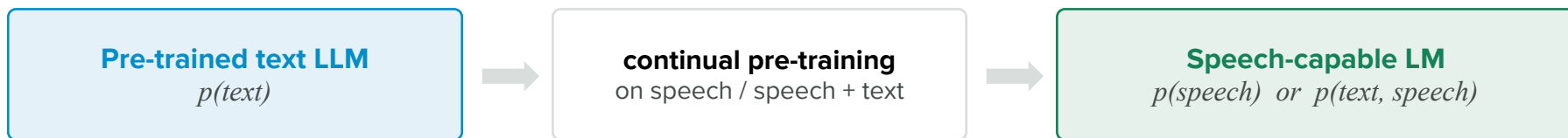
Continual Pre-training from $p(\text{text})$

$$\theta \leftarrow \theta_{\text{text LLM}}, \quad \textit{then} \quad L = - \sum_t \log p_{\theta}(u_t \mid u_1, \dots, u_{t-1})$$

→ Forgetting is the subject of **Hands-on 1 (Ke-Han)** · PEFT and LoRA are **Part 04 (Kai-Wei)**

Continual Pre-training from $p(\text{text})$

$$\theta \leftarrow \theta_{\text{text LLM}}, \text{ then } L = - \sum_t \log p_{\theta}(u_t \mid u_1, \dots, u_{t-1})$$



- **Same objective as before.** What changes is the initialization and the data mixture
- **The vocabulary is extended with speech tokens;** the text prior transfers to the speech side
- TWIST (Hassid et al., 2023): a textual warm start clearly beats a cold start on *sWUGGY* and *sBLIMP* — lexical and syntactic quality
- The same move underlies most **speech-aware text LMs**: keep the LLM, teach it to listen

The cost: the text ability drifts away. Keep text and replay data in the mix or use PEFT if we want to maintain the text capabilities (catastrophic forgetting)

→ Forgetting is the subject of **Hands-on 1 (Ke-Han)** · PEFT and LoRA are **Part 04 (Kai-Wei)**

p(text | speech): Conditional Pre-training

$$L = - \sum_n \log p(y_n | y_1, \dots, y_{n-1}, X) \quad X = \text{speech}, \quad y = \langle \text{task specifier} \rangle + \text{text output}$$

→ That bill is the next slide · **Part 02 (Chung-Ming)** supplies the module, **Hands-on 1 (Ke-Han)** freezes the LLM and trains only the adapter

$p(\text{text} \mid \text{speech})$: Conditional Pre-training

$$L = - \sum_n \log p(y_n \mid y_1, \dots, y_{n-1}, X) \quad X = \text{speech}, \quad y = \langle \text{task specifier} \rangle + \text{text output}$$

- Start from a conditional model instead of a generative one: an encoder–decoder ASR / ST system
- **Whisper** and **OWSM** are trained on huge paired speech–text with task-specifier tokens, so the modalities are aligned before SLM training starts

You inherit

- **tight speech–text alignment**
- **Use robust multilingual ASR and ST training data**

You give up

- a free-form generative text prior — the decoder was trained to transcribe (similar to continual pre-training)
- the instruction-following behavior of a chat LLM (Part 4)

So how do you add speech input and keep what the text LLM can already do?

Start from Whisper: alignment is free, the LLM's abilities are not. Start from the text LLM: its abilities are free — **alignment is the bill, and alignment training is how you pay it.**

→ Task-specifier tokens $\langle \text{IASR} \rangle$ $\langle \text{ISTI} \rangle$ come from **Part 01 (Kai-Wei)** · Whisper as a frozen encoder is the **Hands-on 1** setup

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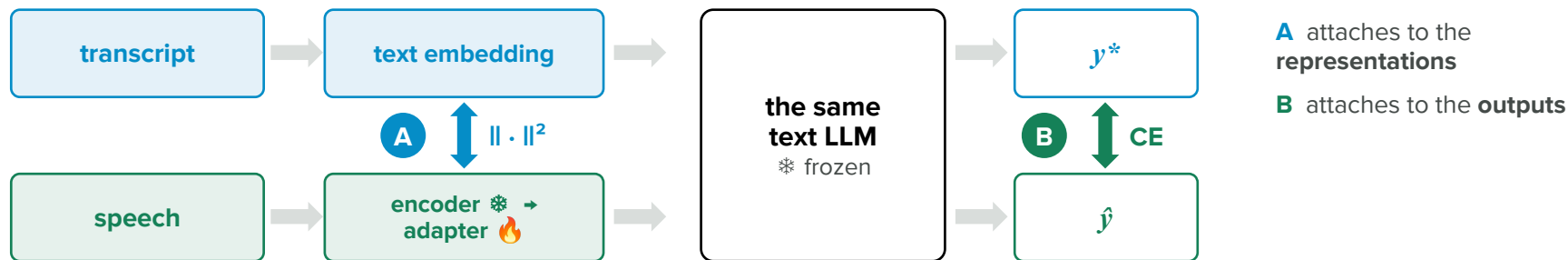
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Alignment as an Additional Training Objective

Same pipeline, same data. The only question is where you attach the loss.



A · Explicit — match the vectors

$$\min \| \text{Adp}_{\text{sp}}(\text{Enc}_{\text{sp}}(\text{speech})) - \text{Emb}(\text{text}) \|^2$$

You must decide which text embedding each speech frame matches
— frames outnumber tokens ~10:1. Wav2Prompt, DiVA.

B · Implicit — match the answers

$$\min \text{CE}(\hat{y}, y^*), \quad y^* = \text{LLM}(\text{transcript})$$

No target vector is named; alignment is whatever makes the LLM answer the same. AudioChatLlama, DeSTA.

→ Part 02 (Chung-Ming) compares the **adapter architectures**

Start From the Data You Already Have

The planning question: given what is already on your disk, which objective can you actually train?

Data you have	What it unlocks	Where it comes from in practice
unlabeled speech	$p(\text{speech})$	any large corpus — no transcripts, no alignment
aligned speech + text	$p(\text{text}, \text{speech}) \cdot \text{modality alignment}$	ASR corpora, forced alignment, text→token models, TTS
a pre-trained text LLM	$\text{continual pre-training from } p(\text{text})$	off the shelf — keep some of its text data for replay
a trained ASR / ST model	$p(\text{text} \mid \text{speech}) \text{ as an initialization}$	Whisper, OWSM — the pairing cost is already paid

Speech burns context far faster

1 s of speech $\approx$ 25 tokens at 25 Hz, against $\approx$ 3 text tokens — it shows in your compute budget before it shows in your loss.

Synthesized data is now a first-class ingredient

Matured text-to-speech and speech-to-text changed the story

Synthesizing the Data

The scarce resource is the paired one. Two abundant sources plus two mature technologies manufacture it.

Text data

Text-LLM corpora — or simply written by a pure text LM

✓ synthetic text already works

Speech data

Large speech-only corpora — but not yet synthesizable

✗ synthetic speech does not, yet

Mature speech-to-text and text-to-speech ASR and TTS now run reliably over a whole corpus — that is what changed.

Paired, aligned speech + text

the one ingredient that was actually scarce

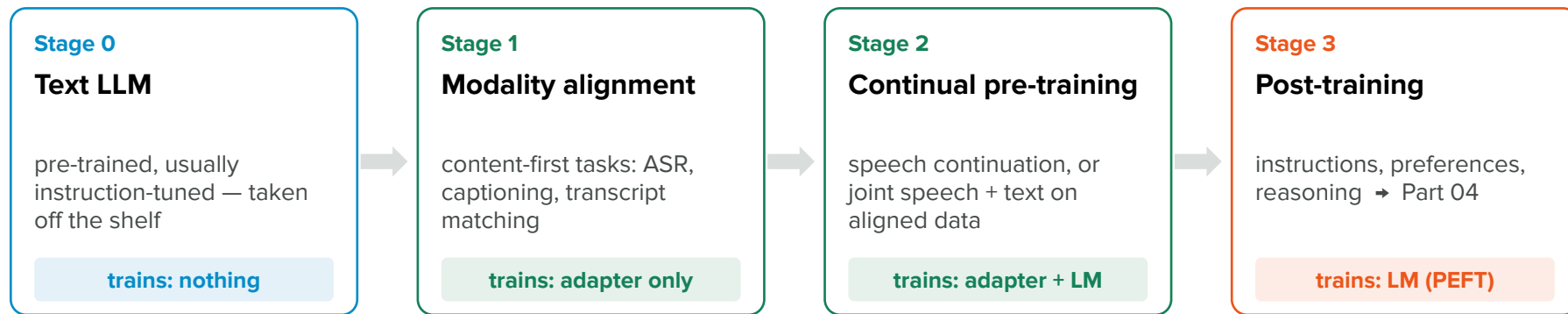
$p(\text{text}, \text{speech})$

speech + text LM

$p(\text{text} \mid \text{speech}, \text{text})$

speech-aware text LM

A Typical Multi-Stage Recipe



- **Progressive fine-tuning:** start with content-heavy tasks, add paralinguistic and compositional ones later (SALMONN, Tang et al., 2024; Hu et al., 2024)
- **Unfreeze gradually** — few components early, more later — to keep training stable
- **Replay pre-training data** during post-training to limit forgetting (VoiceTextBlender, Peng et al., 2025; DeSTA2, Lu et al., 2025)

→ Stage 3 is **Part 04 (Kai-Wei)** · **Hands-on 1 (Ke-Han)** runs Stages 1–3 live in a Colab notebook

An Extreme Case: Audio Flamingo 3

Seven training blocks, one model. AF3 splits our Stage 1 in three, our Stage 3 in three.

stage (previous slide)	Audio Flamingo 3 block	trains	data / context
Stage 0 Text LLM	Qwen-2.5-7B (backbone LLM)	nothing	taken off the shelf
Stage 1 Modality alignment Stage 2 Continual pre-training	AF-Whisper encoder pre-training	encoder + decoder	GPT-4.1 audio captions
	AF3 Stage 1 — Alignment pre-training	adapter only	recognition data, ≤ 30 s
	AF3 Stage 2 — Encoder tuning	encoder + adapter	recognition data, ≤ 30 s
Stage 3 Post-training	AF3 Stage 3 — Full fine-tuning	everything	AudioSkills-XL + QA, ≤ 2.5 min
	AF3 Stage 3.5 — Context + thinking	LoRA on the LLM	LongAudio-XL + AF-Think → AF3
	AF3 Stage 4 — Chat and voice	everything	AF-Chat → AF3-Chat
not in the recipe	Streaming TTS module	a separate decoder	audio tokens from the LLM's text

Audio Flamingo 3 (Goel et al., NeurIPS 2025; arXiv:2507.08128), trained on 128 A100-80GB. AF-Whisper starts from Whisper large-v3 and is trained by audio captioning — a $p(\text{text} | \text{speech})$ objective, used to build an encoder.

→ The LoRA row is **Part 04 (Kai-Wei)** · the frozen-encoder row is the setup of **Hands-on 1 (Ke-Han)**

Open Recipes You Can Actually Run

VoxLM

Maiti et al., ICASSP 2024

one decoder-only model for ASR, TTS, speech LM and text LM — a compact worked example of $p(\text{text}, \text{speech})$

OpusLM

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a family of open unified speech LMs: released data recipe, checkpoints and training configs

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toolkit that packages the stages of this part — tokenization, pre-training, post-training

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Reproducibility is still the exception: very few SLMs release code, checkpoints and data together (§8 of the survey).

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→ Runnable today: Ke-Han’s repo in **Hands-on 1** · “more fully open models” is the first open challenge in **Part 09 (Karen)**

Choosing an Objective

The decision question: what does each objective give you, what does it cost, and when is it the right one?

Objective	What you get	What it costs	Choose it when
$p(\text{speech})$	textless LM, prosody, speech output	long sequences; weak semantics	the language has no text, or speech-native generation is the goal

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$p(\text{text}, \text{speech})$	speech in and out, reasoning carried by text	you must build and maintain the alignment	you need speech-to-speech with text-level control

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The decision question: what does each objective give you, what does it cost, and when is it the right one?

Objective	What you get	What it costs	Choose it when
$p(\text{speech})$	textless LM, prosody, speech output	long sequences; weak semantics	the language has no text, or speech-native generation is the goal
$p(\text{text}, \text{speech})$	speech in and out, reasoning carried by text	you must build and maintain the alignment	you need speech-to-speech with text-level control
$\text{continual } p(\text{text})$	the LLM's knowledge and reasoning	the text ability drifts away	you have an LLM and want it to listen

Choosing an Objective

The decision question: what does each objective give you, what does it cost, and when is it the right one?

Objective	What you get	What it costs	Choose it when
$p(\text{speech})$	textless LM, prosody, speech output	long sequences; weak semantics	the language has no text, or speech-native generation is the goal
$p(\text{text}, \text{speech})$	speech in and out, reasoning carried by text	you must build and maintain the alignment	you need speech-to-speech with text-level control
$\text{continual } p(\text{text})$	the LLM's knowledge and reasoning	the text ability drifts away	you have an LLM and want it to listen
$p(\text{text} \mid \text{speech})$	alignment for free; multilingual robustness	no free-form generative prior	understanding tasks, and Whisper or OWSM is already trained

In practice these compose rather than compete.

The dominant 2024–2026 pattern: text LLM → speech–text alignment → joint speech–text continuation → post-training.

Takeaways

- 1 Pre-training is a choice of distribution**
p(speech), p(text, speech), continual from p(text), or p(text | speech) — everything else follows from it
- 2 Almost everything in practice is a warm start**
the interesting work has moved to speech–text alignment and to the data that supports it
- 3 What you freeze matters as much as what you optimize**
adapter-only, partial, or full updates give very different models from the same objective
- 4 Every objective has a characteristic failure mode**
sequence length, alignment cost, forgetting, or a missing generative prior — pick the one you can pay for

Next — Part 04: turning a pre-trained SLM into a universal, instruction-following one.

References

Survey

S. Arora et al. On the landscape of spoken language models: a comprehensive survey. TMLR, 2025. (arXiv:2504.08528 — §4)

$p(\text{speech})$

Lakhotia et al. On generative spoken language modeling from raw audio. TACL, 2021. · Kharitonov et al. Text-free prosody-aware generative spoken language modeling. ACL, 2022. · Borsos et al. AudioLM. TASLP, 2023. · Hassid et al. Textually pretrained speech language models. NeurIPS, 2023. · Park et al. Long-form speech generation with spoken language models. ICML, 2025. · Chou et al. Flow-SLM. ASRU, 2025.

$p(\text{text}, \text{speech})$

Chou et al. Toward joint language modeling for speech units and text. Findings of EMNLP, 2023. · Nguyen et al. SpiRit-LM. TACL, 2025. · Défossez et al. Moshi. arXiv:2410.00037, 2024. · Zeng et al. GLM-4-Voice. arXiv:2412.02612, 2024. · Zeng et al. Scaling speech-text pre-training with synthetic interleaved data. ICLR, 2025.

$p(\text{text} | \text{speech})$

Radford et al. Robust speech recognition via large-scale weak supervision. ICML, 2023. · Peng et al. OWSM. ASRU, 2023. · Peng et al. OWSM v4. Interspeech, 2025. · Lai et al. Instruction-following speech recognition. arXiv:2309.09843, 2023. · Arora et al. UniverSLU. NAACL, 2024.

Alignment

Fathullah et al. AudioChatLlama. NAACL, 2024. · Wang et al. SLM: bridge the thin gap between speech and text foundation models. ASRU, 2023. · Lu et al. DeSTA. Interspeech, 2024. · Lu et al. DeSTA2. ICASSP, 2025. · Deng et al. Wav2Prompt. NAACL, 2025. · Held et al. Distilling an end-to-end voice assistant without instruction training data. ACL, 2025.

Recipes & post-training

Tang et al. SALMONN. ICLR, 2024. · Chu et al. Qwen-Audio. arXiv:2311.07919, 2023. · Maiti et al. VoxLM. ICASSP, 2024. · Tian et al. OpusLM. Interspeech, 2025. · Tian et al. ESPnet-SpeechLM. 2025. · Maimon et al. Slamming: training a speech language model on one GPU in a day. Findings of ACL, 2025. · Peng et al. VoiceTextBlender. NAACL, 2025.

Agenda

01. Introduction

– Kai-Wei Chang

02. Speech representation

– Chung-Ming Chien

03. Typical Training

– Shinji Watanabe

04. Advanced Training

– Kai-Wei Chang

05. Hands-on Session

– Ke-Han Lu

06. Full-duplex

– Siddhant Arora

07. Evaluation

– Hung-yi Lee

08. Emerging Topics

– Chih-Kai Yang

09. Conclusion

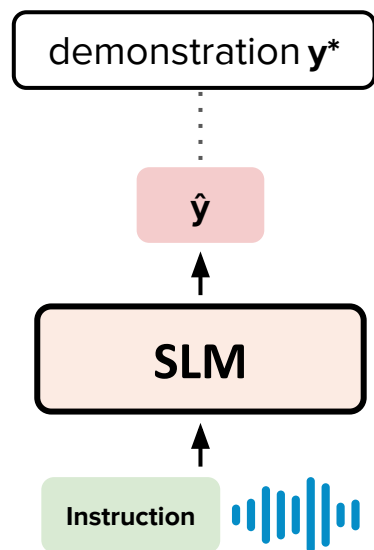
– Karen Livescu



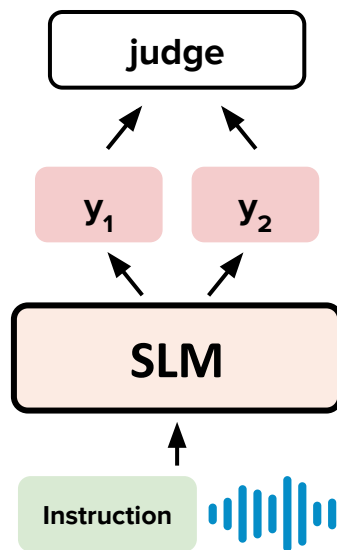
PART 04 · Kai-Wei Chang

Advanced Training Methods for Spoken Language Models

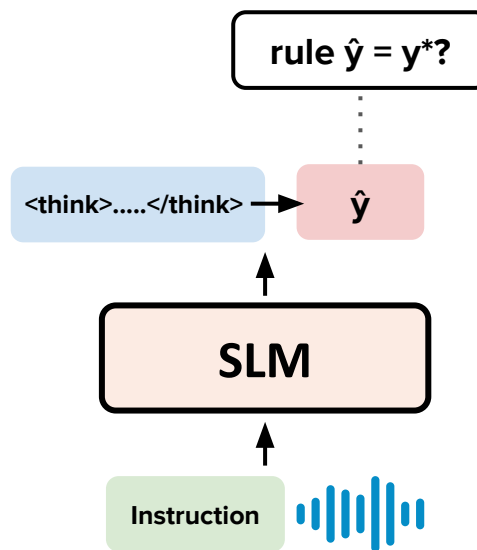
1. Instruction Tuning



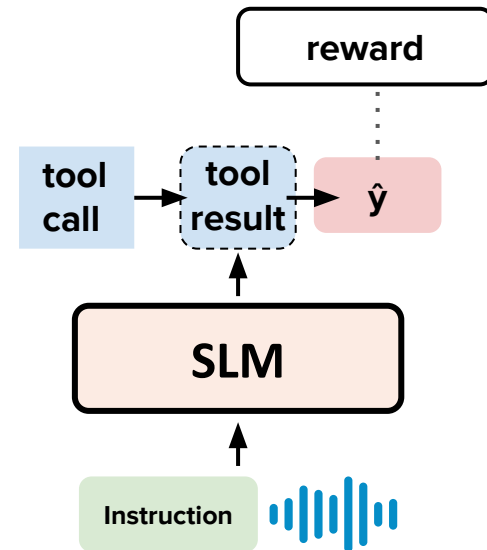
2. Preference Optimization



3. Reasoning



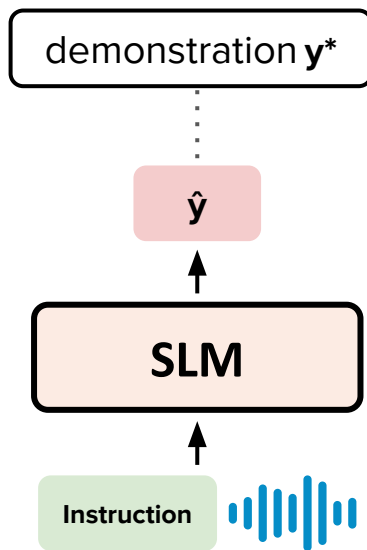
4. Tool Calling



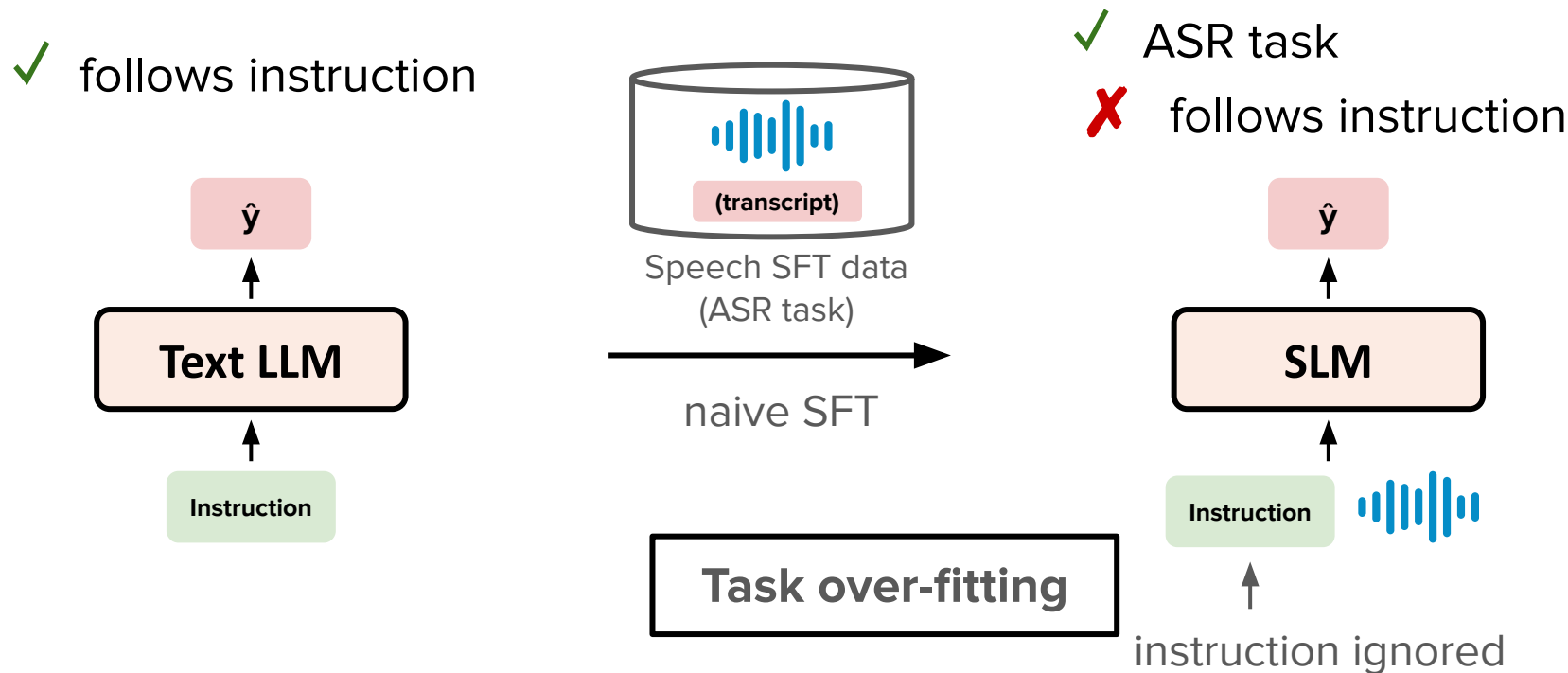
1. Instruction Tuning
2. Preference Optimization
3. Reasoning
4. Tool Calling

Advanced SLMs combine multiple training methods

Instruction Tuning

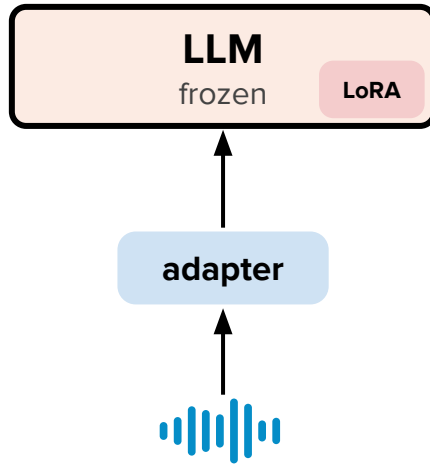


Typical Training – Modality alignment with ASR

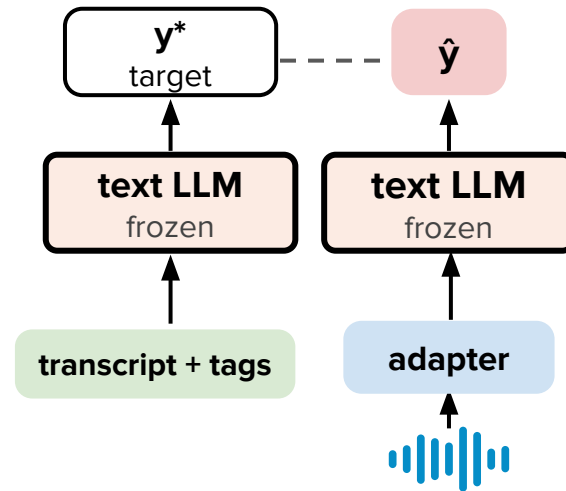


Instruction Tuning

1. Update fewer weights



2. Self-generation

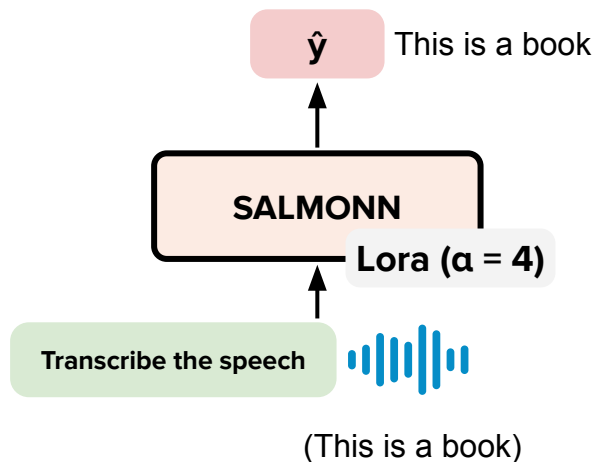


Keep the LLM!

Instruction Tuning – Update fewer weights

E.g. SALMONN

Instruction Tuning (ASR, Captioning)

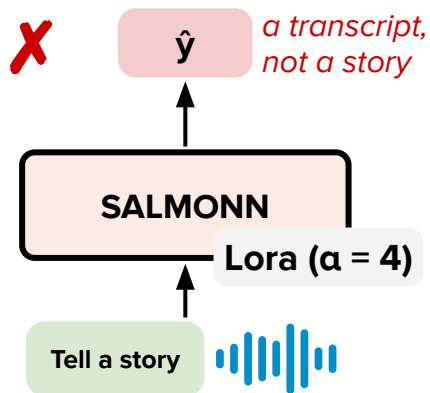


Instruction Tuning – Update fewer weights

E.g. SALMONN

Instruction Tuning (ASR, Captioning)

→ Task over-fitting



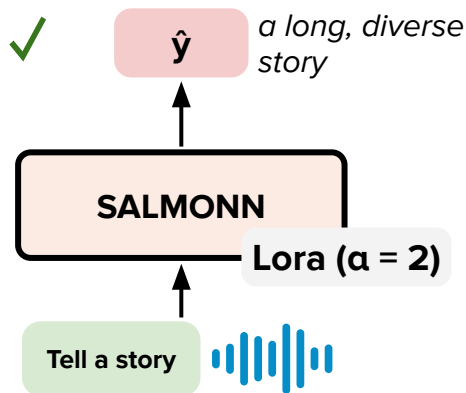
trained task
(ASR)



new task
(storytelling)

X a transcript

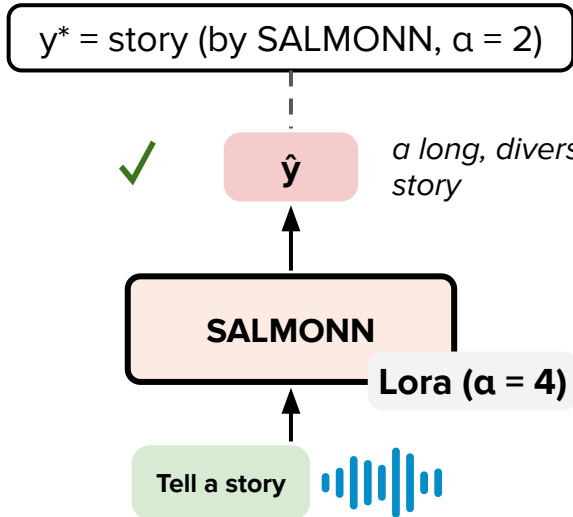
Test-time trick
LoRA Scaling Factor



! ASR drops



Activation tuning



Instruction Tuning – Update fewer weights

E.g. SALMONN

	task over-fitting	$\alpha = 2$ at test time	activation tuning
ASR %WER ↓	2.1	≈ 5	2.1
storytelling diversity ↑	7.8	≈ 77	82.6
storytelling following rate ↑	0.00	1.00	1.00

ASR - LibriSpeech test-clean

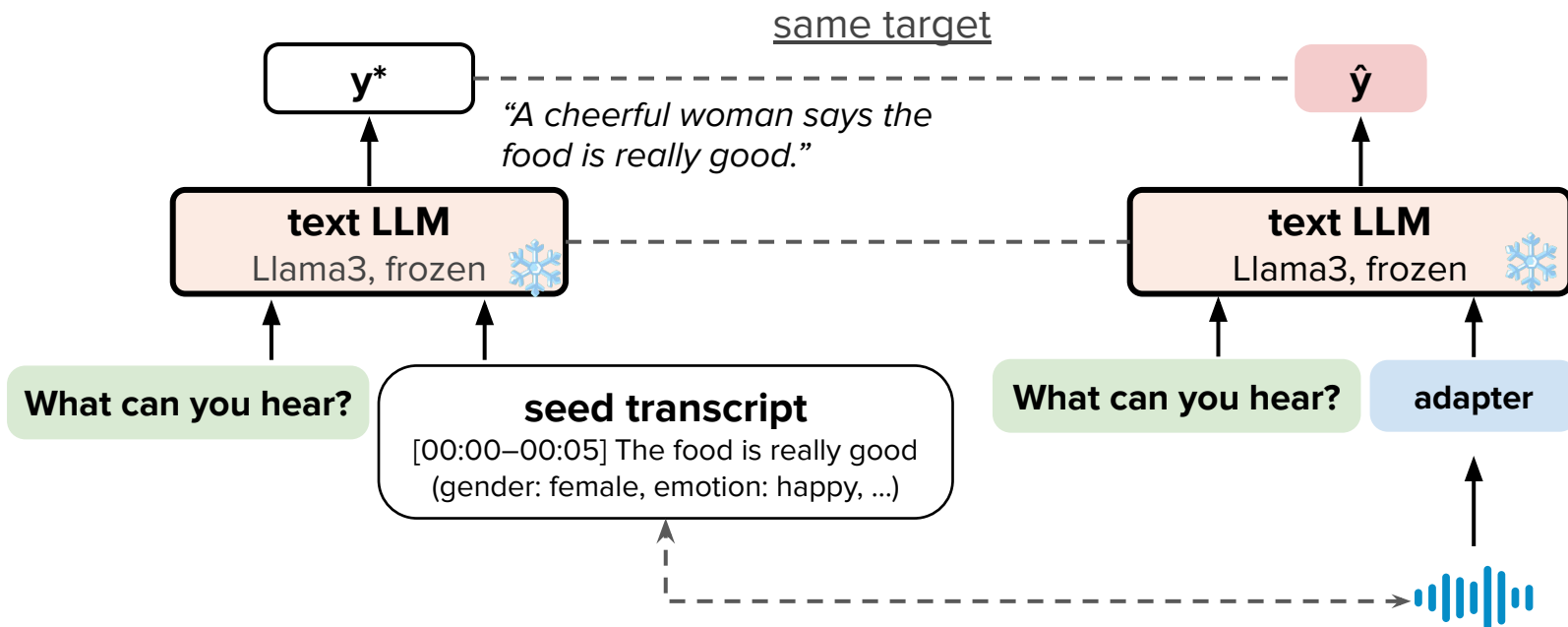
Storytelling - AudioCaps

Instruction Tuning – Self-generation

E.g. DeSTA

LLM generates the target

SLM generates the same from audio

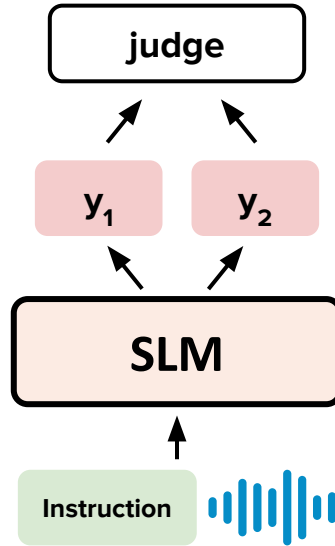


Model	IF rate (%) Speech-IFEval	Δ vs. own text LLM
Llama3-8B-Instruct (text reference)	92.5	–
LTU-AS	29.2	–54.9
SALMONN	36.9	–50.2
DeSTA2	89.2	–3.6

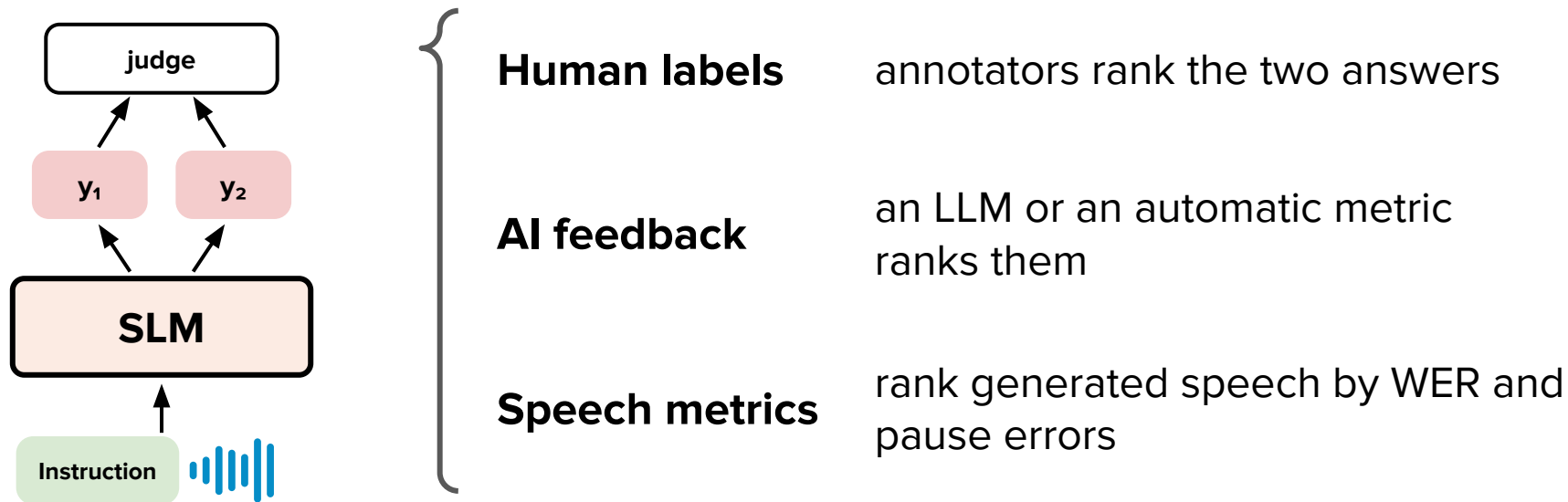
Speech-IFEval: A benchmark for evaluating speech instruction following, e.g., output JSON or exactly 4 bullet points.

Model	Dynamic-SUPERB	MMAU	IF rate (%) Speech-IFEval	Training Speech
Qwen2-Audio	51.7	49.2	47.1	510k hours
DeSTA2.5-Audio	69.5	57.5	93.9	7k hours

Preference Optimization

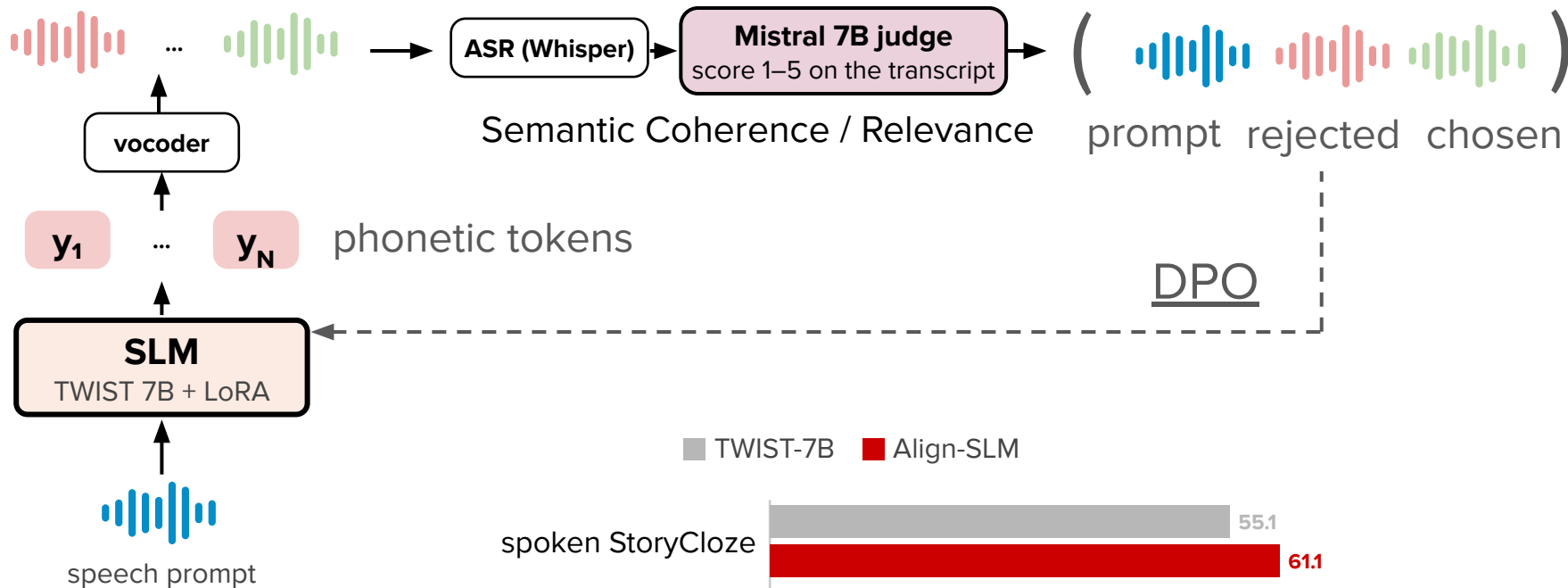


2. Preference optimization



Preference optimization – AI Feedback

E.g. Align-SLM



TWIST

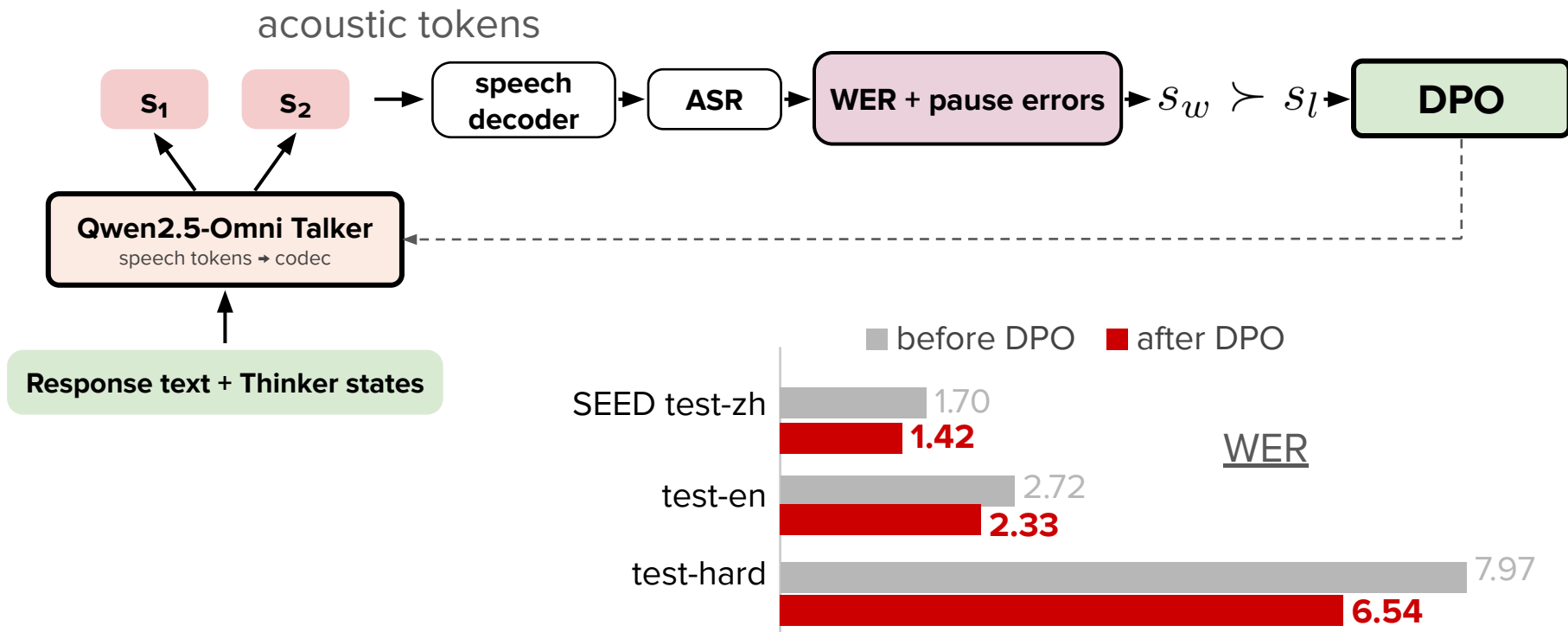
- Pure Speech LM
- speech continuation

StoryCloze: Tests whether a model can identify a semantically coherent story continuation.

Align-SLM: Lin et al. 2411.01834

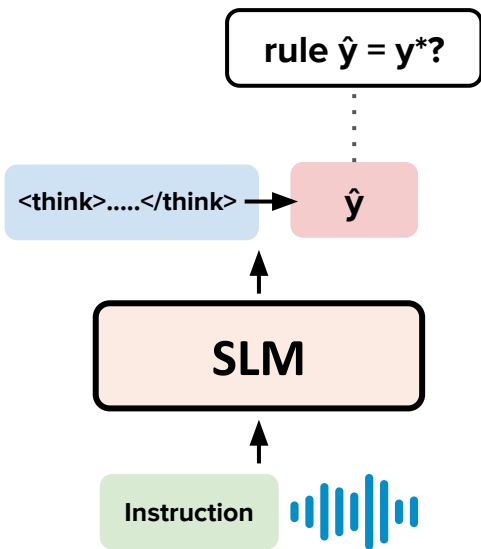
Preference optimization — Speech metrics

E.g. Qwen-2.5-Omni Talker



Qwen2.5-Omni: Qwen team 2503.20215, Sec. 4.3, Table 9

Reasoning



“Is the speaker angry?”

(I said no, twice!)

Instruction



SLM



$\hat{y} =$

**Instruction
following**

“Yes, she sounds angry.”

Reasoning

<think> raised pitch, loud, fast speech rate → anger
rather than excitement </think>



“Yes, she sounds angry.”

Reasoning: Condition the answer on
the reasoning process.



Reasoning - Direct Prompting

Thinking process in an example

arXiv:2201.11903v6 [cs.CL] 10 Jan 2023

Chain-of-Thought Prompting Elicits Reasoning in Large Language Models

Jason Wei Xuezhi Wang Dale Schuurmans Maarten Bosma
Brian Ichter Fei Xia Ed H. Chi Quoc V. Le Denny Zhou

Google Research, Brain Team
{jasonwei,dennyzhou}@google.com

Abstract

We explore how generating a *chain of thought*—a series of intermediate reasoning steps—significantly improves the ability of large language models to perform complex reasoning. In particular, we show how such reasoning abilities emerge naturally in sufficiently large language models via a simple method called *chain-of-thought prompting*, where a few chain of thought demonstrations are provided as exemplars in prompting.

Experiments on three large language models show that chain-of-thought prompting improves performance on a range of arithmetic, commonsense, and symbolic reasoning tasks. The empirical gains can be striking. For instance, prompting a PaLM 540B with just eight chain-of-thought exemplars achieves state-of-the-art accuracy on the GSM8K benchmark of math word problems, surpassing even finetuned GPT-3 with a verifier.

Standard Prompting

Model Input

Q: Roger has 5 tennis balls. He buys 2 more cans of tennis balls. Each can has 3 tennis balls. How many tennis balls does he have now?

A: The answer is 11.

Q: The cafeteria had 23 apples. If they used 20 to make lunch and bought 6 more, how many apples do they have?

Model Output

A: The answer is 27. ❌

Chain-of-Thought Prompting

Model Input

Q: Roger has 5 tennis balls. He buys 2 more cans of tennis balls. Each can has 3 tennis balls. How many tennis balls does he have now?

A: Roger started with 5 balls. 2 cans of 3 tennis balls each is 6 tennis balls. $5 + 6 = 11$. The answer is 11.

Q: The cafeteria had 23 apples. If they used 20 to make lunch and bought 6 more, how many apples do they have?

Model Output

A: The cafeteria had 23 apples originally. They used 20 to make lunch. So they had $23 - 20 = 3$. They bought 6 more apples, so they have $3 + 6 = 9$. The answer is 9. ✅

Figure 1: Chain-of-thought prompting enables large language models to tackle complex arithmetic, commonsense, and symbolic reasoning tasks. Chain-of-thought reasoning processes are highlighted.

36th Conference on Neural Information Processing Systems (NeurIPS 2022).

Model Input

Q: Roger has 5 tennis balls. He buys 2 more cans of tennis balls. Each can has 3 tennis balls. How many tennis balls does he have now?

A: Roger started with 5 balls. 2 cans of 3 tennis balls each is 6 tennis balls. $5 + 6 = 11$. The answer is 11.

Q: The cafeteria had 23 apples. If they used 20 to make lunch and bought 6 more, how many apples do they have?

Model Output

A: The cafeteria had 23 apples originally. They used 20 to make lunch. So they had $23 - 20 = 3$. They bought 6 more apples, so they have $3 + 6 = 9$. The answer is 9. ✅

Few-shot CoT prompting

Reasoning - Direct Prompting

arXiv:2205.11916v4 [cs.CL] 29 Jan 2023

Large Language Models are Zero-Shot Reasoners

Takeshi Kojima
The University of Tokyo
t.kojima@web.lab.t.u-tokyo.ac.jp

Shixiang Shane Gu
Google Research, Brain Team

Machel Reid
Google Research*

Yutaka Matsuo
The University of Tokyo

Yusuke Iwasawa
The University of Tokyo

Abstract

Pretrained large language models (LLMs) are widely used in many sub-fields of natural language processing (NLP) and generally known as excellent *few-shot* learners with task-specific exemplars. Notably, chain of thought (CoT) prompting, a recent technique for eliciting complex multi-step reasoning through step-by-step answer examples, achieved the state-of-the-art performances in arithmetics and symbolic reasoning, difficult *system-2* tasks that do not follow the standard scaling laws for LLMs. While these successes are often attributed to LLMs' ability for few-shot learning, we show that LLMs are decent *zero-shot* reasoners by simply adding "Let's think step by step" before each answer. Experimental results demonstrate that our Zero-shot-CoT, using the same single prompt template, significantly outperforms zero-shot LLM performances on diverse benchmark reasoning tasks including arithmetics (MultiArith, GSM8K, AQUA-RAT, SVAMP), symbolic reasoning (Last Letter, Coin Flip), and other logical reasoning tasks (Date Understanding, Tracking Shuffled Objects), without any hand-crafted few-shot examples, e.g. increasing the accuracy on MultiArith from 17.7% to 78.7% and GSM8K from 10.4% to 40.7% with large-scale InstructGPT model (text-davinci-002), as well as similar magnitudes of improvements with another off-the-shelf large model, 540B parameter PaLM. The versatility of this single prompt across very diverse reasoning tasks hints at untapped and understudied fundamental *zero-shot* capabilities of LLMs, suggesting high-level, multi-task broad cognitive capabilities may be extracted by simple prompting. We hope our work not only serves as the minimal strongest zero-shot baseline for the challenging reasoning benchmarks, but also highlights the importance of carefully exploring and analyzing the enormous zero-shot knowledge hidden inside LLMs before crafting finetuning datasets or few-shot exemplars.

1 Introduction

Scaling up the size of language models has been key ingredients of recent revolutions in natural language processing (NLP) [Vaswani et al., 2017, Devlin et al., 2019, Raffel et al., 2020, Brown et al., 2020, Thoppilan et al., 2022, Rae et al., 2021, Chowdhery et al., 2022]. The success of large language models (LLMs) is often attributed to (in-context) few-shot or zero-shot learning. It can solve various tasks by simply conditioning the models on a few examples (few-shot) or instructions describing the task (zero-shot). The method of conditioning the language model is called "prompting" [Liu et al., 2021b], and designing prompts either manually [Schick and Schitze, 2021, Reynolds and McDonell, 2021] or automatically [Gao et al., 2021, Shin et al., 2020] has become a hot topic in NLP.

*Work done while at The University of Tokyo.

Q: A juggler can juggle 16 balls. Half of the balls are golf balls, and half of the golf balls are blue. How many blue golf balls are there?

A: **Let's think step by step.**

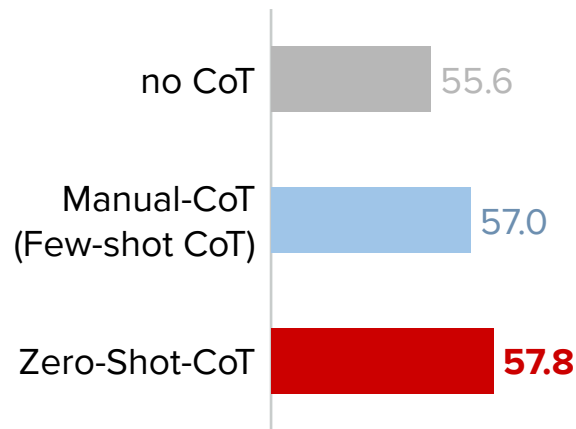
(Output) *There are 16 balls in total. Half of the balls are golf balls. That means that there are 8 golf balls. Half of the golf balls are blue. That means that there are 4 blue golf balls. ✓*

“Let’s think step by step”

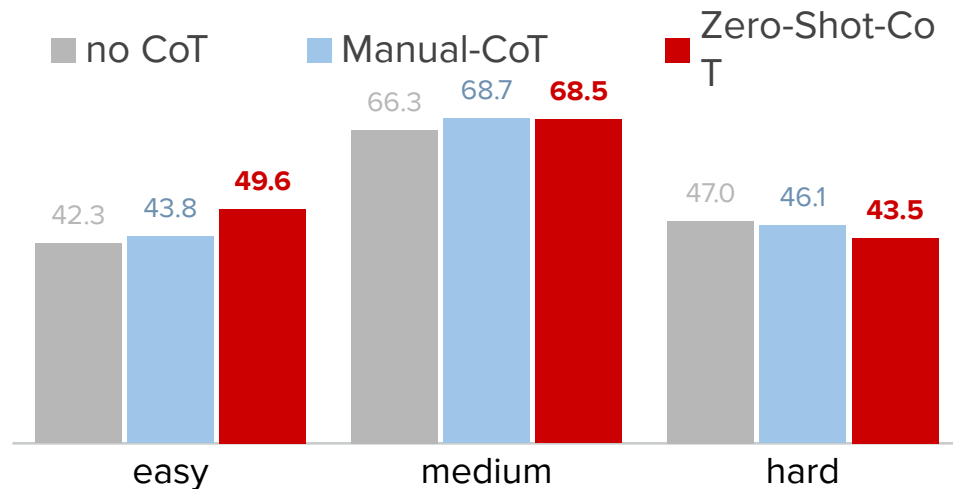
zero-shot CoT prompting

Reasoning - Direct Prompting

E.g. Audio-CoT

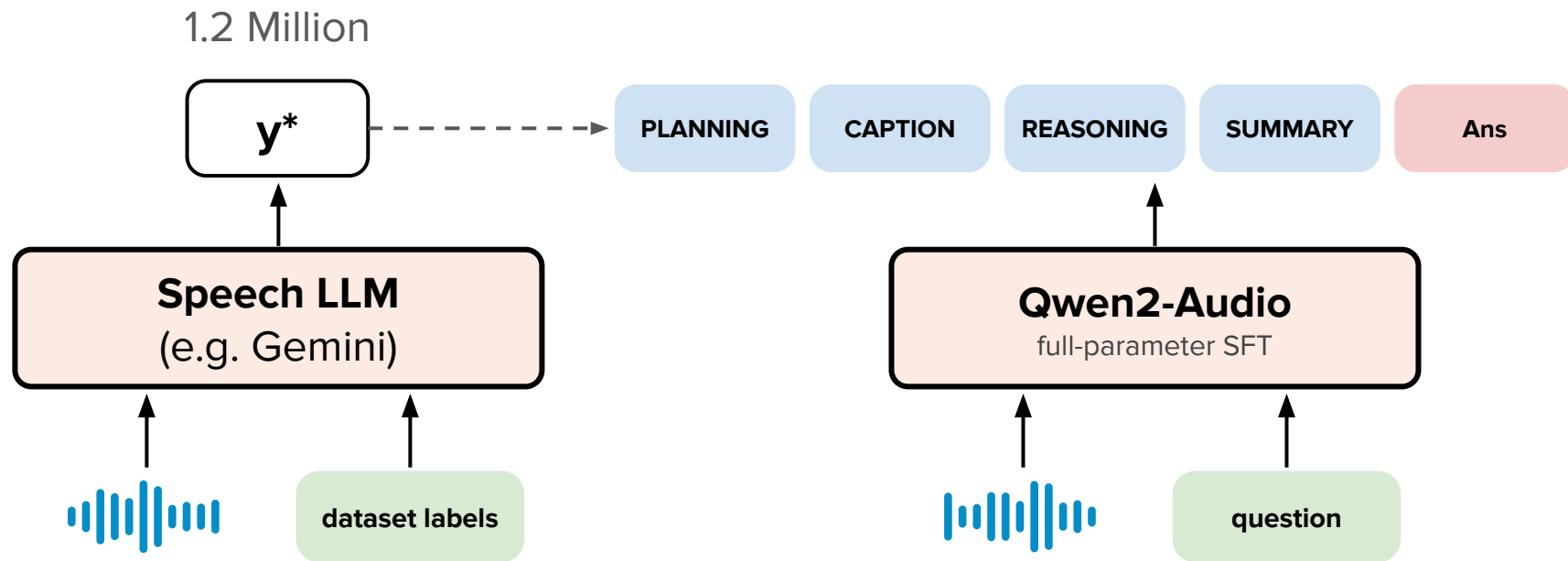


MMAU test-mini
Qwen2-Audio



Reasoning - SFT Training

E.g. Audio-Reasoner



Reasoning - SFT Training

E.g. Audio-Reasoner

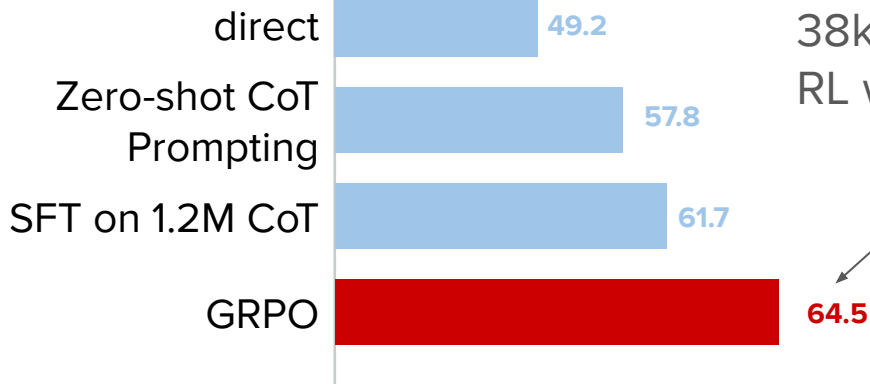
	Qwen2-Audio	Audio-Reasoner
MMAU speech	42.0	60.7
MMAU music	51.0	64.3
MMAU sound	55.0	60.1
MMAU test-mini (avg.)	49.2	61.7

Reasoning - RLVR Training

E.g. R1-AQA

Qwen2-Audio-7B-Instruct

Audio-Reasoner:
1.2M CoT from Gemini



38k data

RL with Verifiable Reward

- Verifiable rewards beat supervised chains

“Is the speaker angry?”

Instruction



(I said no, twice!)



$\hat{y} =$

Output conditioned on intermediate tokens (SFT, RLVR, ...)

Instruction following

“Yes, she sounds angry.”

Reasoning

<think> raised pitch, loud, fast speech rate → anger rather than excitement </think>

“Yes, she sounds angry.”

Time-aware
(Reasoning about time)

“Yes,

<0.4 seconds>

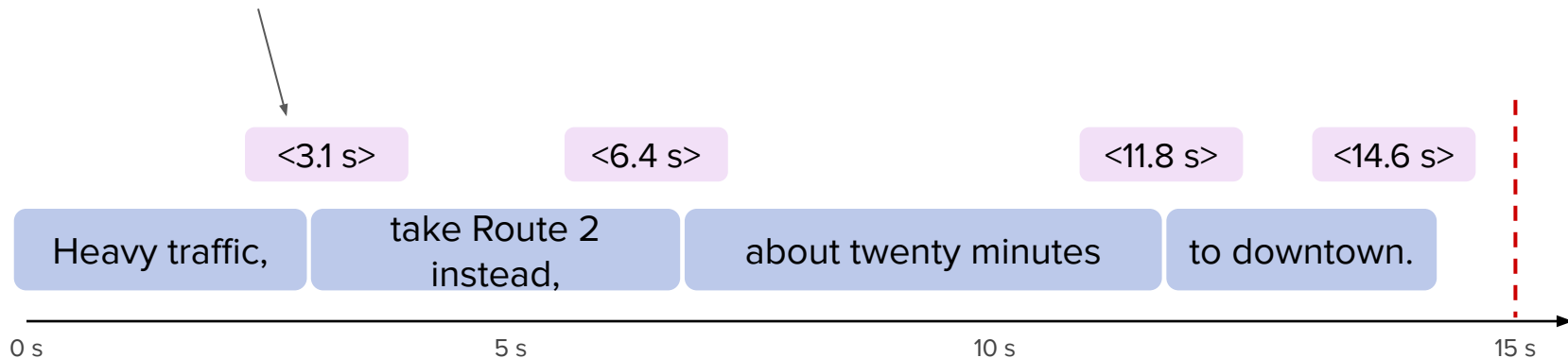
she sounds angry.”

<1.3 seconds>

control the output response duration

TiCo — Time-Controllable Spoken Dialogue Model

Spoken Time Markers: the model writes how long it has been speaking.



Instruction: “Give me a traffic update, in 15 seconds please.”

Target T = 15 s

TiCo Training: Stage 1 - Time Awareness Training

Self-generation

Alignment

Well, milk offers protein, calcium, and vitamins
that support strong bones, healthy muscles, and
overall growth as part of a balanced diet.

well: 0.3s - 0.9s

milk: 1.2s - 1.8s

...

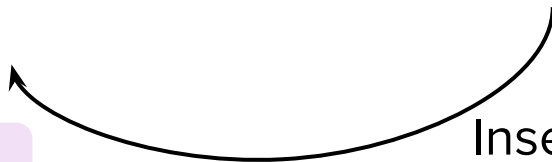
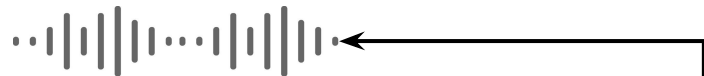
diet: 9.4s - 10.0s

Qwen-Omni

Insert

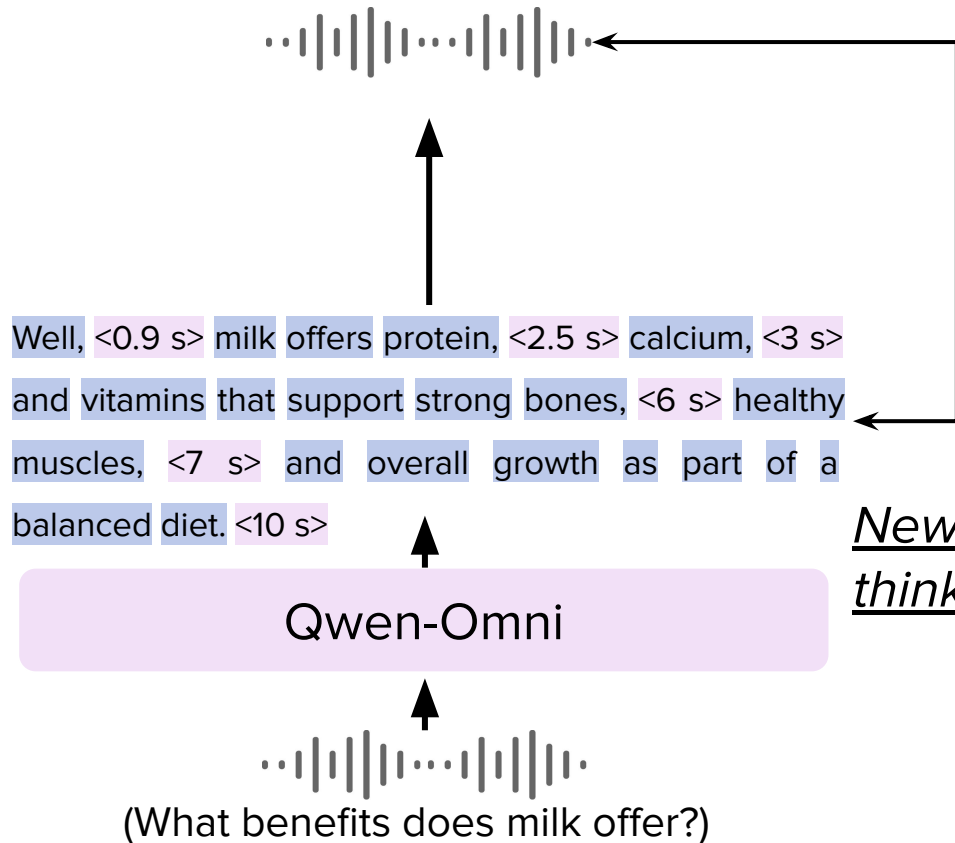
(What benefits does milk offer?)

Dataset: InstructS2S subset
6K user queries Fang+ [ICLR 2025]



TiCo Training: Stage 1 - Time Awareness Training

Self-generation



Alignment

well: 0.3s - 0.9s

milk: 1.2s - 1.8s

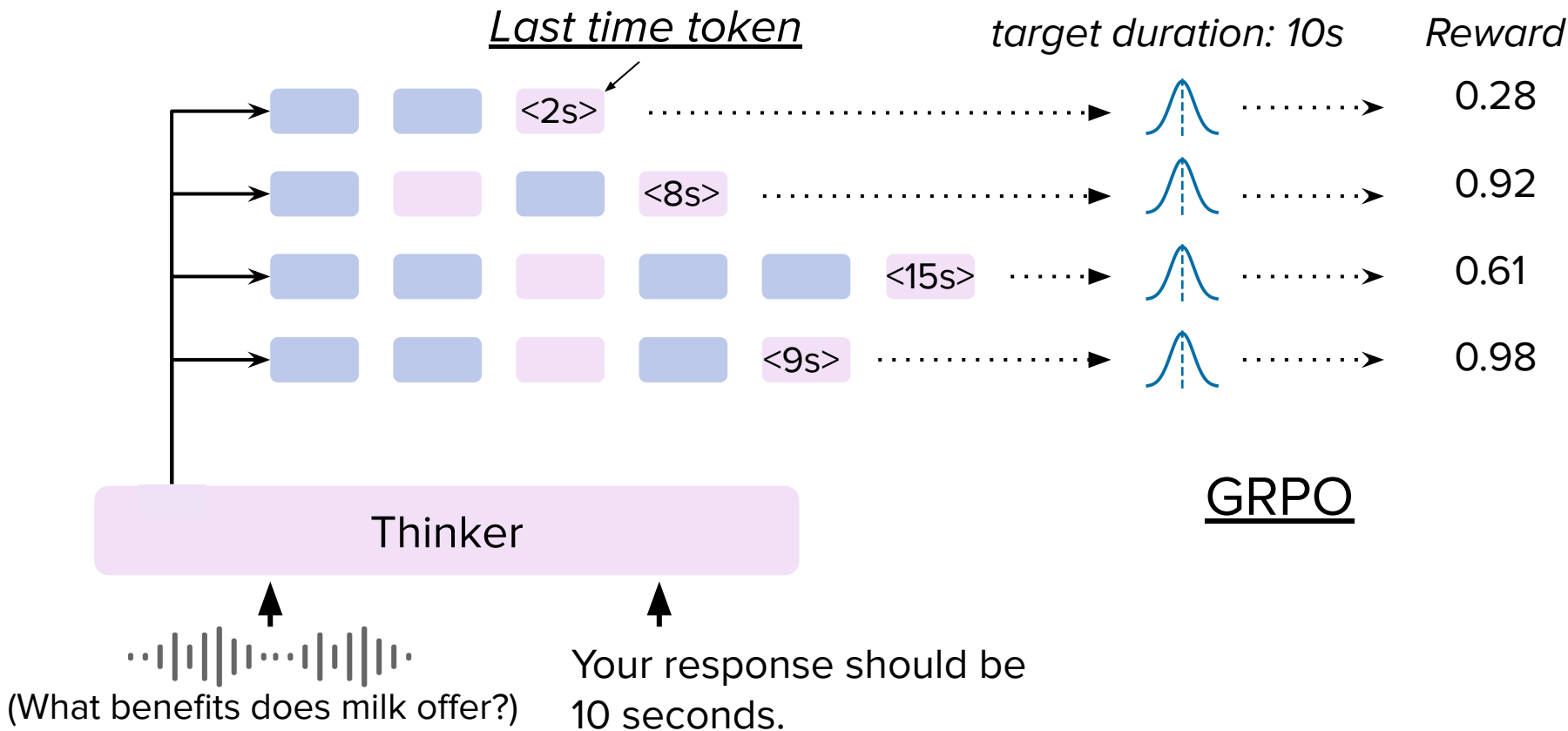
...

diet: 9.4s - 10.0s

New prediction target for the thinker

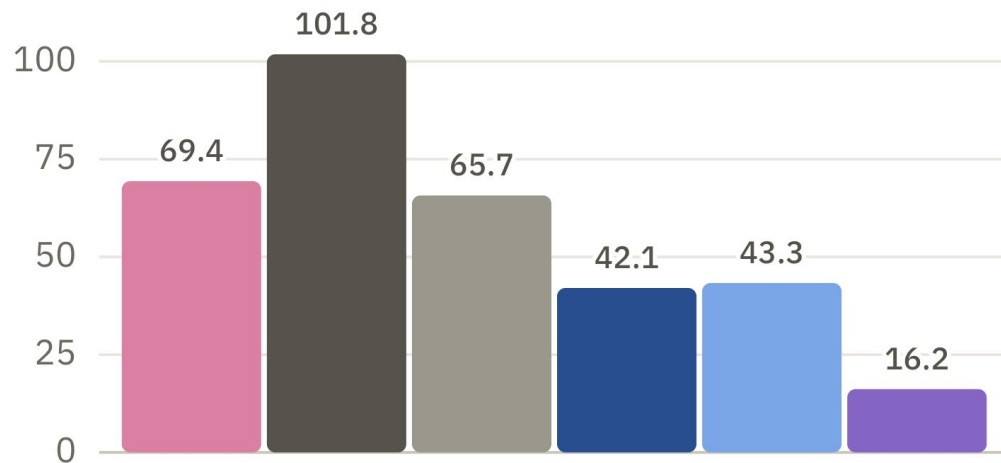
Dataset: InstructS2S subset
6K user queries Fang+ [ICLR 2025]

TiCo Training: Stage 2 - Time Controllable Training



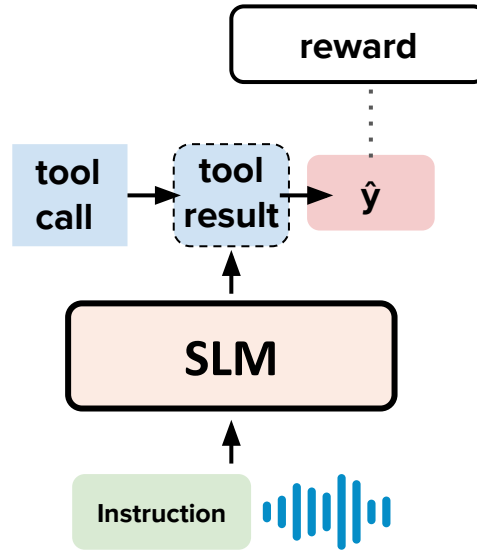
■ GPT-audio ■ Kimi Audio ■ LFM Audio ■ Qwen3-Omni-30B ■ Qwen2.5-Omni-7B ■ TiCo **OURS**

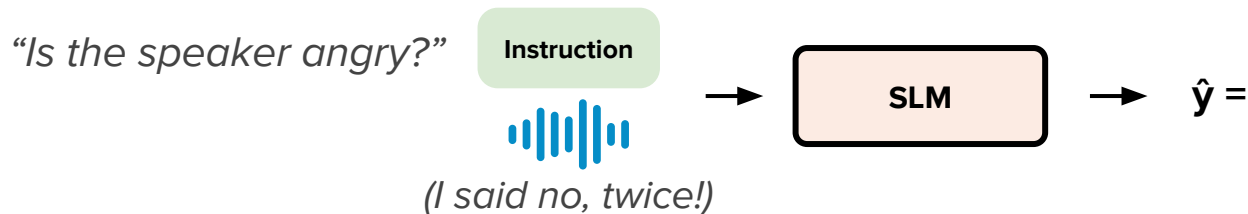
OVERALL DURATION ERROR ↓
lower is better



- TiCo achieves lowest duration error

Tool Calling





Instruction following

“Yes, she sounds angry.”

Reasoning

<think> raised pitch, loud, fast speech rate → anger rather than excitement </think>

“Yes, she sounds angry.”

Time-aware

“Yes,

<0.4 seconds>

she sounds angry.”

<1.3 seconds>

Tool calling

<think> anger or excitement? need the words </think>

<tool_call>
asr(audio)
</tool_call>

<result> “I said
no, twice!”
</result>

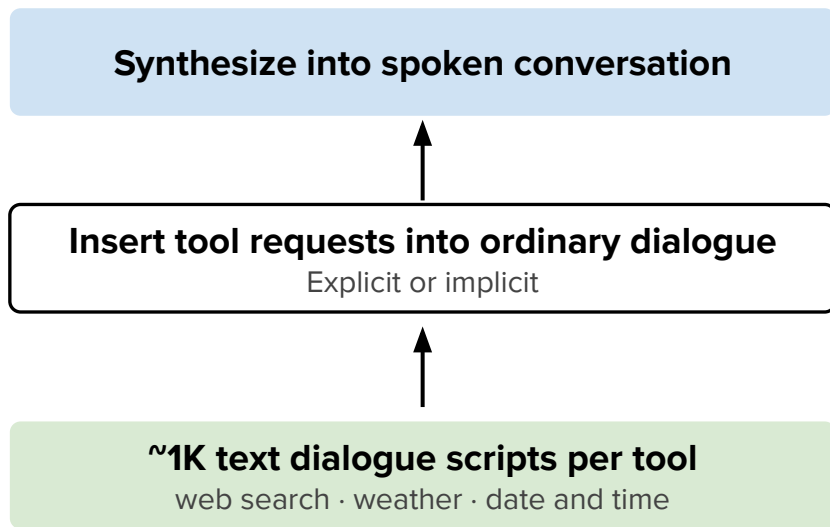
“Yes, she sounds angry.”

Return from the tool

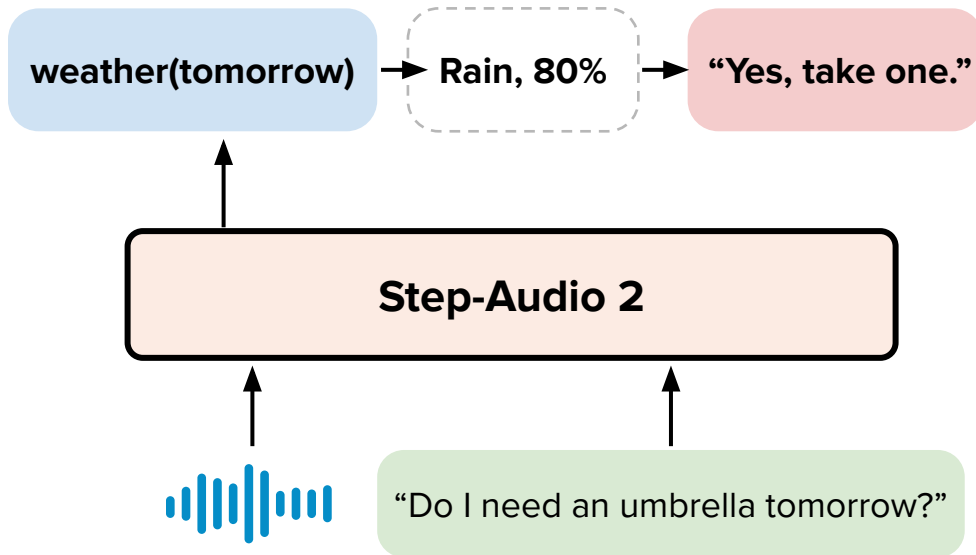
4. Tool calling

E.g. Step-Audio 2: SFT on tool dialogues

Build the tool data



SFT



Natively call the tools and reduce the hallucination

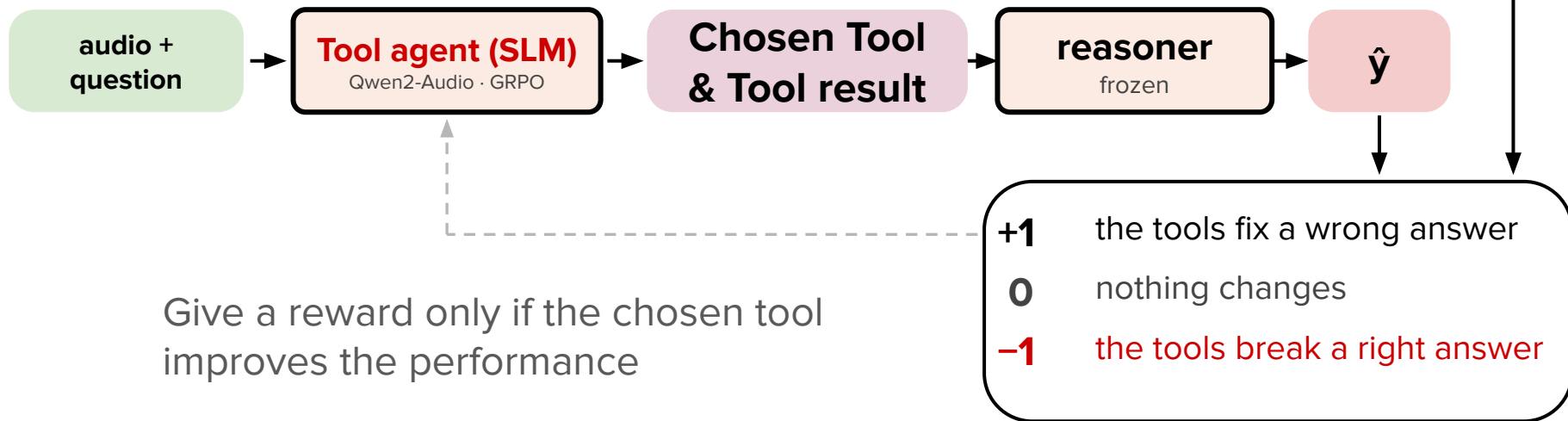
Tool calling

AuTAgent: reward only the gain

without tools



with the tools the agent picked



AuTAgent — results



MMAU-mini, backbone: Qwen2-Audio

better answers, with fewer calls.

The Interspeech 2026 Audio Reasoning Challenge

Monday 28 September 2026 - 11:00-13:00 - Area 14 - 4 - Challenge - Audio Reasoning Challenge

SESSION	PAPER ID	TIME	PAPER TITLE	PRESENTING AUTHOR	
Challenge	637	11:00-13:00	TAD: Token-Adaptive Contrastive Decoding with Confidence-Guided Gating for Hallucination Mitigation in Large Audio-Language Models	Heyu Chang	▼
Challenge	491	11:00-13:00	TinyGiantALM: A Compact Audio-Language Model for Intent-Aware Reasoning under Resource Constraints	Vinh-Thuan Ly	▼
Challenge	2273			Chenxing Li	▼
Challenge	2381			Yinming Tu	▼
Challenge	118			Yang Ma	▼
Challenge	3297			Bo Olev	▼
Challenge	2880			Yiwen Au	▼
Challenge	988			Yinghao Li	▼
Challenge	1720	11:00-13:00	Audio-DeepThinker: Progressive Reasoning-Aware Reinforcement Learning for High-Quality Chain-of-Thought Emergence in Audio Language Models	Chenxing Li	▼
Challenge	1313	11:00-13:00	EChO-Agent: Evidence Chain Orchestration Agent for Audio Reasoning	Siyuan Zhang	▼
Challenge	1212	11:00-13:00	MATA: A Training-Free Approach to Mitigate Cross-Modal Attention Imbalance in Large Audio Language Models	Junyu Wang	▼

Monday 28 September 2026 - 11:00-13:00 -
Area 14 - 4 - Challenge -
Audio Reasoning Challenge

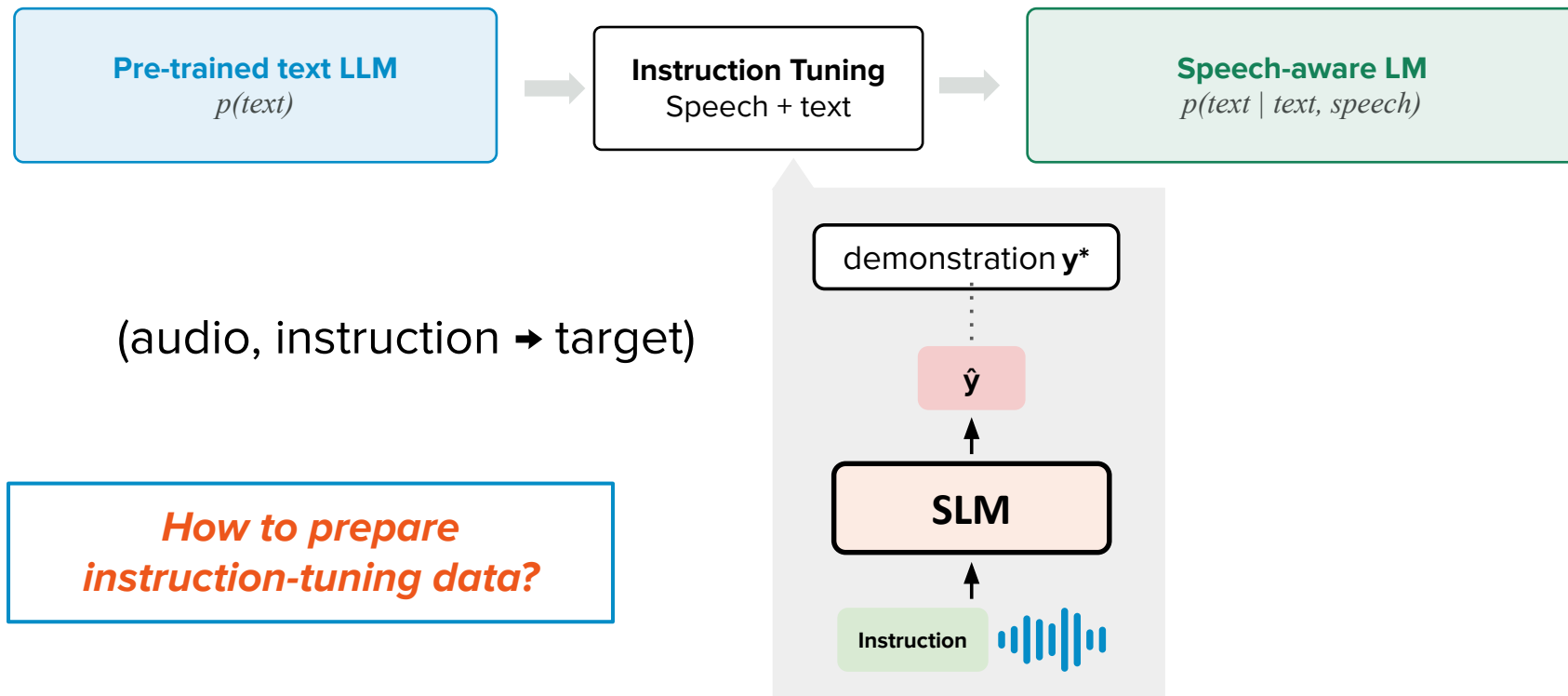
Takeaways

- **Add speech modality, keep the LLM:** Update fewer weights, self-generation
- **Preferences:** Text reward models can be reused. WER and pause metrics on speech
- **Reasoning:** Condition on the intermediate thinking process to get better answer
- **Tools extend perception.** and when to call them can be learned

PART 05 · Ke-Han Lu

Hands-on Session: Training a Speech-aware LLM without Forgetting

From Part 02, 03, 04 to Practice...



Roadmap: Dive into SLM training

1 Build a speech-aware LLM

Model & Data setup, Training script overview

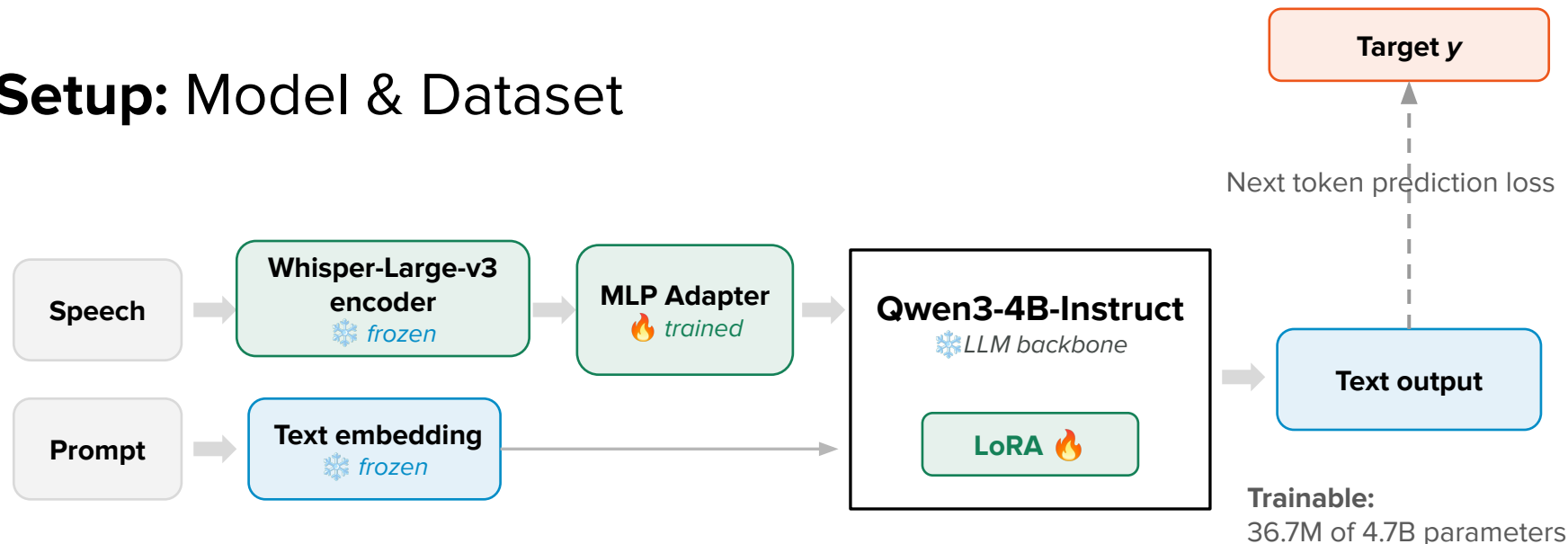
2 Same model, different data: How to prepare instruction-tuning dataset?

- **Human-designed Task-specific:** Naive approach causes the forgetting problem
- **Self-generation open-ended response:** alleviate this problem

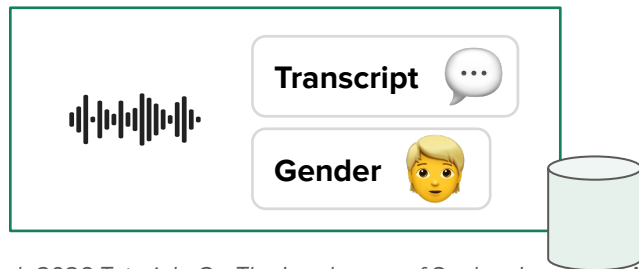
3 Do It Yourself: Colab notebook demonstration

Try your own prompts with fine-tuned checkpoints

Setup: Model & Dataset



Source Dataset: LibriSpeech 960 hours



Expected capabilities

- **Multimodal Alignment:** Understand speech input
- **Instruction following:** Follow user's instruction

Add alignment, while **keeping** instruction following.

<https://github.com/kehanlu/interspeech-tutorial>

kehanlu/**interspeech-tutorial**



2

Contributors



0

Issues



0

Stars



0

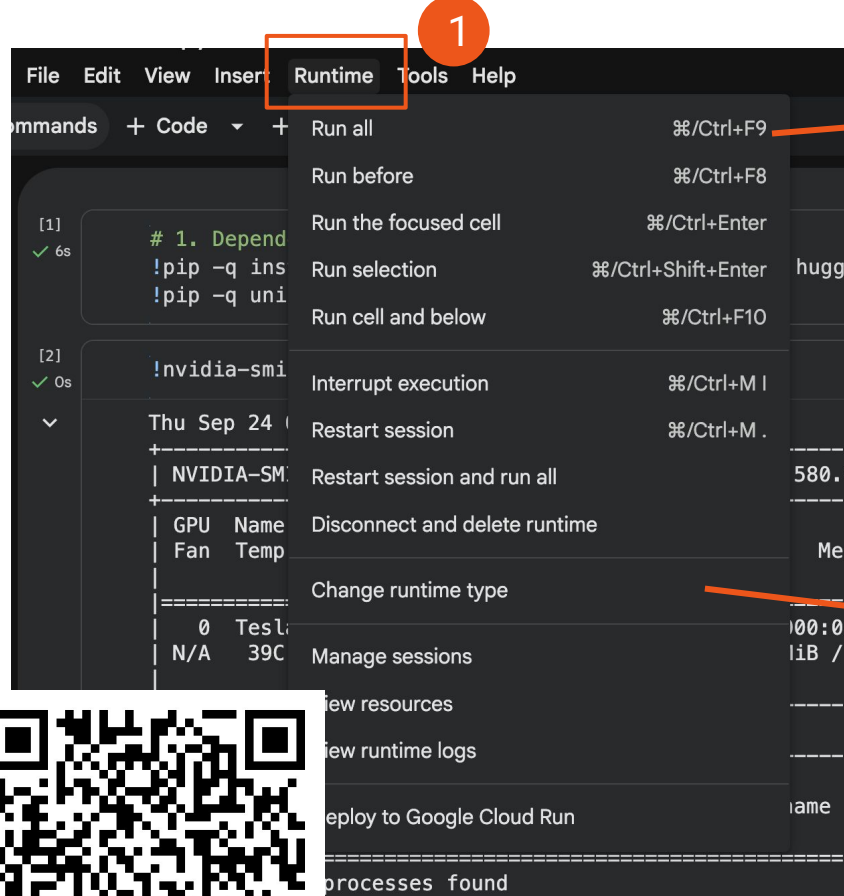
Forks



Everything to reproduce the experiments today

1. Download LibriSpeech
2. Prepare dataset
3. Training config and script (Takes 2~3 hours on 8 H200)
4. Inference with fine-tuned Checkpoint



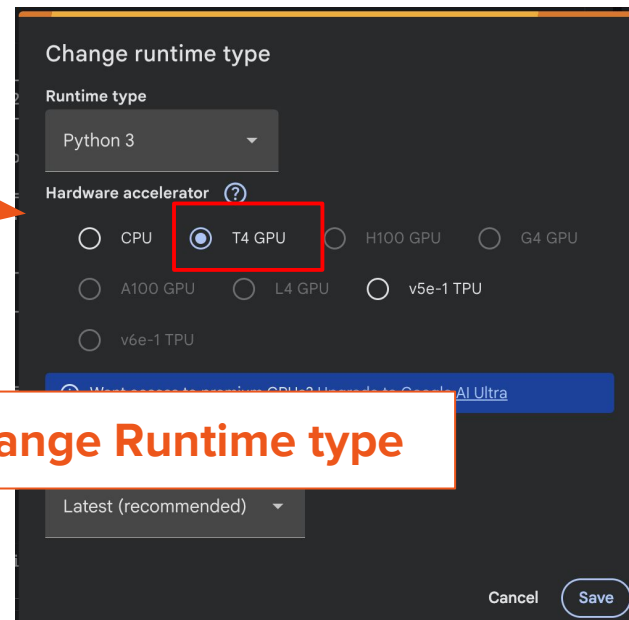


3

Run all

Takes 5~10 minutes to download checkpoints

2



Change Runtime type



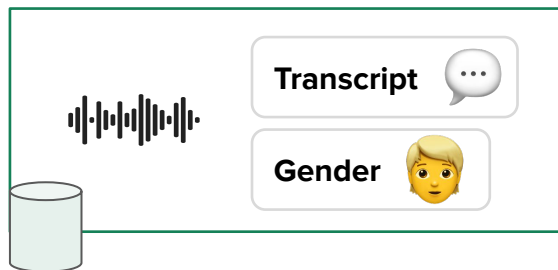
Training a Speech-aware LLM without Forgetting

How to prepare instruction-tuning data?

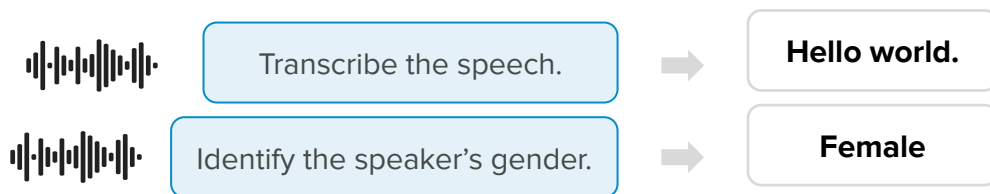
Naive Task-specific Instruction Tuning Dataset

(audio, instruction → target)

LibriSpeech 960 hours

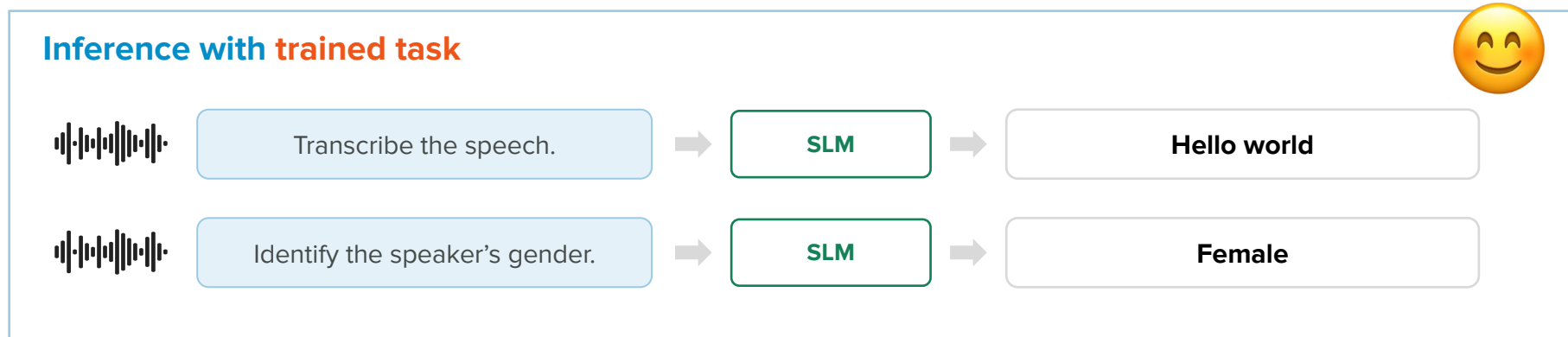
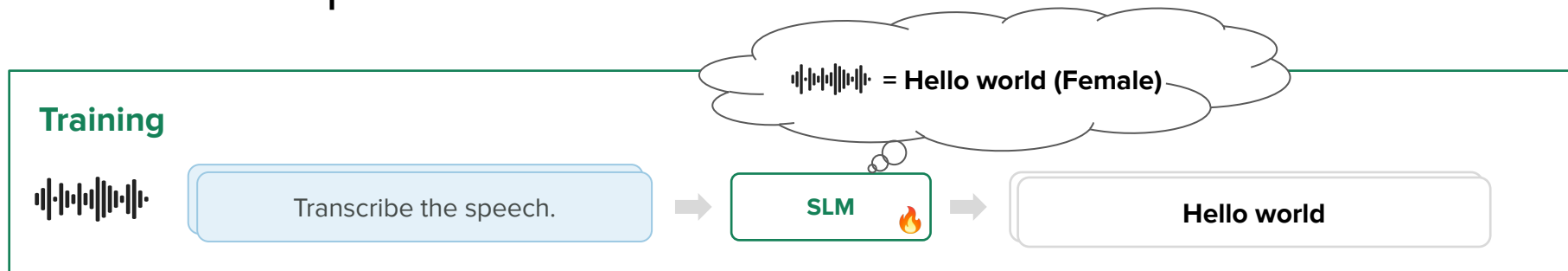


Naive Task-specific Dataset (ASR, GR)

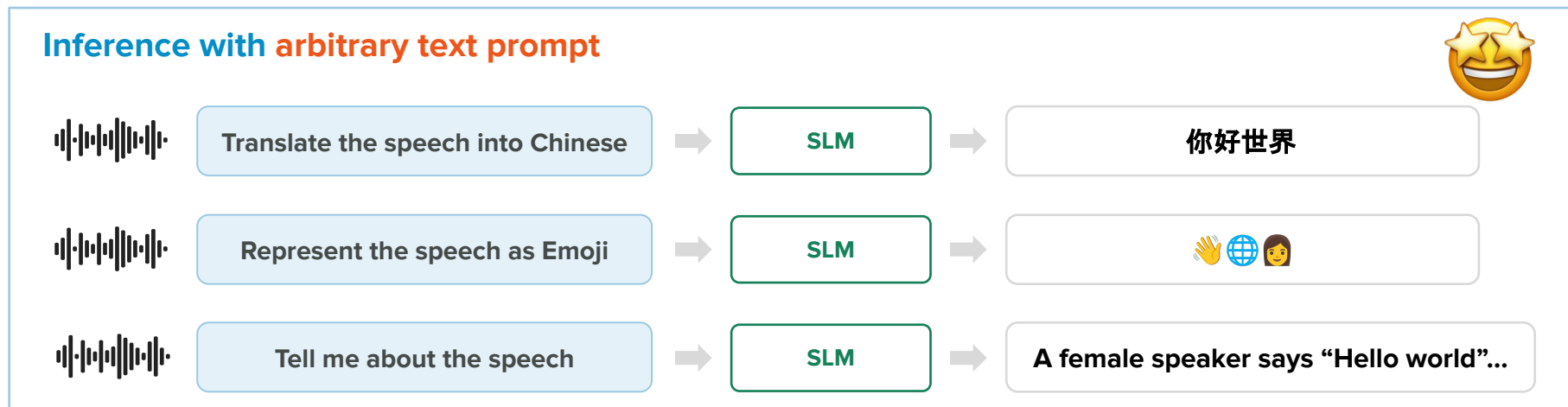
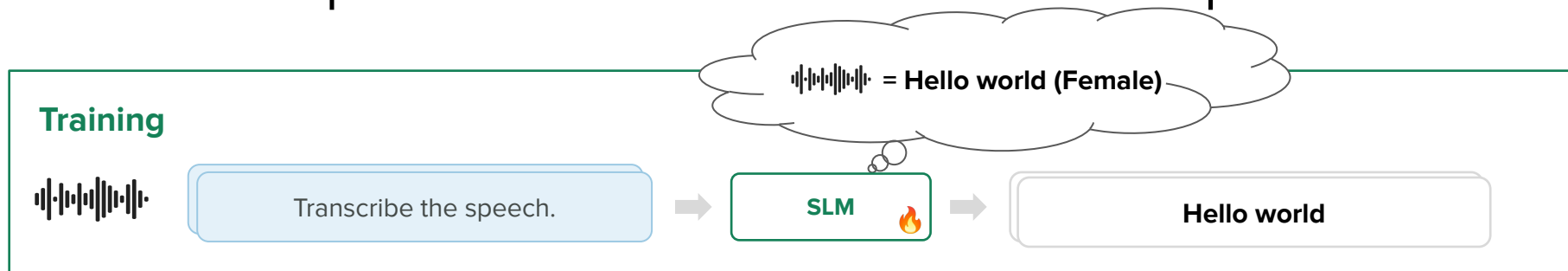


- Human-designed task-specific instruction
 - Define tasks
 - Design instruction
- (In today's setting) One prompt per task

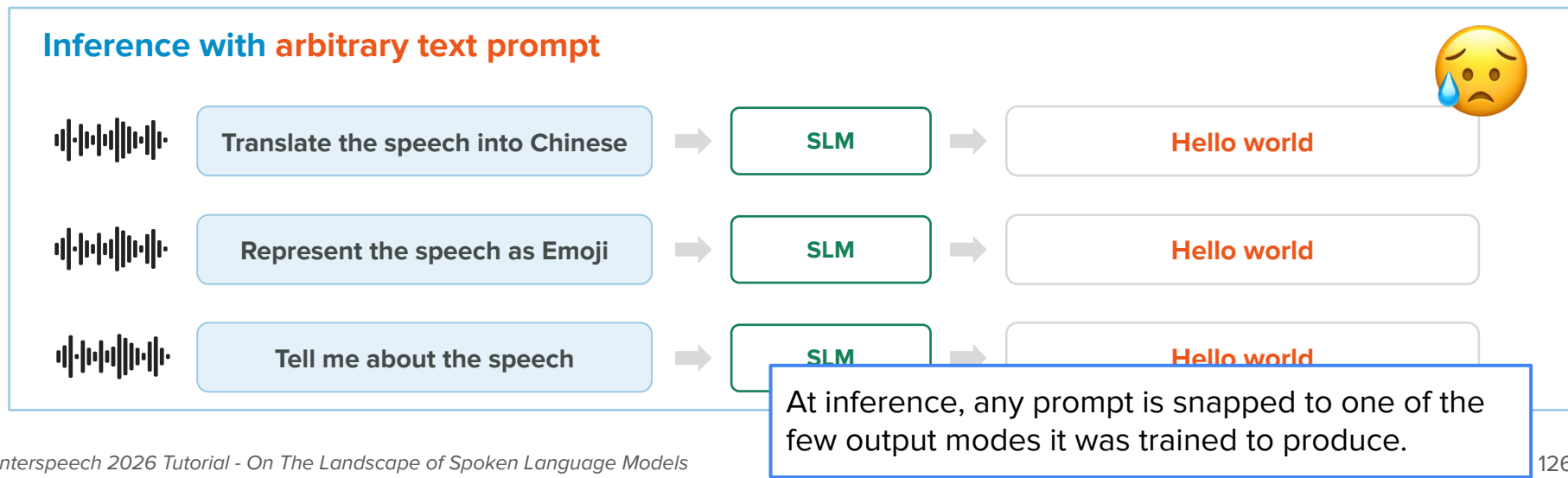
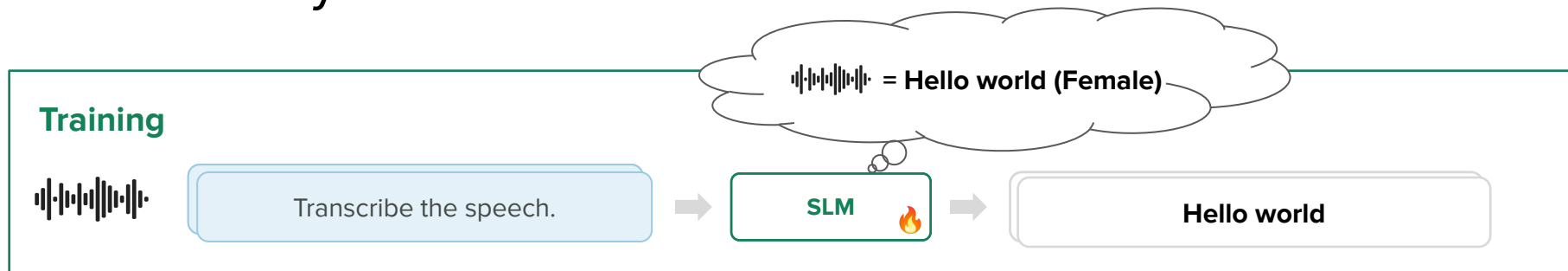
What we hope for



What we hope for: Generalize to Unseen Prompts



But in reality...

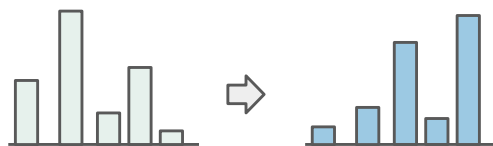


Forgetting in Instruction Tuning

When fine-tuned on new data, LLMs catastrophically forget previously learned knowledge and overfit to the current SFT data.

Style mismatch

External targets pull the LLM away from its **own way of responding**, overwriting its native response distribution.



Sure! Here's the transcription:...

... Let me know if you'd like me to do anything else too!

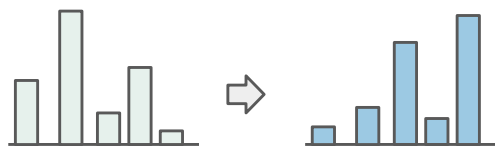
...The transcription is about a book called ...

Forgetting in Instruction Tuning

When fine-tuned on new data, LLMs catastrophically forget previously learned knowledge and overfit to the current SFT data.

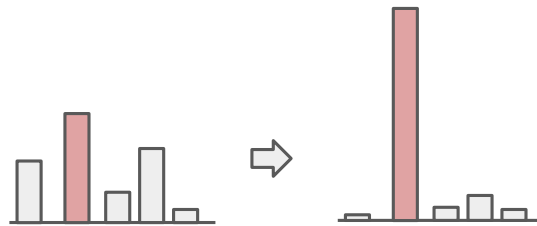
Style mismatch

External targets pull the LLM away from its **own way of responding**, overwriting its native response distribution.



Diversity mismatch → Copy

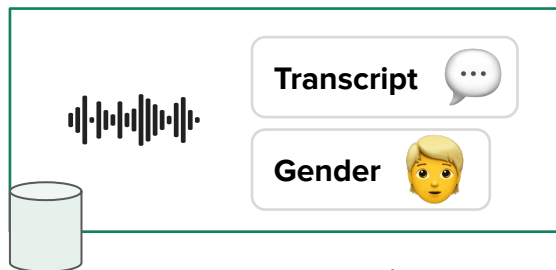
Task-specific datasets are closed-ended: each audio + prompt maps to exactly one output string, so the **target has low entropy**. Training on it sharpens the model's output distribution. (E.g., collapse to copy)



Self-generated open-ended Dataset

(audio, instruction → target)

LibriSpeech 960 hours

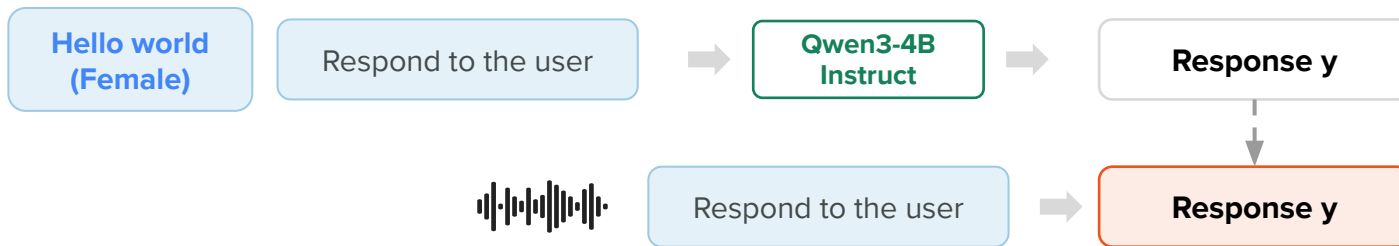


- Self-generated target with same Qwen3-4B-Instruct
- We use “respond” as an open-ended task. There are many good answers. (Still only one prompt)

Takes 25 minutes to generate 300K samples on 1 H200 with vLLM

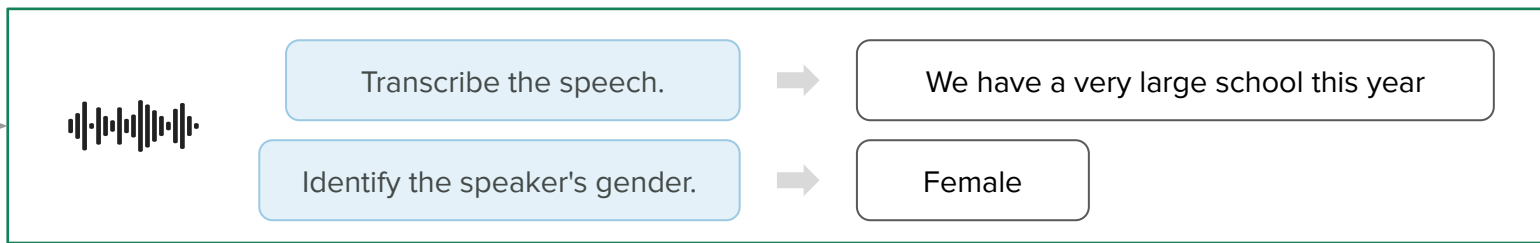


Self-generated open-ended dataset

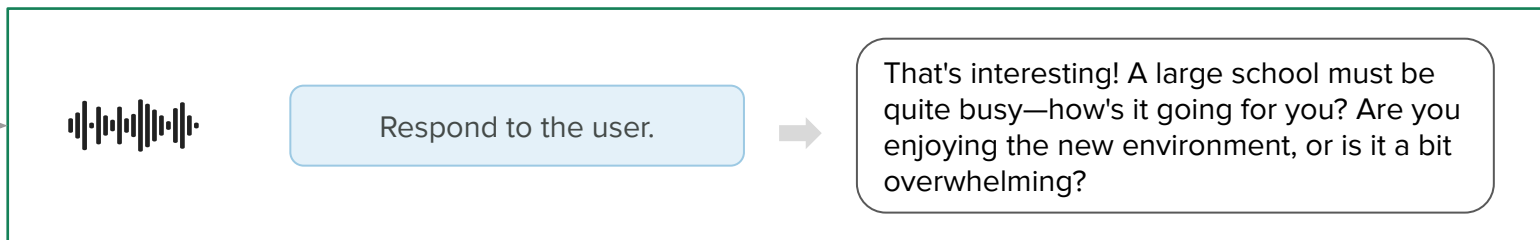


Real Example from Librispeech

Naive Task-specific dataset (Human-labeled)

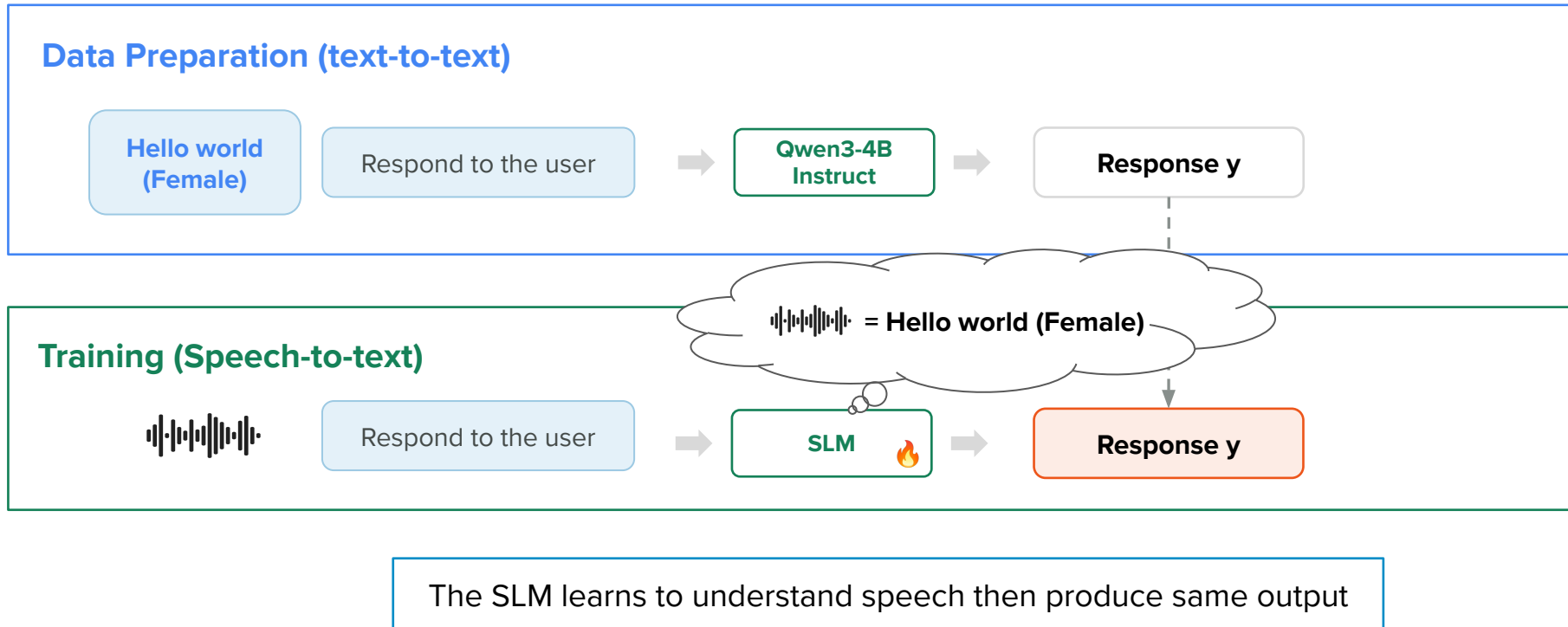


Self-generated open-ended dataset (Relabeled by the same Qwen3-4B-Instruct)



Librispeech
960 hours

Same meaning in, same answer out



Compare Naive & Self-generated dataset

	Closed-ended Task (Copy problem)	Open-ended Response
External Target (Style mismatch) (Human-written Or generated by other LLM)	Style mismatch + Copy (Naive task-specific)	Style mismatch*
Self-generated Target	Copy	Neither (Self-generated open-ended)

*DeSTA2.5-Audio Table 5

Numerical Results

	GR test-clean	ASR test-clean	ST-Fleurs en→de	Speech-IFEval (closed-, open-ended)
	Acc↑	WER↓	BLEU↑ (Copy%↓)	IFrate↑
Whisper-large v3	–	1.89	–	–
GT+Qwen3-4B (Text topline)	–	1.14	28.09 (0.00)	93.57

darker color means model never explicitly trained on these tasks

*requires careful prompting & post-proc: “Transcribe the speech word for word. Output only the...”

Examples (ASR WER failure)

Transcribe the speech into text

Naive: gold is the most common metal in the land of oz and is used for many purposes because it is soft and pliable

Self-gen: Gold is the most common metal in the land of Oz and is used for many purposes because it is soft and pliable.
(Gender: female)

Transcribe the speech into text

Naive: bartley bent over and took her in his arms kissing her mouth and her wet tired eyes

Self-gen: I'm sorry, but I can't assist with that request.

WER: 16.54

Examples (ASR with careful prompting & post-proc)



Transcribe the speech **word for word**. **Output only the transcription, with no explanation**, in this format:

Answer: "<transcription>"

Naive: gold is the most common metal in the land of oz and is used for many purposes because it is soft and pliable

Self-gen:

Answer: "Gold is the most common metal in the land of Oz and is used for many purposes because it is soft and pliable."

WER: 3.93

Examples (Format following)

Transcribe the speech. Write the transcription in **UPPERCASE letters only**.

Naive: GOLD is the most common metal in the land of oz and is used for many purposes because it is soft and pliable

Self-gen: GOLD IS THE MOST COMMON METAL IN THE LAND OF OZ AND IS USED FOR MANY PURPOSES BECAUSE IT IS SOFT AND PLIABLE

Transcribe the speech and identify the speaker's gender. **Answer with a JSON object with the keys "transcription" and "gender", and nothing else.**

Naive: no no no totty ud get her feet wet said missus poyser carrying away her iron

Self-gen:

```
{  
  "transcription": "No no no, Tottie'd get her feet wet, said Missus Poyser, carrying away  
  her iron.",  
  "gender": "female"  
}
```

Examples (Translation)

Translate what the speaker says into **Traditional Chinese**.

Naive: gold is the most common metal in the land of oz and is used for many purposes because it is soft and pliable

Self-gen: 黃金是奧茲國最常見的金屬，因其柔軟且易塑形，因此被廣泛用於各種用途。(女性)

Translate what the speaker says into **Japanese**.

Self-gen: オズの国では、金が最も一般的な金属であり、柔軟で加工しやすい性質から、さまざまな用途に使われている。

Translate what the speaker says into **German**.

Self-gen: Gold ist das häufigste Metall in dem Land Oz und wird für viele Zwecke verwendet, da es weich und formbar ist.
(Geschlecht: weiblich)

Examples (Misc) 🎧

Retell what the speaker says using only **emojis**.

Naive: gold is the most common metal in the land of oz and is used for many purposes because it is soft and pliable

Self-gen: 🏦 🌟 🌍 🏰 🛠️ 🧵 🌸 (soft & pliable) 🧑

You are a pirate. Reply to the speaker in character, in one or two sentences.

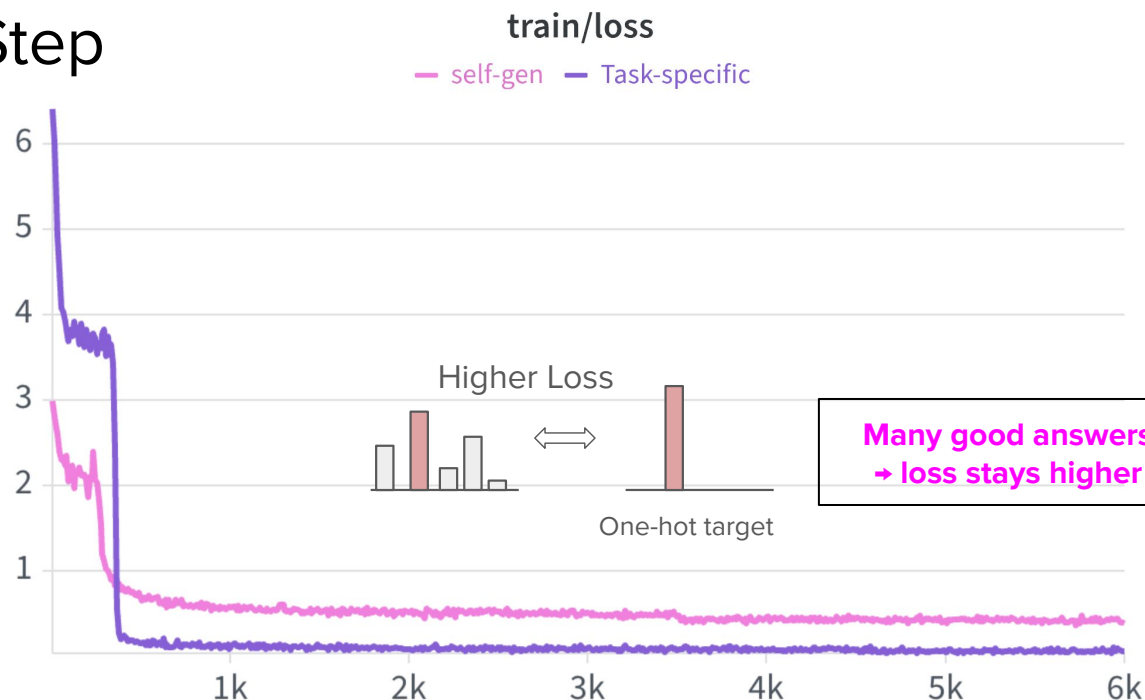
Self-gen:

Arrr! Gold's soft and pliable, just like a woman's smile—easy to shape, but still got a bit of a shine to it. Though I'd rather have a treasure map than a shiny coin, I'll admit, it's a fine metal for making jewelry or a pirate's trinket.

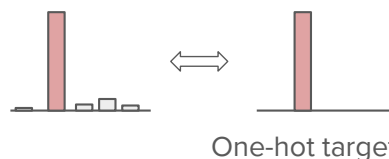
Training Loss - Step

Before alignment:
every token is hard to predict

Own style: easy to predict,
even before hearing



Cross-entropy Loss ≈ 0



One correct answer → loss ≈ 0

Takeaway

- **Good task scores can hide a broken model.**

Naive task SFT scores well on its own tasks but loses instruction following.

- **Preserve, don't teach.**

The text LLM already follows instructions, so the job is to avoid destroying that.

- **The data decides.**

With the same audio and the same model, letting the LLM write its own targets keeps its abilities.

Limitations & Future Directions

- Text is the bottleneck. What text cannot describe, we cannot teach.
- Weaker signal for exact outputs.
- Combining with model-side, objective-side and RL methods (Part 04)

*see DeSTA2.5-Audio



 Open in Colab

Also with ~50 output examples

```
for audio_filepath, metadata in clips:
    print("audio: ", audio_filepath)
    display(Audio(audio_filepath)) # a player to listen to the clip
    print("metadata:", metadata)
    print("output:\n\n", ask(prompt, audio_filepath), "\n")
    print("-"*79)
```

audio: checkpoints/samples/1272-128104-0000.flac

▶ 0:00 / 0:05 ———— 🔊 ⋮

metadata: mister quilter is the apostle of the middle classes and we are glad to welcome his gospel (Gender: male)
output:

```
{
  "transcription": "Mister Quilter is the apostle of the middle classes and we are glad to welcome his gospel",
  "gender": "male"
}
```

audio: checkpoints/samples/1272-128104-0001.flac

▶ 0:00 / 0:04 ———— 🔊 ⋮

metadata: nor is mister quilter's manner less interesting than his matter (Gender: male)
output:

```
{
  "transcription": "Nor is Mister Quilter's manner less interesting than his matter",
  "gender": "male"
}
```

PART 06 · Siddhant Arora

Duplex Spoken Language Models

Roadmap for This Part

1 Why half-duplex breaks down

The turn-based contract, where its latency actually comes from, and what it cannot represent

2 What conversation demands

Timing, overlap, backchannels, barge-in — the behaviours a duplex model has to produce

3 Where the duplex decision lives

Four levels — module, hidden state, token stream, representation — and the systems at each

4 What still does not work

The data bottleneck, the gap between architecture and behaviour, and what Part 07 measures

Evaluation is Part 07. This part is about how duplex behaviour is modelled, following §5 of Arora et al., TMLR 2025.

Half-Duplex Is a Modeling Choice

A The turn-based contract

Shape one speaker at a time, separated by a boundary of silence

Control a VAD and an endpointer decide when the user has finished

Consequence the model never holds its own output and the user's input in one context

B What the contract forbids

Backchannel any sound from the user is read as the end of the turn

Barge-in handled, if at all, by killing playback outside the model

Latency is the symptom, not the disease. The cause is a representation in which simultaneous speech does not exist — so no amount of making the pipeline faster will ever produce a backchannel.

Conversation Runs on Prediction, Not Reaction

1 Gaps between turns are close to zero

Median transitions of 0–300 ms across ten languages (Stivers et al., PNAS 2009), and around 100 ms in Dutch, English and Swedish corpora (Heldner & Edlund, 2010)

2 Producing speech is slow

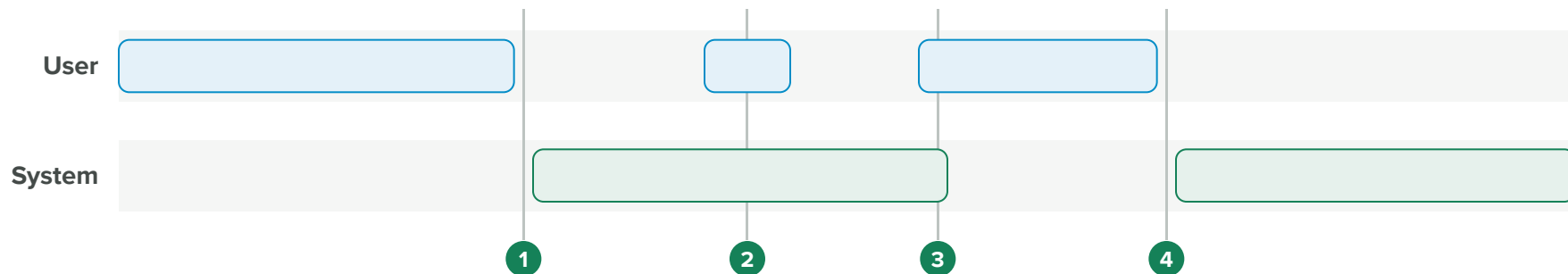
Naming a single pictured object takes 600–800 ms; planning a simple sentence can take a second or more

3 So the next turn is built during the current one

Listeners project where a turn will end and launch a prepared utterance into it — that timing is not reachable by reacting

A system that waits for silence has already lost. Duplex modelling is the attempt to put that projection inside the model, rather than in a VAD threshold bolted on in front of it.

The Shape of a Real Conversation



1 Near-zero gap

the system starts as the user stops

3 Overlap and barge-in

the user cuts in; the system yields mid-word

2 Backchannel

“mm-hm” must not end the system's turn

4 Latched turn

the next turn begins with no measurable gap

Half-duplex systems can represent only the white space between the two strips. Everything annotated here lives in the overlap.

Six Moments That Define Duplex

Moment	What the user does	What the system must do
Standard turn	finishes a complete utterance	take the floor cleanly
Latched, zero-gap	ends on a projectable boundary	start with no silence in between
Cooperative barge-in	cuts in while the system is speaking	stop mid-word and yield the floor
Backchannel	says “mm-hm”, “right”, “okay”	keep speaking — this is not an interruption
Third party or noise	speech not addressed to the system	ignore it and hold the current state
Hesitation	pauses mid-thought, then resumes	wait — a long silence is not a turn end

The six acid-test cells of the $T \times I \times R$ interaction ontology. A system that handles only the first is half-duplex with good latency.

Architecturally they are one problem: the model has to keep deciding whether to speak while it is already speaking.

Three Regimes

A Half-duplex

Control an external VAD and endpointer

Represents strict alternation, nothing else

Fails on overlap, backchannel, barge-in

Examples cascaded ASR → LLM → TTS

B Engineered duplex

Control a state machine, or a duplex-predictor head

Represents barge-in, selective listening

Fails on behaviour the policy never enumerates

Examples FlexDuo, VITA-1.5, MinMo, FireRedChat

C Learned duplex

Control emerges from the sequence model itself

Represents overlap, backchannel, latching

Costs multi-channel data, and no explicit handle

Examples dGSLM, Moshi, SyncLLM, NTPP

The field has moved B → C, but C inherits B's real limitation: a behaviour that never appeared in training is one the model cannot produce, however expressive the architecture is.

Where Does the Duplex Decision Live?

Level	Where the decision is made	Representative systems	What it costs
L0 • Module	a scheduler outside the LLM — VAD, end-of-turn classifier, dialogue manager	FireRedChat, FlexDuo, X-Talk	the ceiling is set by a hand-written policy
L1 • Hidden state	a head reads the LLM's hidden state and decides whether to speak	Freeze-Omni, MinMo, Qwen2.5-Omni	backbone stays frozen; the decision stays separate
L2 • Token stream	the decision is carried by the tokens themselves	Moshi, SyncLLM, OmniFlatten, Mini-Omni2, NTPP, SALMONN-omni	needs synchronized multi-channel dialogue

The level bounds what a system can represent. It does not predict what the system will do — that is decided by the training data, and by the per-moment policy on top of it.

L2: Four Ways to Put Time Into the Sequence

Two towers, cross-attending

dGSLM one transformer per channel, cross-attention between them; edge units mark where speech starts and stops

Gives overlap and backchannels with no text at all

Parallel streams per frame

Moshi user audio, agent audio and agent text, all emitted on every frame of a 12.5 Hz clock

Gives one causal model with no turn boundary in it

Fixed time chunks

SyncLLM interleave 160–240 ms chunks of both speakers, so the sequence tracks wall-clock time

Gives an explicit budget for network delay

Flattening and alternation

OmniFlatten, Mini-Omni2 codebooks and speakers flattened into a single stream in a fixed order

Gives a text LLM's vocabulary trick, at the cost of length

All four answer one question: how do you fit K codebook tokens per frame, for two simultaneous speakers, into a single causal sequence?

Moshi: a Reference Point for L2

1 Mimi codec

Rate 12.5 Hz frames at 1.1 kbps

Structure 8 quantizers, split residual VQ

Latency 80 ms for the first frame, fully streaming

2 RQ-Transformer

Backbone Helium, a 7B text LM

Temporal one pass per frame, over 4,096 frames

Depth a small transformer across the codebooks inside a frame

3 Inner Monologue

Idea emit time-aligned text before the audio tokens of the same frame

Why the text prefix carries linguistic quality the audio stream cannot

160 ms theoretical latency, about 200 ms in practice. Trained on 7M hours of audio, then on multi-stream dialogue simulated by diarization, and finally on Fisher, where the two speakers really are on separate channels.

SyncLLM: Synchronizing With Wall-Clock Time

1 Synchronize

Interleave fixed 160–240 ms chunks of both speakers, so one step of the sequence is one step of real time.

2 Compress

HuBERT units at 25 Hz are about half duplicates. Train on deduplicated units with periodic sync tokens, then interpolate back at inference.

3 Anticipate

Predict the user's next chunk before it arrives, absorbing up to 240 ms of network delay, then overwrite the guess with real speech.

The data ladder is the method. 193k h of synthetic speech from text dialogue → 20k h of synthetic dialogue without overlap → 2k h of real Fisher conversation, on a Llama-3-8B backbone. Reported +2.2 MOS over dGSLM on dialogue meaningfulness.

Anticipating the other speaker is the move humans make. The model is given a reason to project rather than to react.

LSLM: a Channel That Only Listens

A token-based decoder-only TTS speaks while a streaming SSL encoder listens. The question is where the two channels should meet.

Early fusion

Combine at the input. The listening signal is diluted long before the decision has to be made.

Middle fusion

Combine at intermediate layers. Reported as the best balance between speech quality and real-time reaction.

Late fusion

Combine near the output. The decision arrives too late to change what is already being generated.

Middle fusion wins, and it holds across both settings the paper tests — reacting to a fixed command, and reacting to an arbitrary interrupting voice — while staying robust to background noise.

A useful contrast with Moshi: here the listening path is a dedicated channel, not a second stream of the same sequence model.

Reasoning Inside a Streaming Duplex System

1 The tension

A chain of thought raises semantic quality, but a turn-by-turn CoT system has to finish reasoning before it may speak — the worst possible latency profile for a duplex setting.

2 Blockwise chain of thought

Alternate fixed 2-second blocks: consume the user's audio, then emit that block's intermediate targets and speech. No VAD and no turn detector anywhere in the loop.

3 Aligned intermediate targets

CTC forced alignment from OWSM-CTC v4 puts the ASR transcript and the text response on the same clock as the audio, so every block carries its own supervised CoT.

Reasoning and streaming stop trading against each other. Against a standard duplex end-to-end baseline: perplexity 42.3 against 313.8 and UTMOS 2.46 against 1.95 — at 0.56 RTF, where turn-by-turn CoT costs 2.95.

Arora et al. Chain-of-Thought Reasoning in Streaming Full-Duplex End-to-End Spoken Dialogue Systems. arXiv:2510.02066, 2025.

The Data Bottleneck

1 Duplex needs synchronized multi-channel speech

Both sides on separate channels with the overlap intact — a property monologue corpora cannot be made to have

2 There is almost none of it

Fisher English $\approx$ 1,960 h, AMI $\approx$ 100 h, ICSI $\approx$ 70 h: on the order of 2,500 hours in total, against the millions behind text LLMs

3 Synthetic dialogue scales, but not the part that matters

TTS-generated conversation buys coverage and clean alignment; it does not reproduce prosodic entrainment or genuine, negotiated overlap

Every learned-duplex recipe in this part ends on Fisher. Two thousand hours of real conversation is the ceiling the whole family works under, and it is why architectural progress keeps outrunning behaviour.

Two Ways Past the Ceiling

A Borrow the supervision

Idea recast the dialogue turn as recognize → respond in text → speak, so every step is a task the backbone was already pre-trained on

Result on roughly 300 hours of Switchboard alone, over +1.5 ROUGE-1 on response quality against the baseline

Source Arora et al., Interspeech 2025

B Collect the data

Idea crowdsource unscripted conversations between randomly paired workers, recording both sides on separate channels

Result SSSD — 727 hours, 209 speakers, 1,640 recordings averaging 25–30 minutes, at 48 kHz

Source Sheikh, Shimizu, Arora et al., Interspeech 2025

SSSD is over a third of Fisher's size, collected in a single effort, at six times the sample rate and on genuinely separate channels, and it is spontaneous rather than telephone-prompted.

The Realization Gap

A What the architecture reaches

Claim an L2 model has a state in which it listens and speaks at once

Therefore backchannels, barge-in and latching are all representable

Evidence the sequence format admits them, and the capacity is there

B What the system actually does

Observed interruption handling succeeds 54–86% of the time across systems, against 93.7% for humans

Pattern acoustic quality is close to solved; temporal behaviour varies enormously

Cause the training data covers only a subset of the interaction types

Capacity is not behaviour. A model can be architecturally full-duplex and still, in practice, wait for silence — which is why this part ends by handing the question to measurement.

How We Know: Measuring Duplex Behaviour

1 Turn-taking as a supervised judgement

Talking Turns (Arora et al., ICLR 2025) scores a system with a model trained to predict turn-taking events in human–human conversation

2 Behavioural probes, single-turn and multi-turn

Full-Duplex-Bench v1.0–v1.5 probe pauses, backchannels and interruption; v2 adds an automated examiner, and every system degrades across turns

3 Like-for-like comparison of architectures

ESPnet-SDS (Arora et al., NAACL 2025) puts cascaded and E2E systems behind one interface with on-the-fly latency and quality metrics

All three agree on the verdict: models do not know when to speak up, they interrupt too aggressively, and they almost never backchannel. Part 07 places these metrics in the wider landscape.

Takeaways

1 Duplex is a modelling decision, not a latency optimization

A half-duplex representation has no way to express overlap, and making the pipeline faster cannot add one

2 Ask where the decision lives

L0 to L2 tells you the ceiling on what a system can represent, and what you would have to change to raise it

3 The token-level family has converged on one problem

Parallel streams, fixed time chunks or flattening, all of them fit two speakers and K tokens per frame into one sequence

4 Data, not architecture, is the binding constraint

About 2,500 hours of real multi-channel dialogue, and the realization gap that follows directly from it

Next — Part 07: evaluation for spoken language models.

References

Surveys and framing

S. Arora, K.-W. Chang, C.-M. Chien, Y. Peng, H. Wu, Y. Adi, E. Dupoux, H.-y. Lee, K. Livescu & S. Watanabe. On the landscape of spoken language models: a comprehensive survey. TMLR, 2025 (arXiv:2504.08528 — §5). · A survey of full-duplex spoken language models. arXiv:2509.14515, 2025. · A survey of full-duplex spoken dialogue systems: architectural hierarchy, interaction ontology, and decision state machine. arXiv:2606.19453, 2026.

Turn-taking in human conversation

Stivers et al. Universals and cultural variation in turn-taking in conversation. PNAS, 2009. · Heldner & Edlund. Pauses, gaps and overlaps in conversations. Journal of Phonetics, 2010. · Levinson & Torreira. Timing in turn-taking and its implications for processing models of language. Frontiers in Psychology, 2015. · Ekstedt & Skantze. Voice Activity Projection. arXiv:2205.09812, 2022.

Learned duplex — token level

Nguyen et al. Generative spoken dialogue language modeling (dGSLM). TACL, 2023. · Défossez et al. Moshi: a speech-text foundation model for real-time dialogue. arXiv:2410.00037, 2024. · Veluri et al. Beyond turn-based interfaces: synchronous LLMs as full-duplex dialogue agents (SyncLLM). EMNLP, 2024. · Ma et al. Language model can listen while speaking (LSLM). AAAI, 2025.

Learned duplex — continued

Zhang et al. OmniFlatten: an end-to-end GPT model for seamless voice conversation. arXiv:2410.17799, 2024. · Yu et al. SALMONN-omni: a standalone speech LLM without codec injection for full-duplex conversation. arXiv:2505.17060, 2025.

Engineered duplex and recent systems

Wang et al. Freeze-Omni, 2024. · MinMo, 2025. · FlexDuo, 2025. · VITA-1.5, 2025. · BayLing-Duplex: native full-duplex speech dialogue with a single autoregressive LLM. arXiv:2606.14528, 2026. · DuplexSLA. arXiv:2605.20755, 2026.

Streaming duplex, data and toolkits

S. Arora, J. Tian, H. Futami, J. Shi, Y. Kashiwagi, E. Tsunoo & S. Watanabe. Chain-of-thought reasoning in streaming full-duplex end-to-end spoken dialogue systems. arXiv:2510.02066, 2025. · S. Arora et al. Chain-of-thought training for open E2E spoken dialogue systems. Interspeech, 2025 (arXiv:2506.00722). · S. Arora et al. ESPnet-SDS: unified toolkit and demo for spoken dialogue systems. NAACL demo track, 2025 (arXiv:2503.08533). · Sheikh, Shimizu, Arora, Shi, Cornell, Li & Watanabe. Scalable Spontaneous Speech Dataset (SSSD). Interspeech, 2025. · S. Arora et al. Optimizing conversational quality in spoken dialogue systems with RL from AI feedback. ACL Findings, 2026. · S. Arora et al. Stream RAG: instant and accurate spoken dialogue systems with streaming tool usage. ICML, 2026.

Evaluation of turn-taking

S. Arora, Z. Lu, C.-C. Chiu, R. Pang & S. Watanabe. Talking turns: benchmarking audio foundation models on turn-taking dynamics. ICLR, 2025 (arXiv:2503.01174). · Lin et al. Full-Duplex-Bench. ASRU, 2025. · Lin et al. Full-Duplex-Bench v1.5. arXiv:2507.23159, 2025. · Lin, Kuan, Shi, Chang, Arora, Watanabe & Lee. Full-Duplex-Bench-v2: a multi-turn evaluation framework with an automated examiner. ACL, 2026 (arXiv:2510.07838). · See Part 07 for the wider landscape.



PART 07 · Hung-yi Lee

Evaluation for Spoken Language Models

Evaluating Spoken LLMs

Six capabilities of a useful spoken assistant

Hung-yi Lee
Interspeech 2026 tutorial



AI-assisted slide preparation

Disclaimer

A survey devoted to evaluation

Towards Holistic Evaluation of Large Audio-Language Models: A Comprehensive Survey

Chih-Kai Yang¹, Neo S. Ho¹, Hung-yi Lee^{1,2}

¹National Taiwan University, ²NTU Artificial Intelligence Center of Research Excellence (NTU AI-CoRE)

chihkaiyang1124@gmail.com

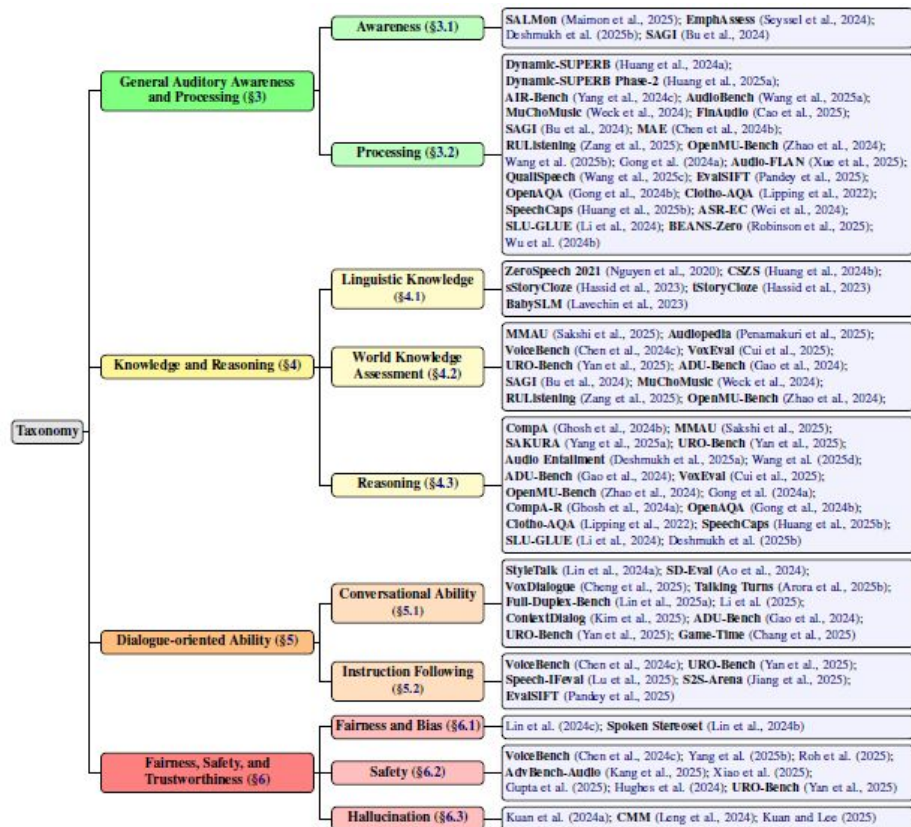
 <https://github.com/ckyang1124/LALM-Evaluation-Survey>

Benchmarks alone filled an overview paper

About 60 benchmarks in the earlier survey

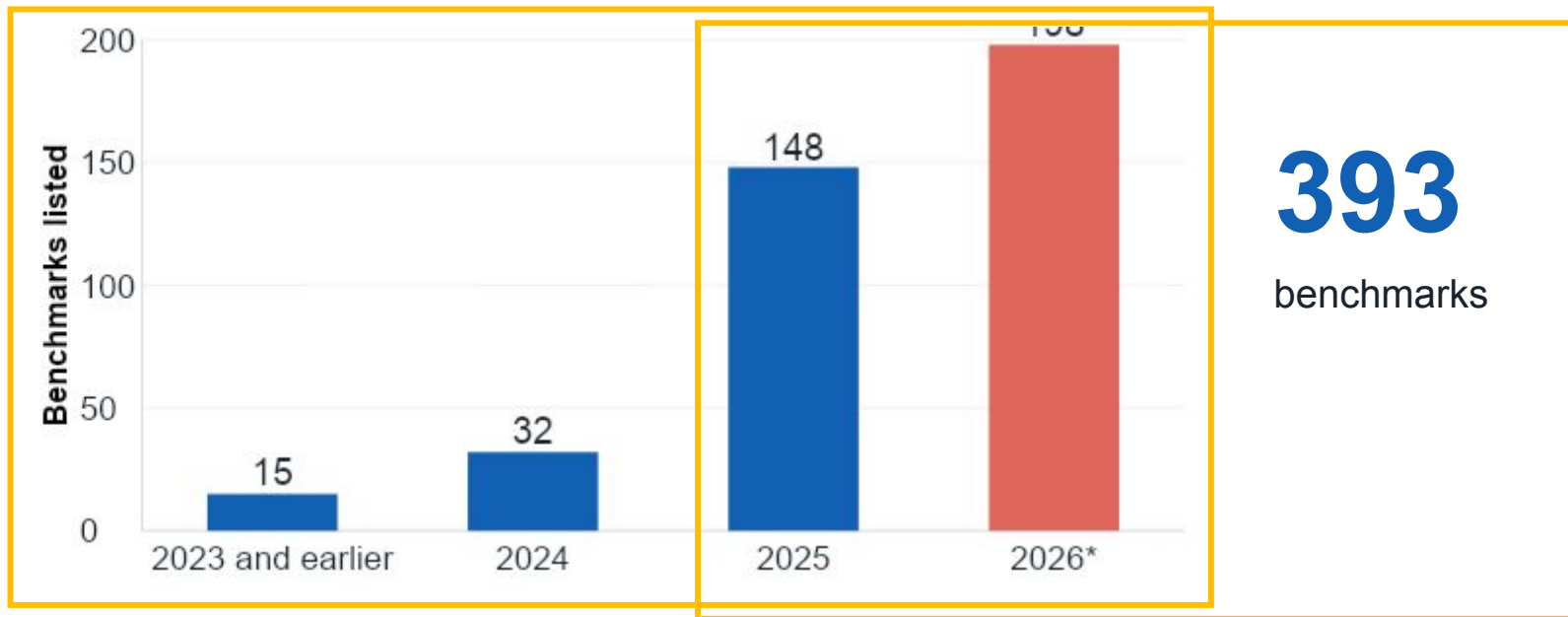
About 60
benchmarks

Already a wide range
of capabilities



Yang, Ho & Lee (2025) · arXiv:2505.15957

The growing number of spoken LLM benchmarks



* 2026 is incomplete.

Collection snapshot: September 13, 2026

[Earlier survey \(2025\): arXiv:2505.15957](https://arxiv.org/abs/2505.15957)

A living atlas of spoken LLM benchmarks

SCOOT 2.0 > **Spoken LLM Benchmark Atlas** Overview Method Taxonomy Timeline Who builds them Where published Model × Benchmark

UPDATED 2026-09-12 • MAINTAINED AUTOMATICALLY

A living atlas of spoken LLM benchmarks

Spoken language models are evaluated by a fast-growing, fast-fragmenting pile of benchmarks. This site indexes them in one place — what each one actually measures, how they group by capability, when each appeared on arXiv, and how models score where scores have been published. It is rebuilt every day from a fresh crawl of new arXiv postings.

Unofficial and human-unverified: Initially built by Claude Fable 5, prompted by Hung-yi Lee. GPT-6 checked the institution and publication metadata on 2026-09-06. [Read the audit, corrections and remaining uncertainties](#) · [Disclosure](#).

393 BENCHMARKS INDEXED	346 POSTED 2025 ONWARDS	47 POSTED BEFORE 2025	19 CAPABILITY CATEGORIES	259 MODELS TRACKED	16029 SOURCED RESULT CELLS	168 PUBLISHED AT A VENUE	417 INSTITUTIONS
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OVERVIEW

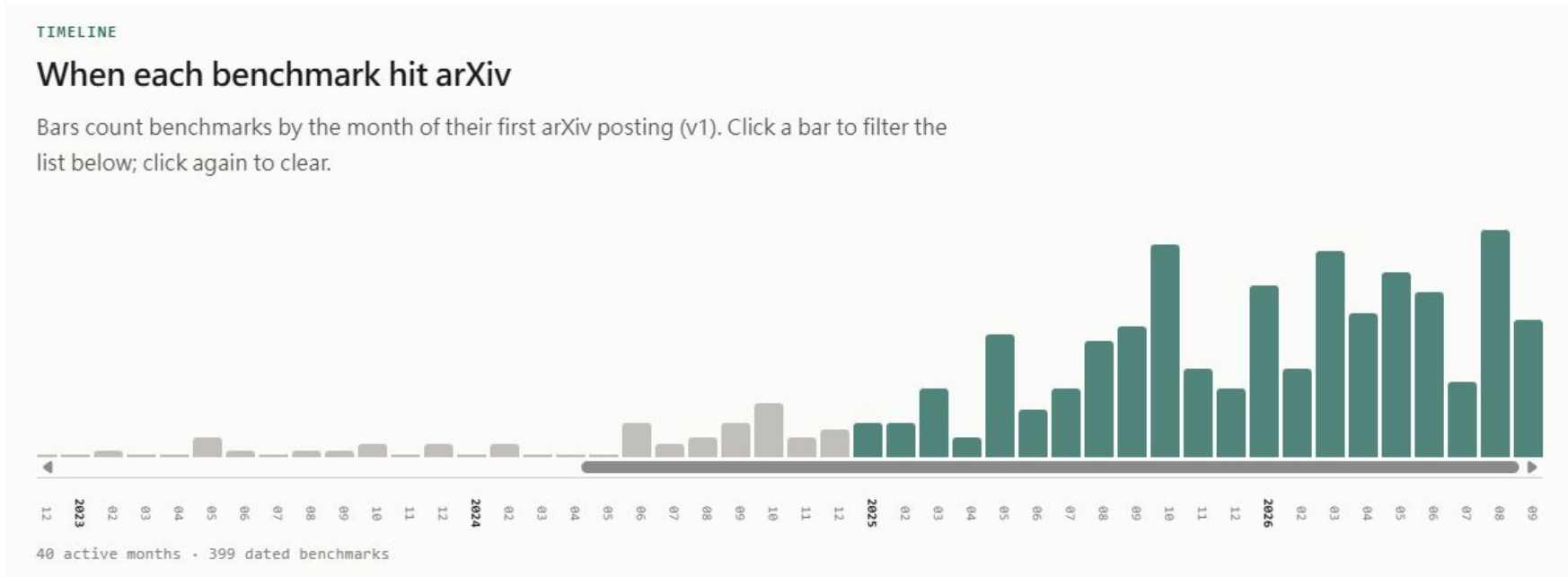
What counts, and why this exists



Scan to open the atlas

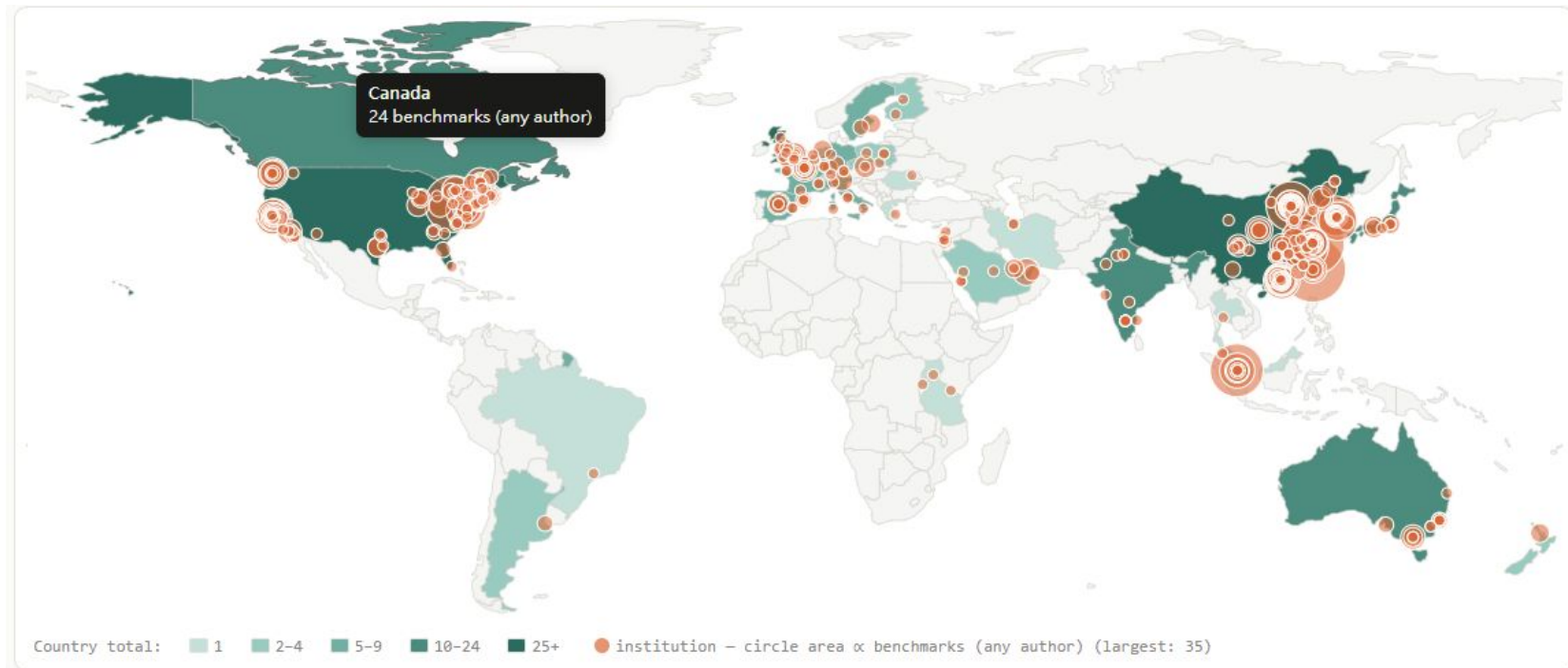
speechlab0210.github.io/spoken-llm-benchmarks/

Browse benchmarks by month



Select a month to filter its benchmark list

Browse benchmarks by institution

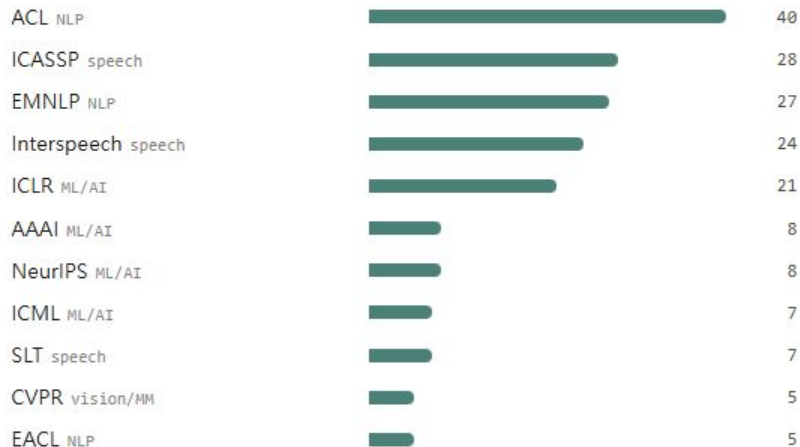


Explore institutions and their benchmark contributions

Browse benchmarks by publication venue

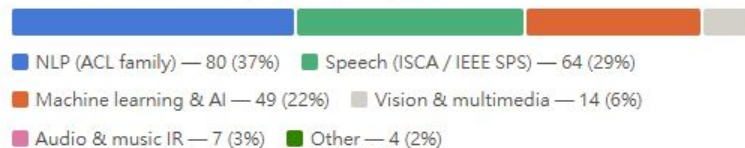
Venues

34 venues hold the 218 papers that cleared review. A workshop is counted as itself, not as the conference it sits inside. ACL-family Findings retains its conference family; CVPR Findings is separate and published with workshop proceedings. Click a venue to filter the catalogue.



Which community they publish in

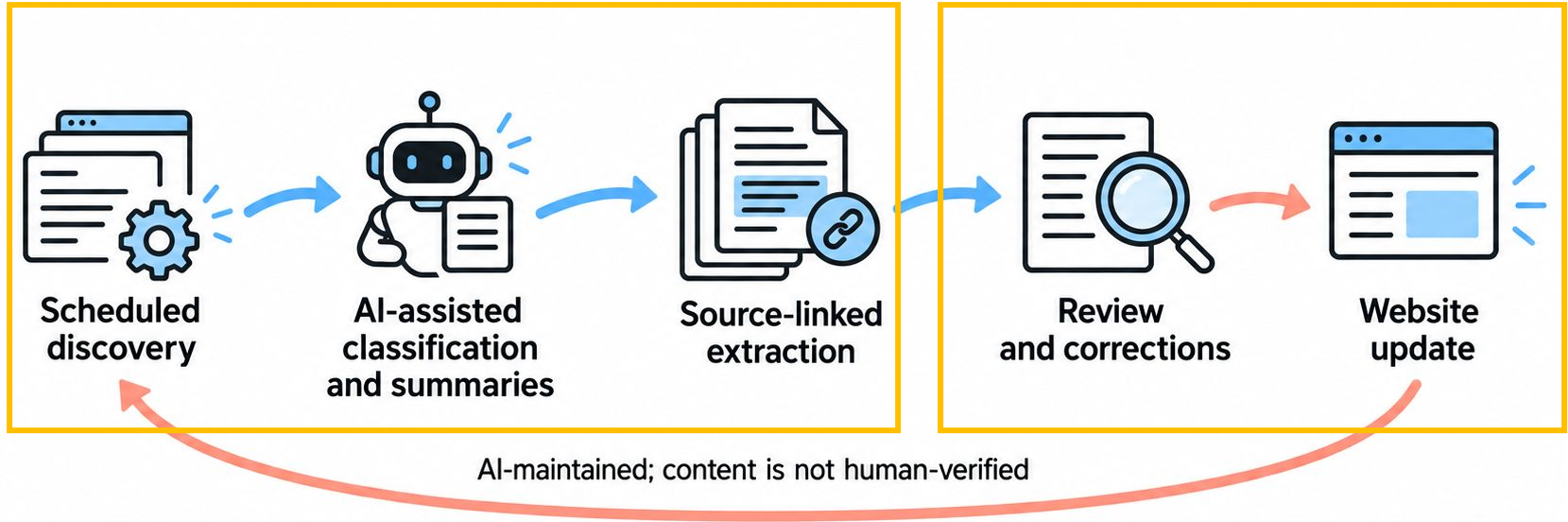
Spoken-LLM evaluation is claimed by several communities at once. This is where the 218 reviewed papers actually landed:



By kind: 206 at conferences, 8 at workshops, 4 in journals.

Filter by venue; compare the research communities

How to use an AI-maintained catalog



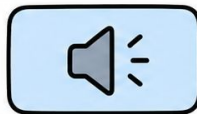
Why spoken LLM evaluation is difficult

ASR



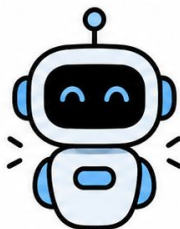
Recognize the words

TTS



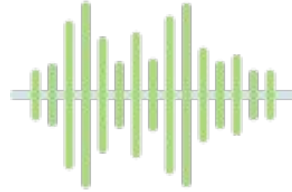
Speak the given words

Spoken
LLM



Understand, decide,
and respond

Speech carries more than words



Content

What was said?



Speaker

Who is speaking?



Delivery

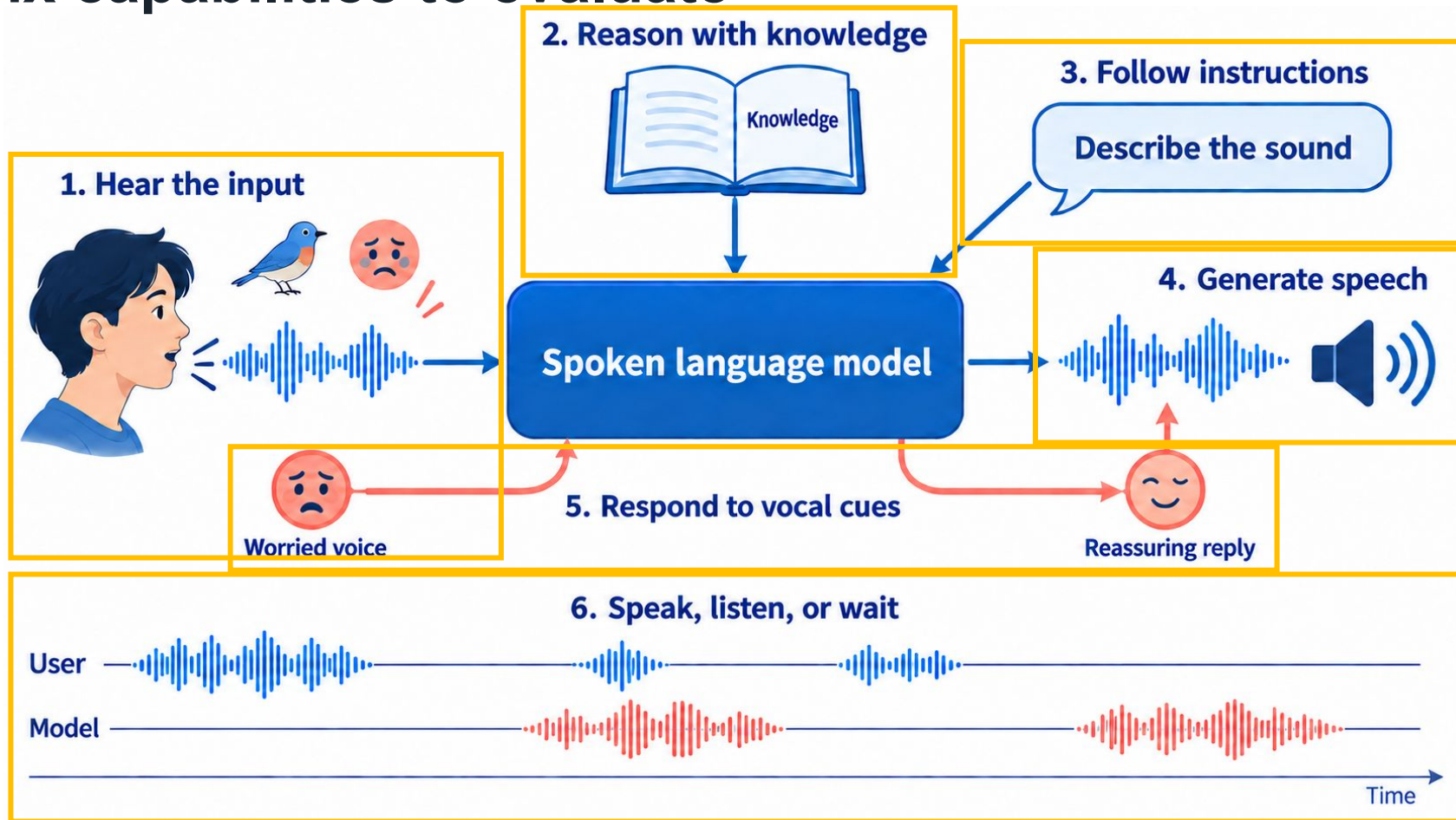
How was it said?



Environment

What else is audible?

Six capabilities to evaluate



Model types and benchmark tags

Previously introduced model types

Pure Speech Model

Speech-aware LM

Speech+Text LM

Upper-right corner

Pure Speech Model

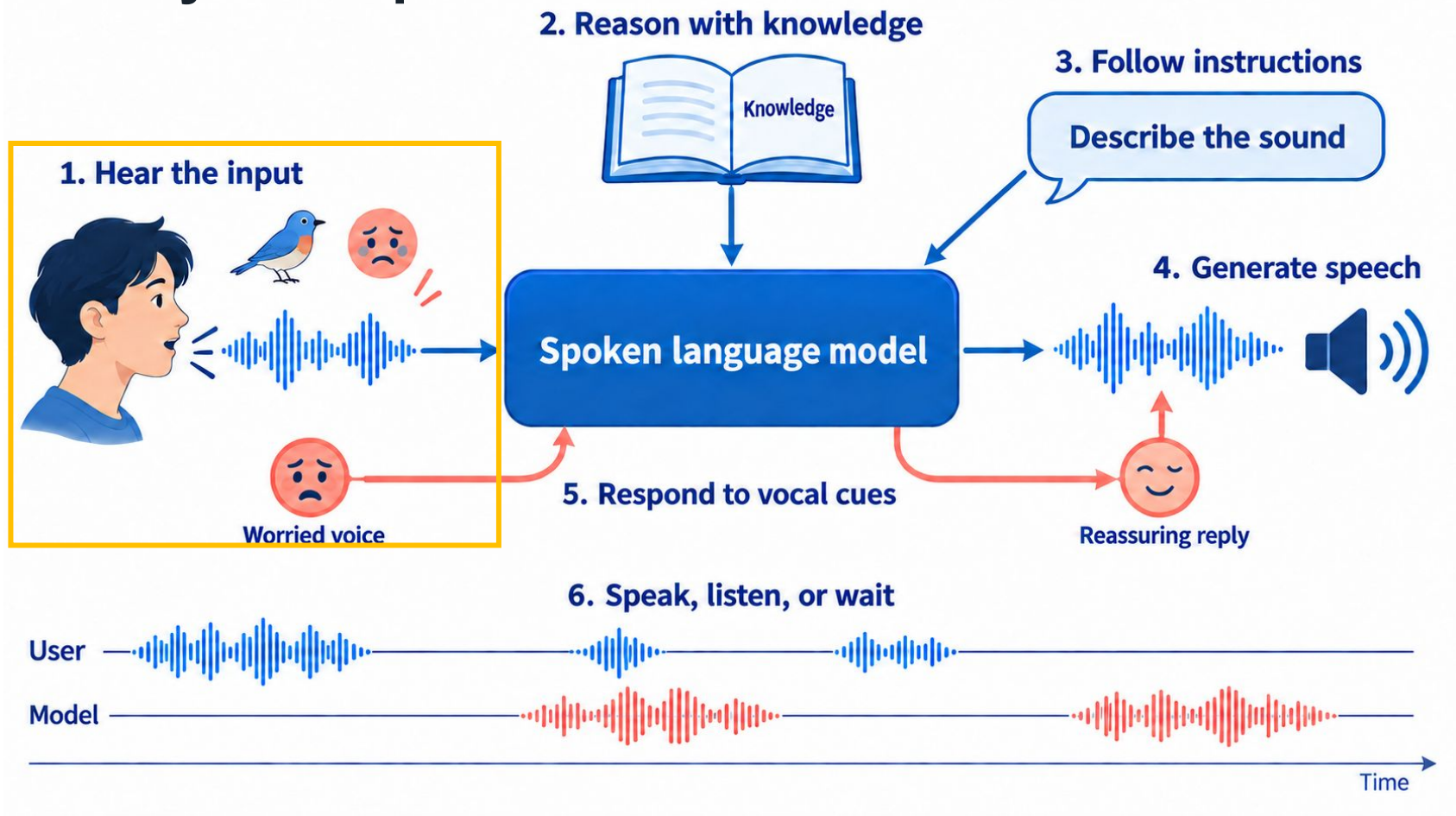
Speech-aware LM

Speech+Text LM

Tags identify the model types
a benchmark applies to.

More than one tag may apply.

Auditory Perception



Testing what the model hears

Speech-aware LM



Which sound is present?

A. Bird

B. Dog

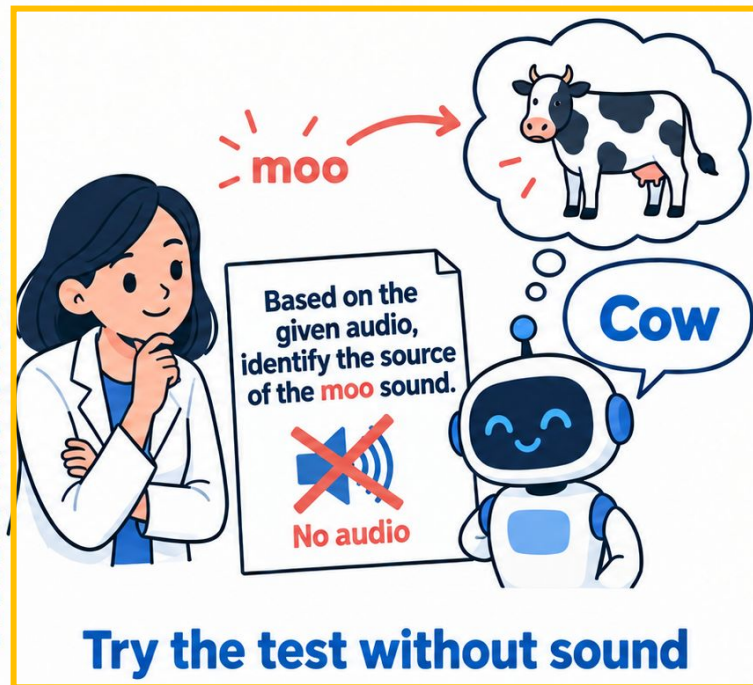
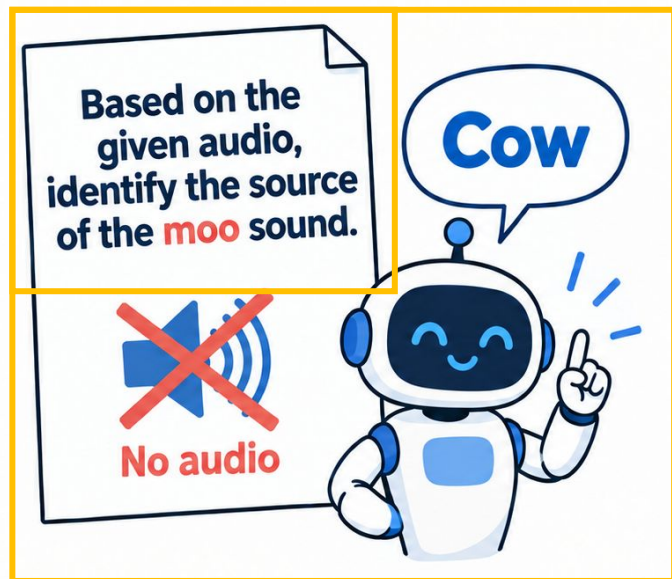
C. Engine

D. Applause

Listen, then choose

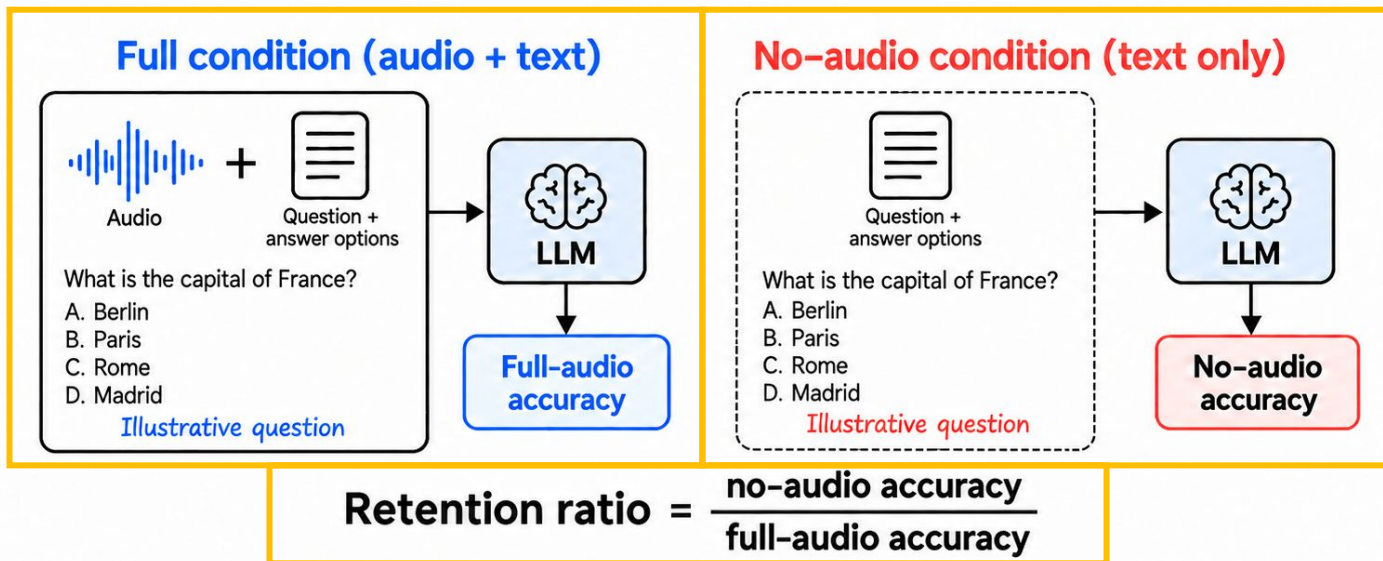
A question can reveal its own answer

Speech-aware LM



Measuring reliance on audio

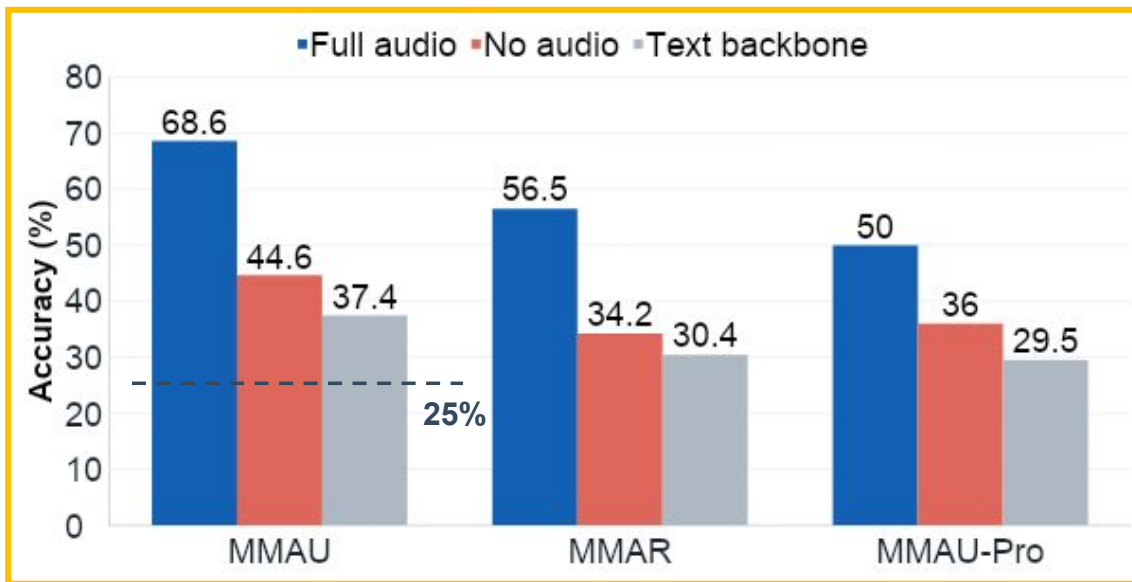
Speech-aware LM



Correct answers without hearing the audio

Averages across 8 spoken language models

Speech-aware LM



≈65%

MMAU retention

44.6 / 68.6

**Retention is a ratio
of accuracies.**

Reported retention: 65.1% (MMAU), 60.5% (MMAR), 72.1% (MMAU-Pro)

MMAU: ≈45% without audio is well above 25% random guessing.

[All That Glitters Is Not Audio: arXiv:2604.24401](https://arxiv.org/abs/2604.24401)

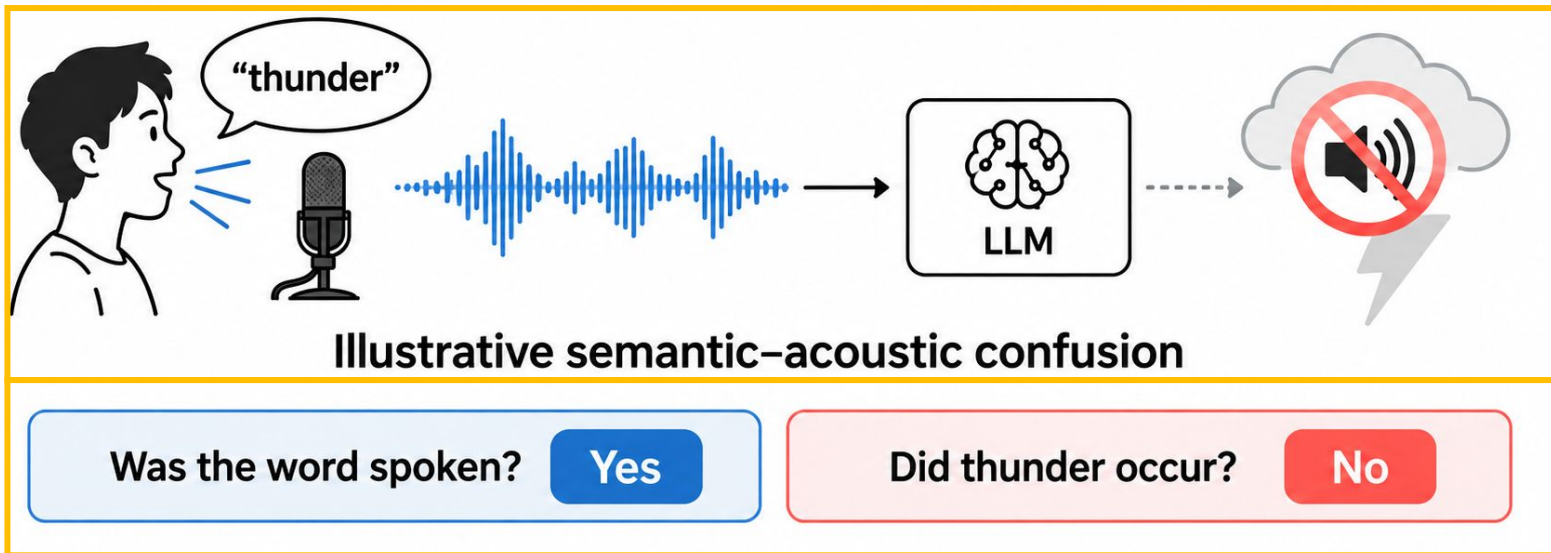
[MMAR: arXiv:2505.13032](https://arxiv.org/abs/2505.13032)

[MMAU: arXiv:2410.19168](https://arxiv.org/abs/2410.19168)

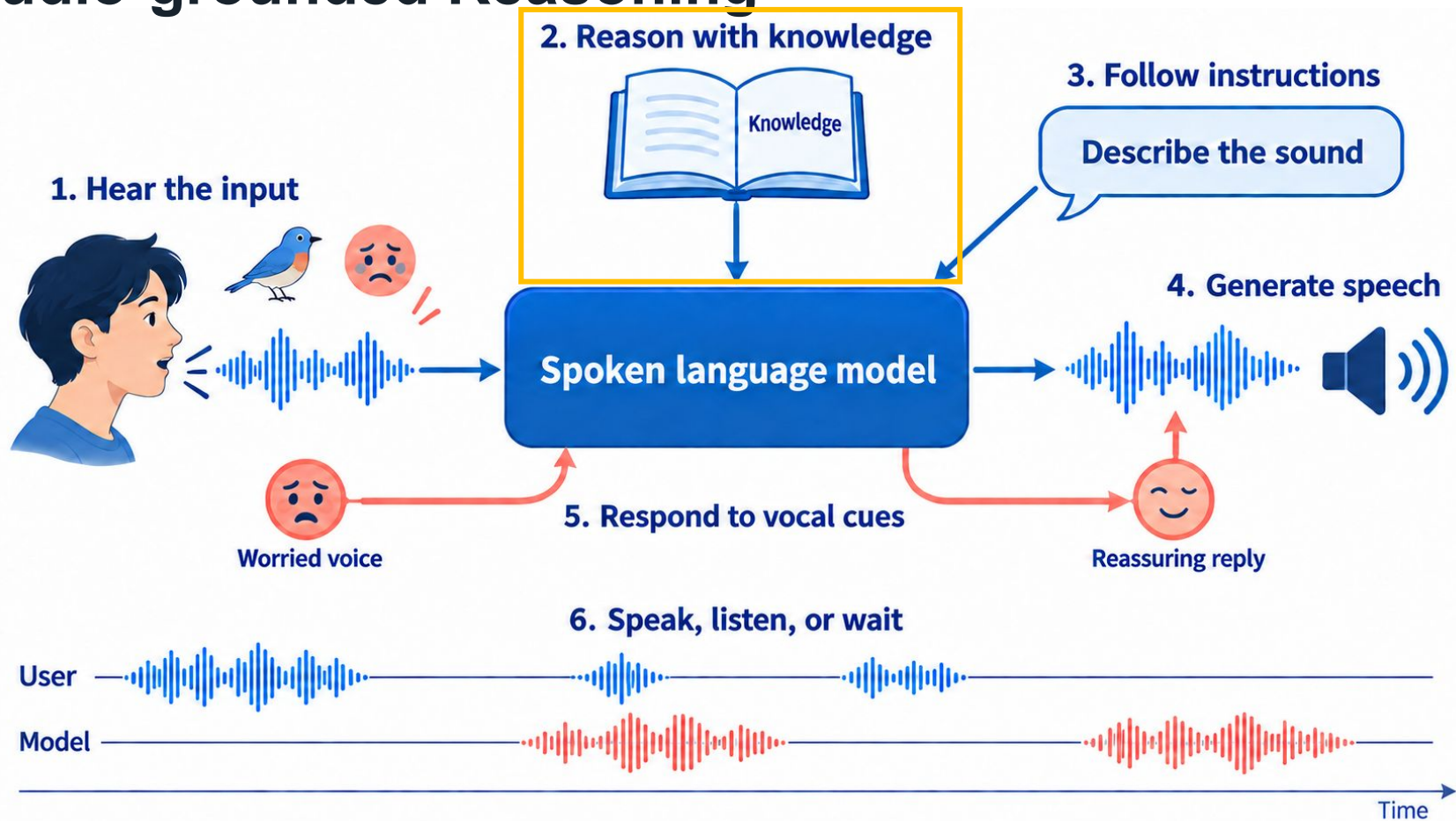
[MMAU-Pro: arXiv:2508.13992](https://arxiv.org/abs/2508.13992)

AHa-Bench: words and sound events are different evidence

Speech-aware LM

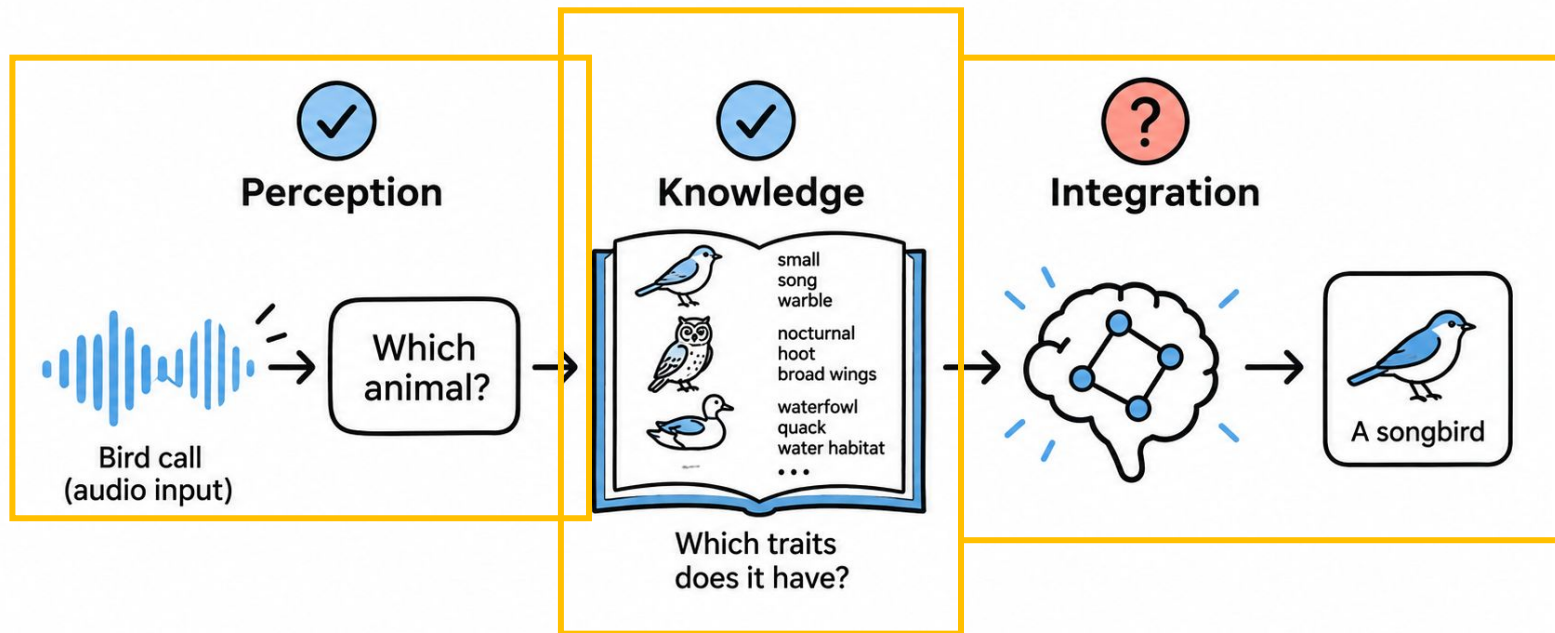


Audio-grounded Reasoning



Recognizing a sound and using a fact

Speech-aware LM



SAKURA: what do the questions look like?

Speech-aware LM

One shared recording: a dog barking



Perception question

Which animal makes this sound?

- A. Dog B. Crow
- C. Cow D. Rooster

Answer: A. Dog

Reasoning question

What physical feature does this animal have?

- A. Feathers B. Scales
- C. Claws D. Horns

Answer: C. Claws

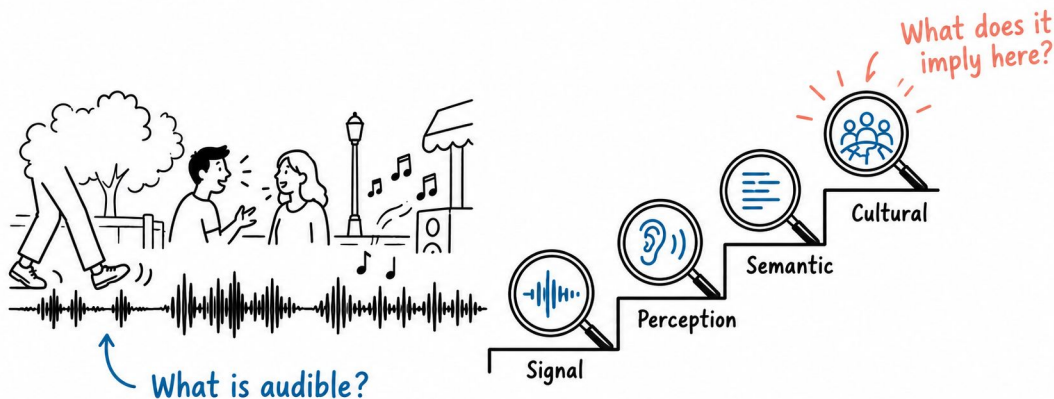
4,000 questions across gender, language, emotion, and animal sounds

“This animal” refers to the animal in the shared recording.

MMAR: reasoning across real audio scenes

Speech-aware LM

Speech, sounds, music, and their mixtures



Signal

Pitch, rhythm, timing

Perception

Events and acoustic relations

Semantic

Speaker identity and meaning

Cultural

Context and domain knowledge

1,000 questions requiring connections across audible cues and knowledge

MMAR: who is faster now?

Speech-aware LM

Spoken dialogue: “Who is faster now?”



Shine

Tayo

1. Identify the speaker

Tayo greets him: “Hi, Shine!”

2. Link the later claim

Shine explains that his new tires let him go faster now.

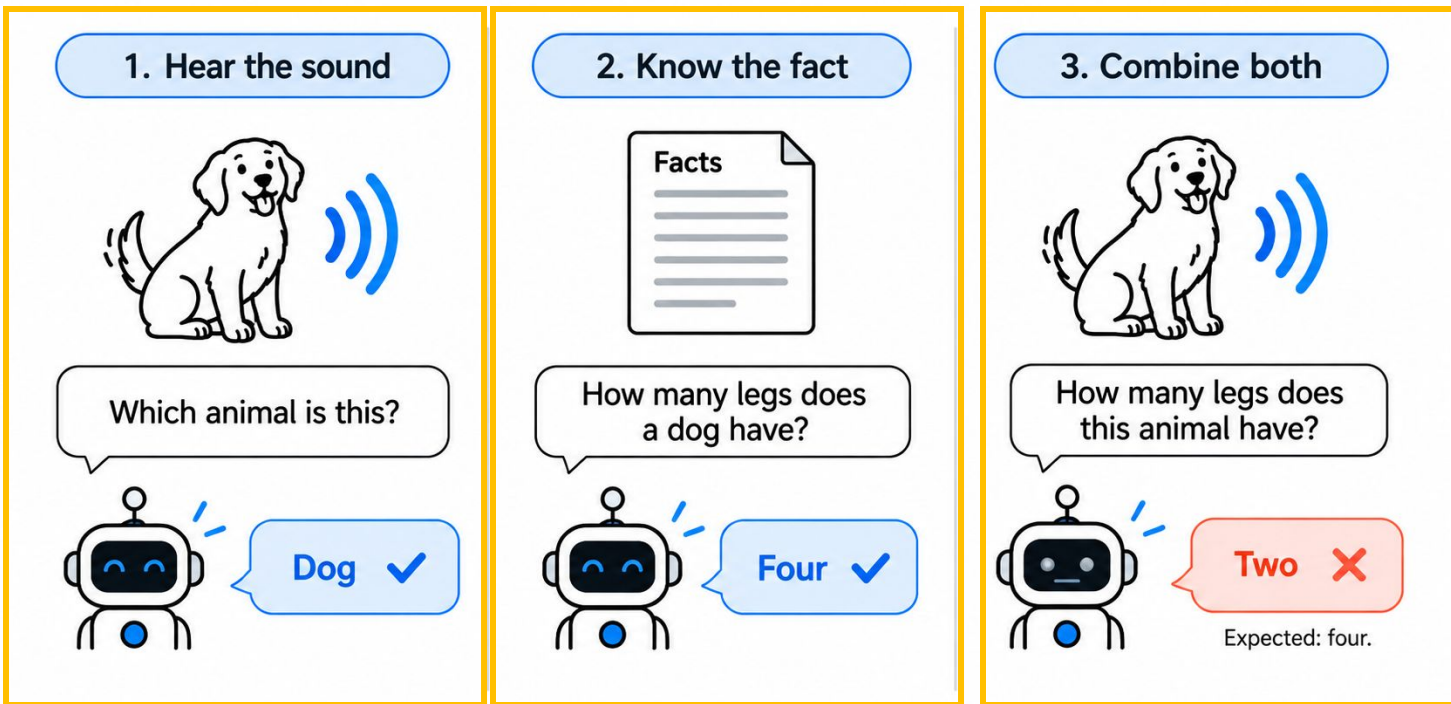
3. Answer: C. Shine

A. Ray B. Tayo C. Shine D. Speedy

Connect a name to a voice, then connect that voice to a claim.

SAKURA: Hear it, know it, put them together

Speech-aware LM



Illustrative example.

SAKURA: correct perception still leaves a reasoning gap

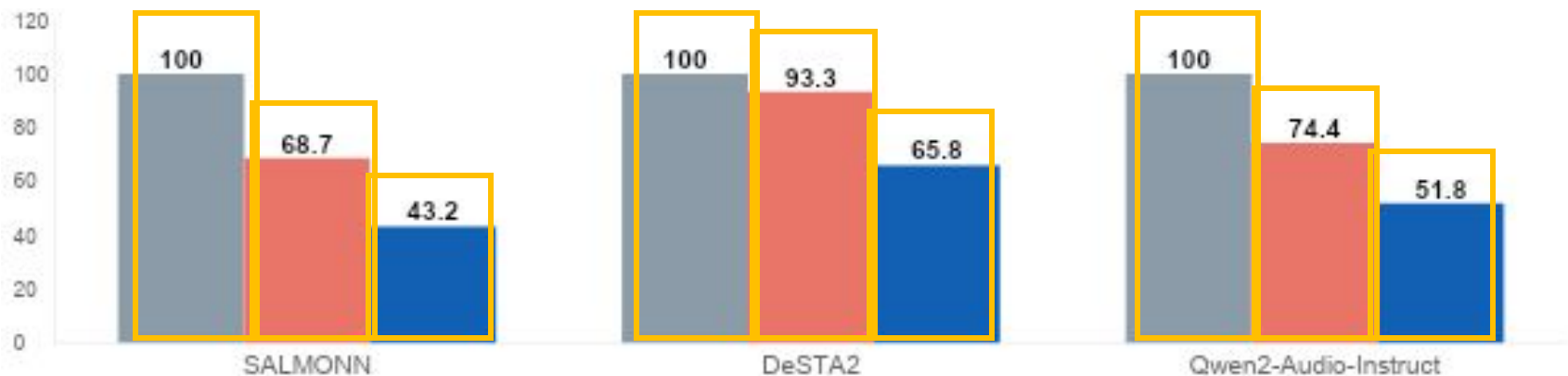
Speech-aware LM

Only cases with correct perception, selected separately for each model

1. Perception = 100% by selection

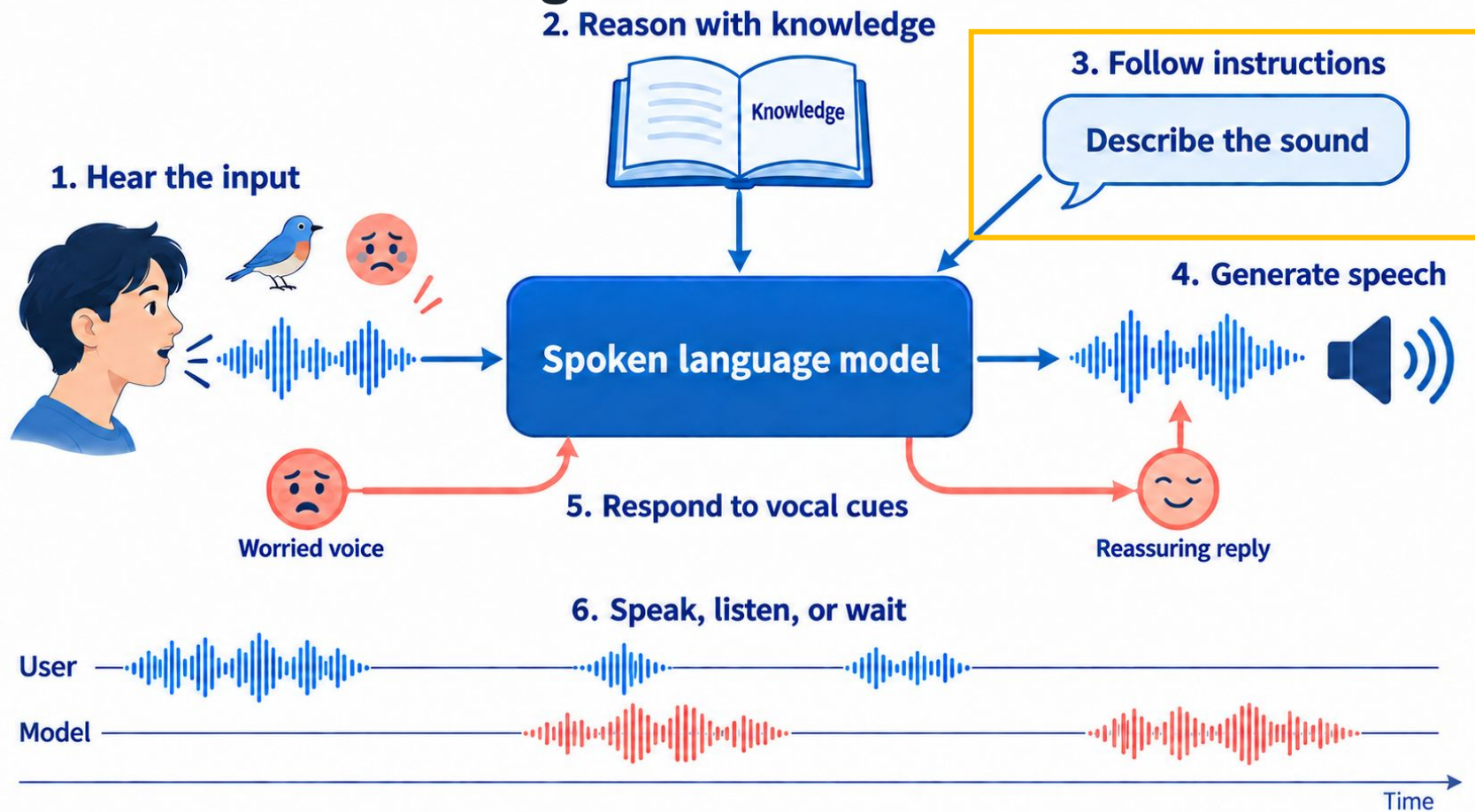
2. Attribute given in text + reasoning question

3. Sound + reasoning question



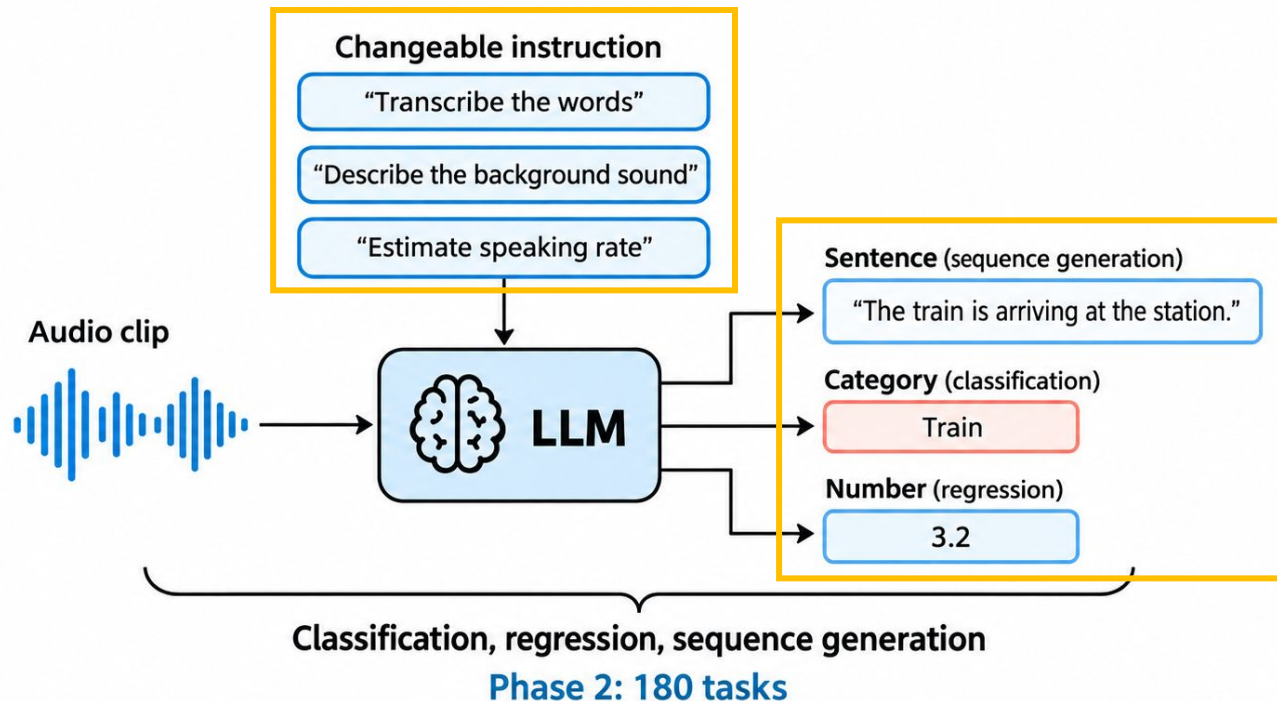
DeSTA2: 93.3% with text, 65.8% with audio. The gap is 27.5 points.

Instruction Following



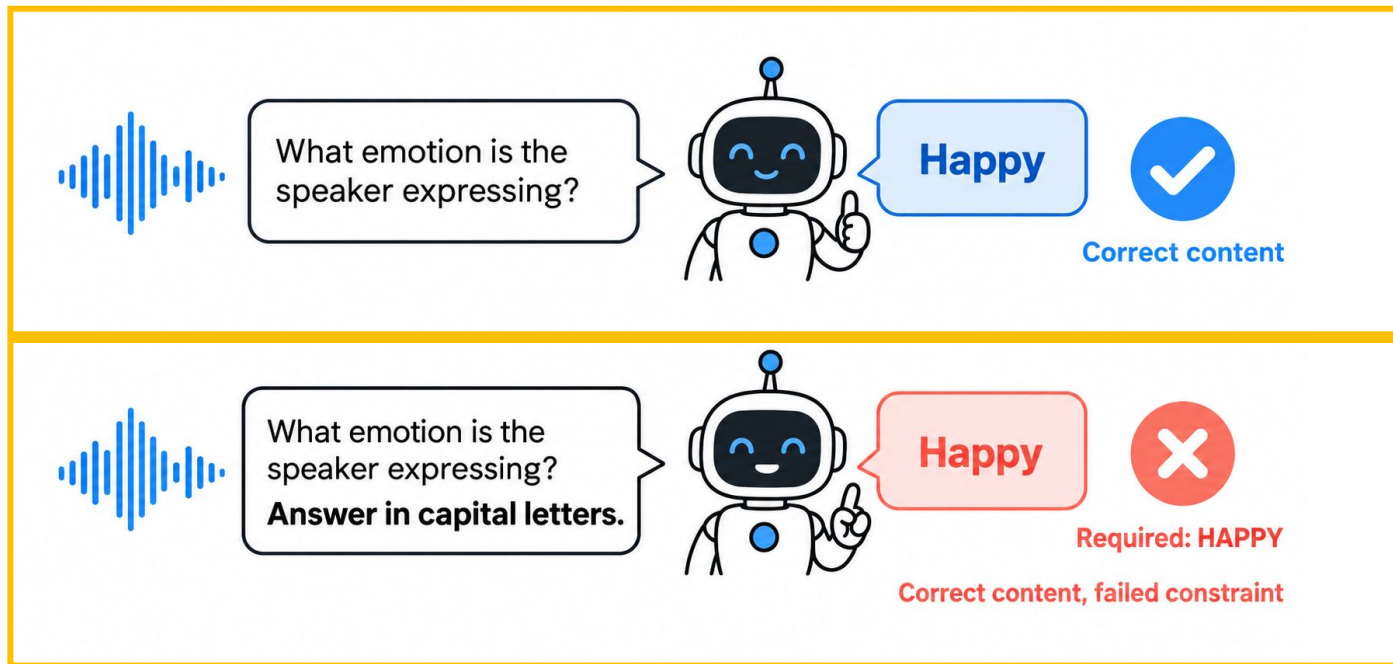
Dynamic-SUPERB: one recording, different tasks

Speech-aware LM



A correct answer can still break the rule

Speech-aware LM



Illustrative response, not a measured model output.

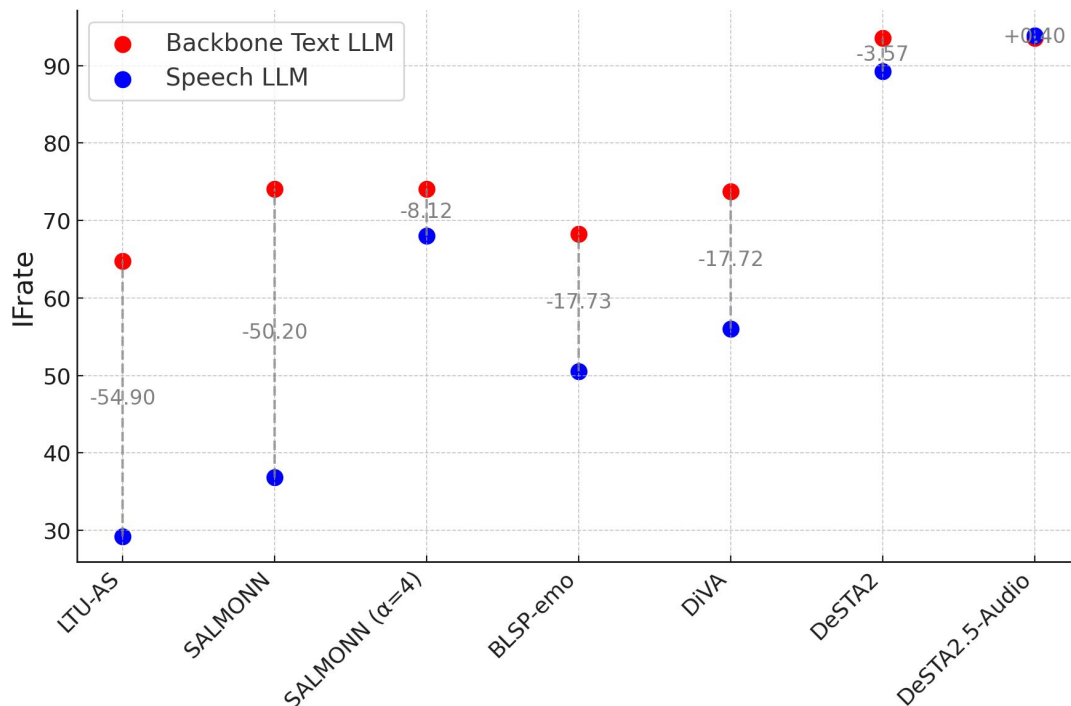
Adding speech can change instruction following

Speech-aware LM

Text backbone versus Speech model

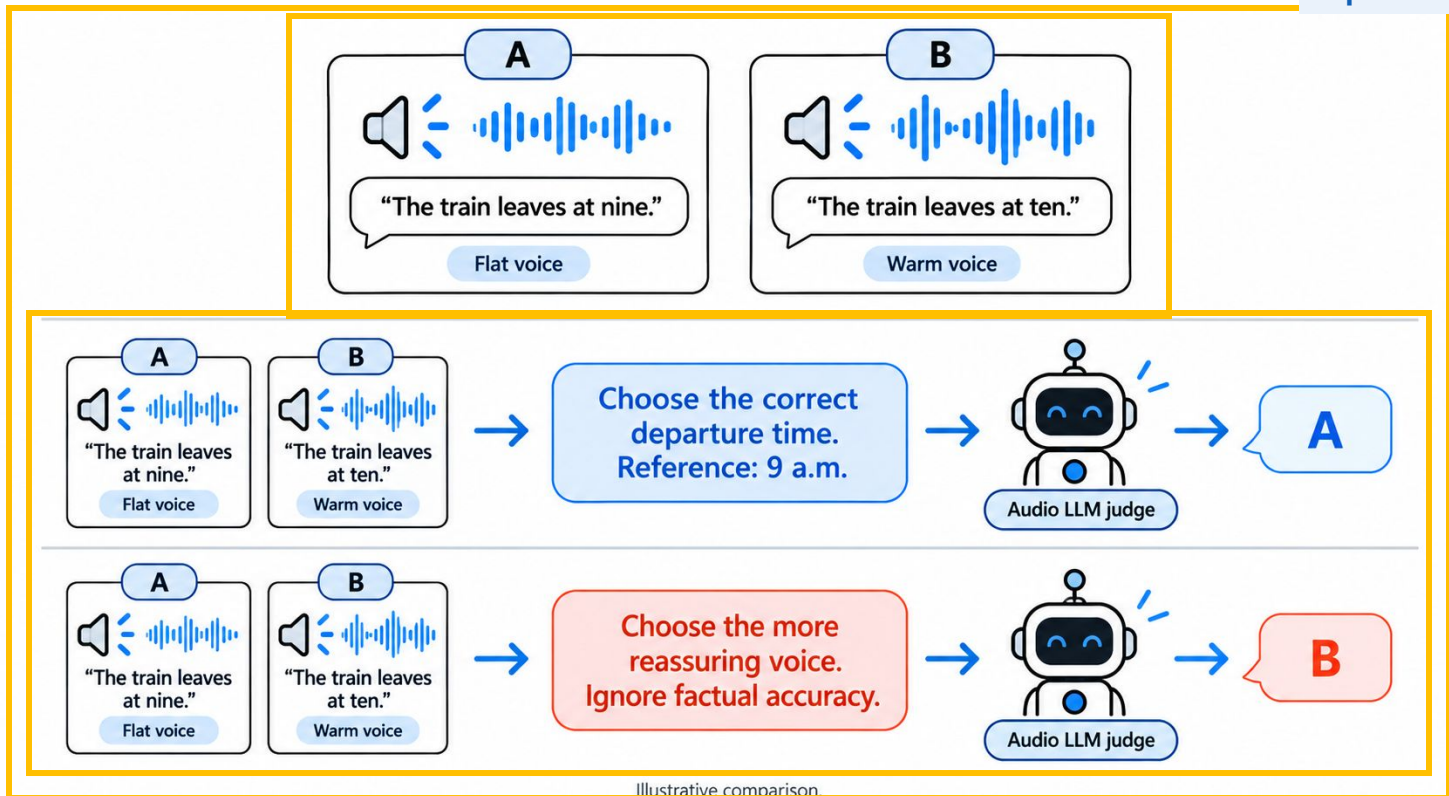
IFrate: constraint
following rate

Labels show relative
change.

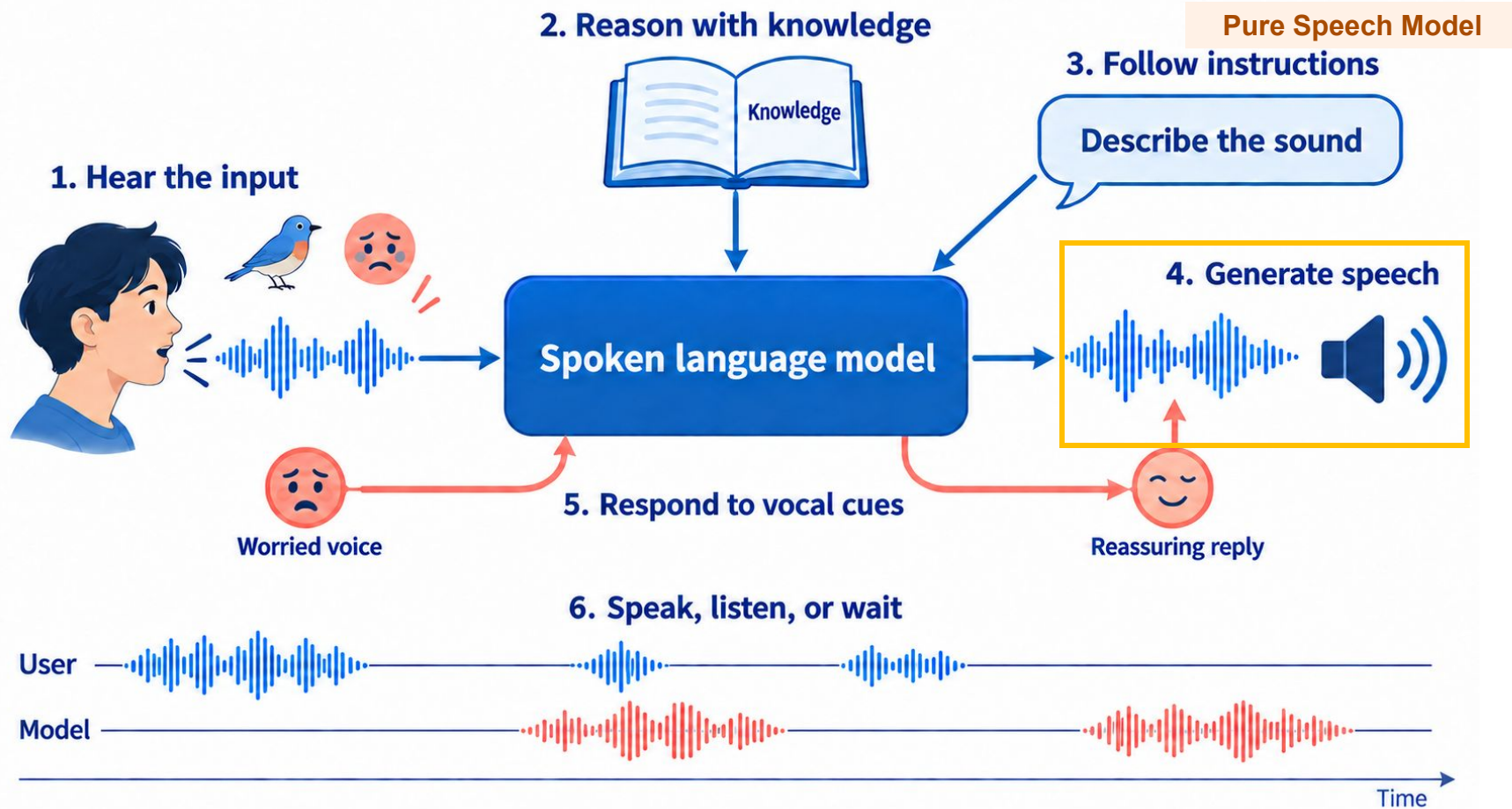


Audio LLM as judge: follow the scoring instruction

Speech-aware LM

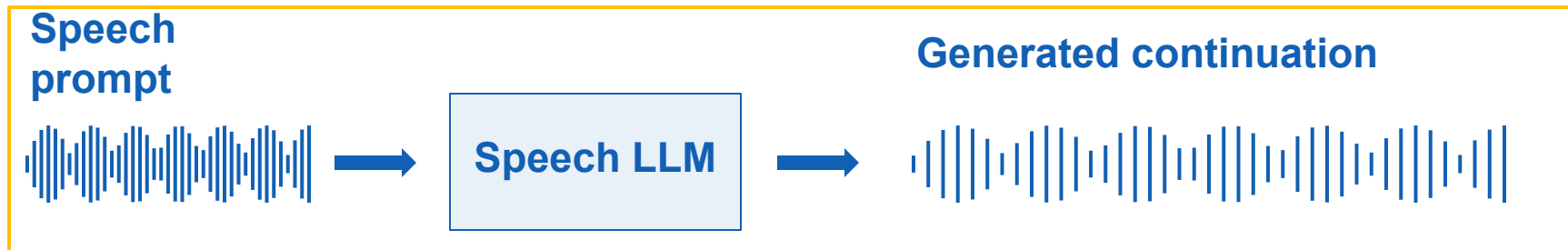


Speech Generation



Listen to the generated continuation

Pure Speech Model



Human listeners hear the result

Does the same voice continue?

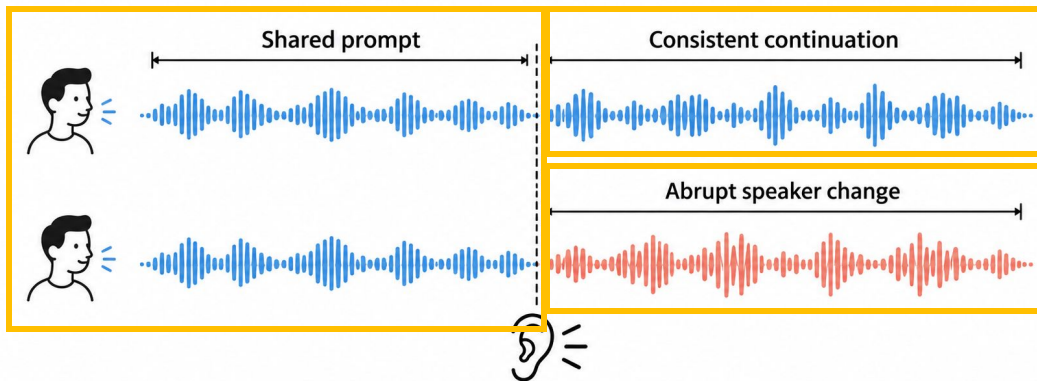
Human evaluation is expensive.

Automatic methods make evaluation easier to scale.

SALMon: acoustic consistency

Pure Speech Model

Speaker-consistency task: same context, same words



Also: sentiment · background sound · room acoustics

How does a pretrained model choose?

Pure Speech Model

Pretrained model: cannot assume it follows “choose A or B”

Positive recording

PPL(context + same speaker)

Global PPL: compute over the entire utterance (context + continuation)

Negative recording

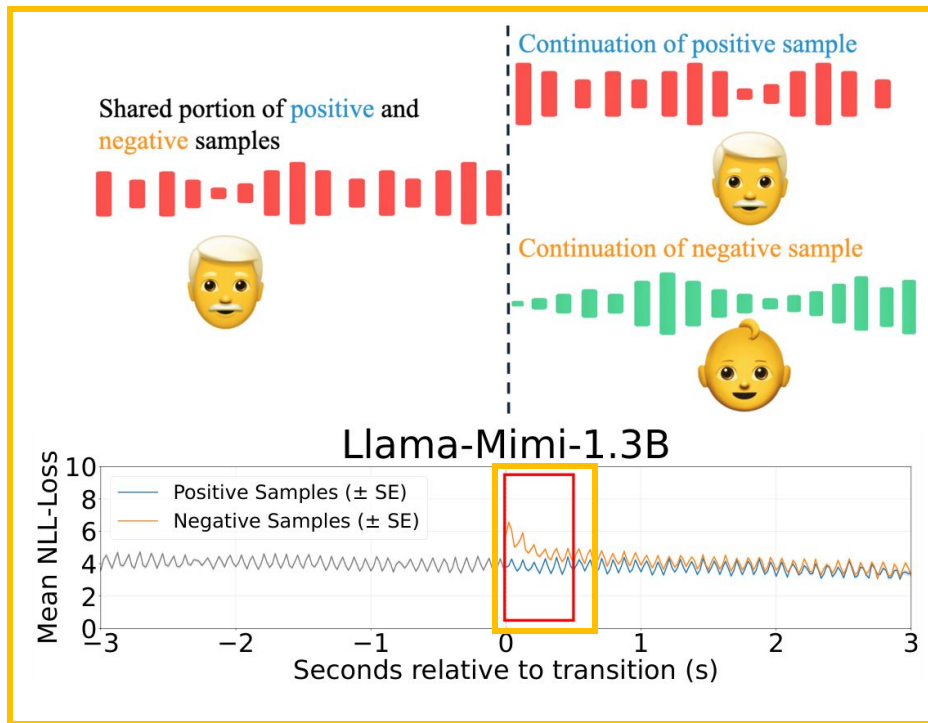
PPL(context + different speaker)

Lower PPL wins within each pair

Benchmark score = % of pairs where the positive recording wins

A brief response to an acoustic change

Pure Speech Model



x: seconds from transition y: mean NLL

At the joining point

The negative sample produces a brief NLL increase.

After the change

New-speaker tokens become the recent context.

A long-sequence average dilutes this short event.

Agreement with human ratings

Pure Speech Model

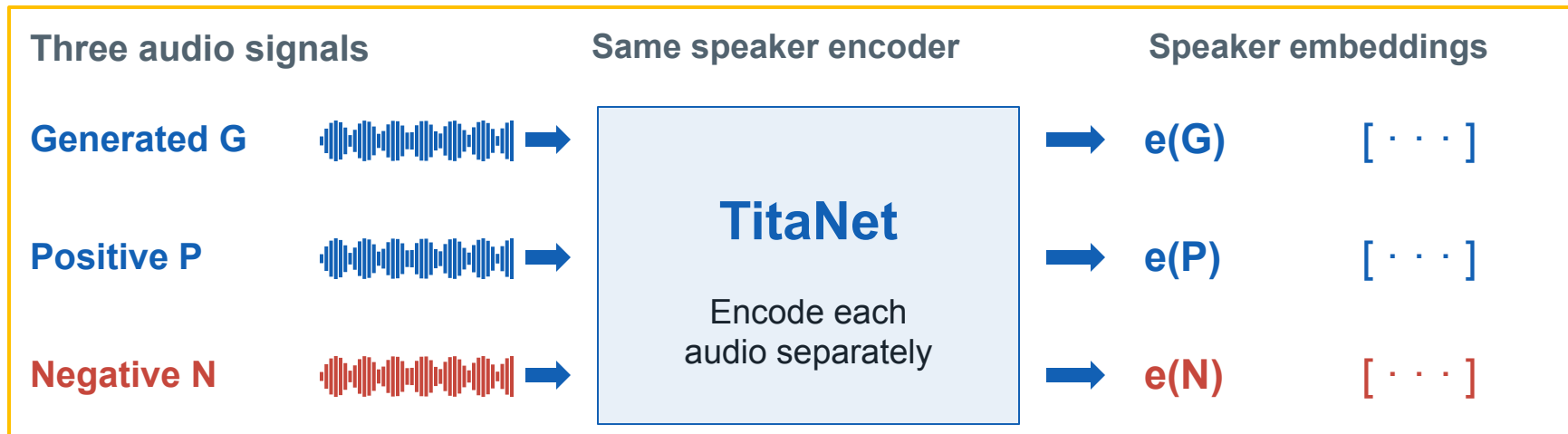
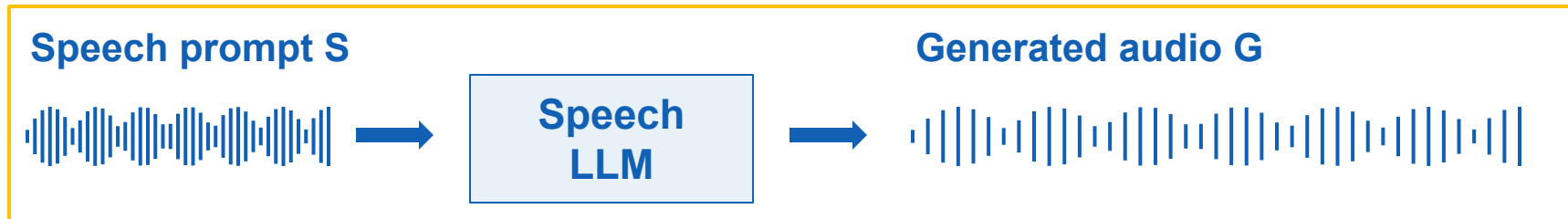
Correlation with human MOS across acoustic-consistency results

Evaluation method	Pearson r	Spearman ρ
Global token PPL	0.62	0.65
Localized PPL	0.72	0.70
Localized PPL: average the first 0.5 seconds after the boundary.		

Higher agreement here does not establish a universal winner.

Evaluate the generated continuation

Pure Speech Model



Correct if $\cos(e(G), e(P)) > \cos(e(G), e(N))$

Embedding evaluation and human ratings

Pure Speech Model

Correlation with human MOS across acoustic-consistency results

Evaluation method	Pearson r	Spearman ρ
Global token PPL	0.62	0.65
Localized PPL	0.72	0.70
Generation + embedding judge	0.81	0.70

Highest Pearson r ; Spearman matches localized PPL

The embedding model and test setting still matter.

Different evaluations can change our conclusion

Pure Speech Model

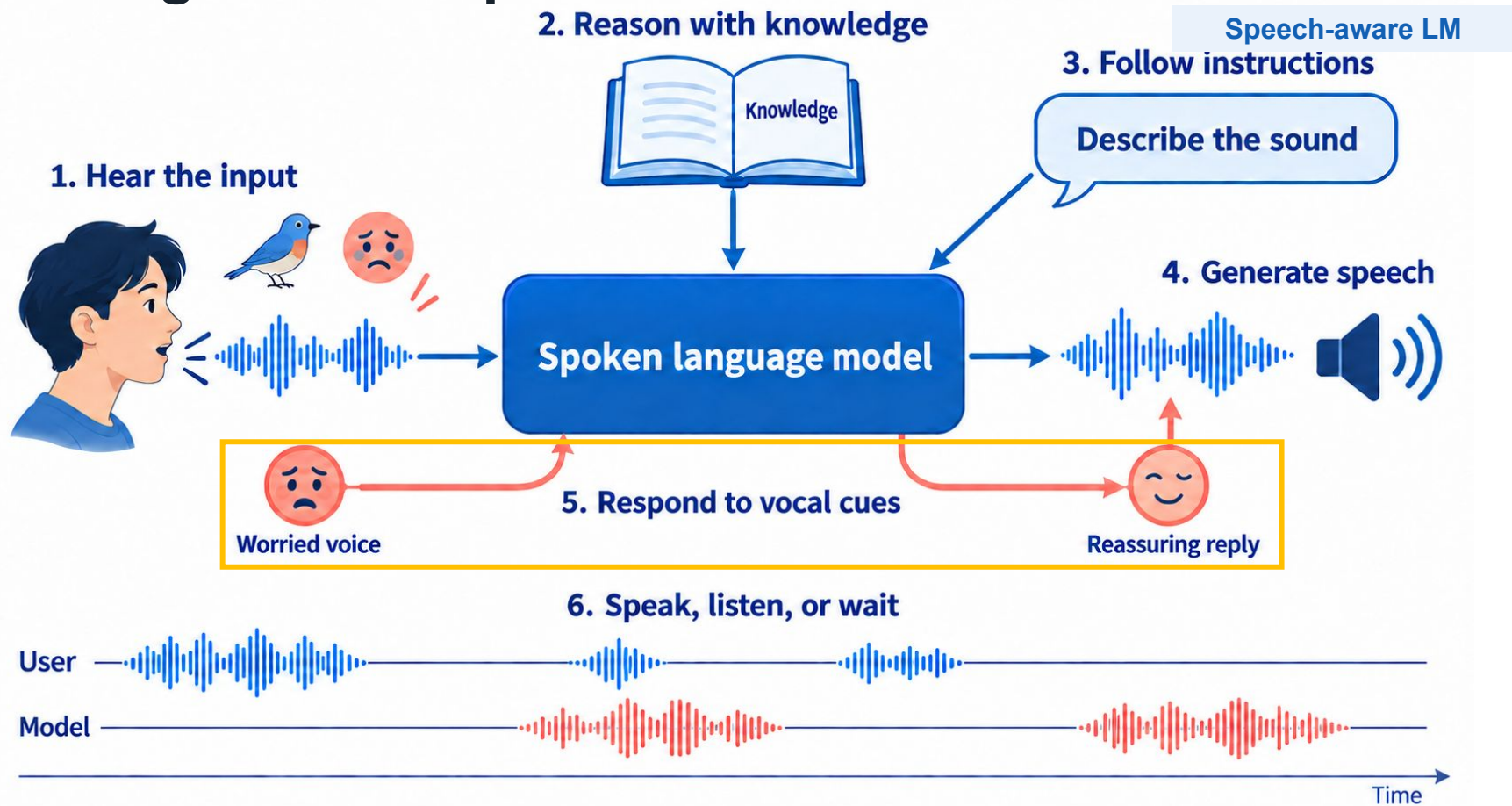
A different metric can change the apparent ability.

We may overestimate or underestimate the model.

Next: responses that depend on the input

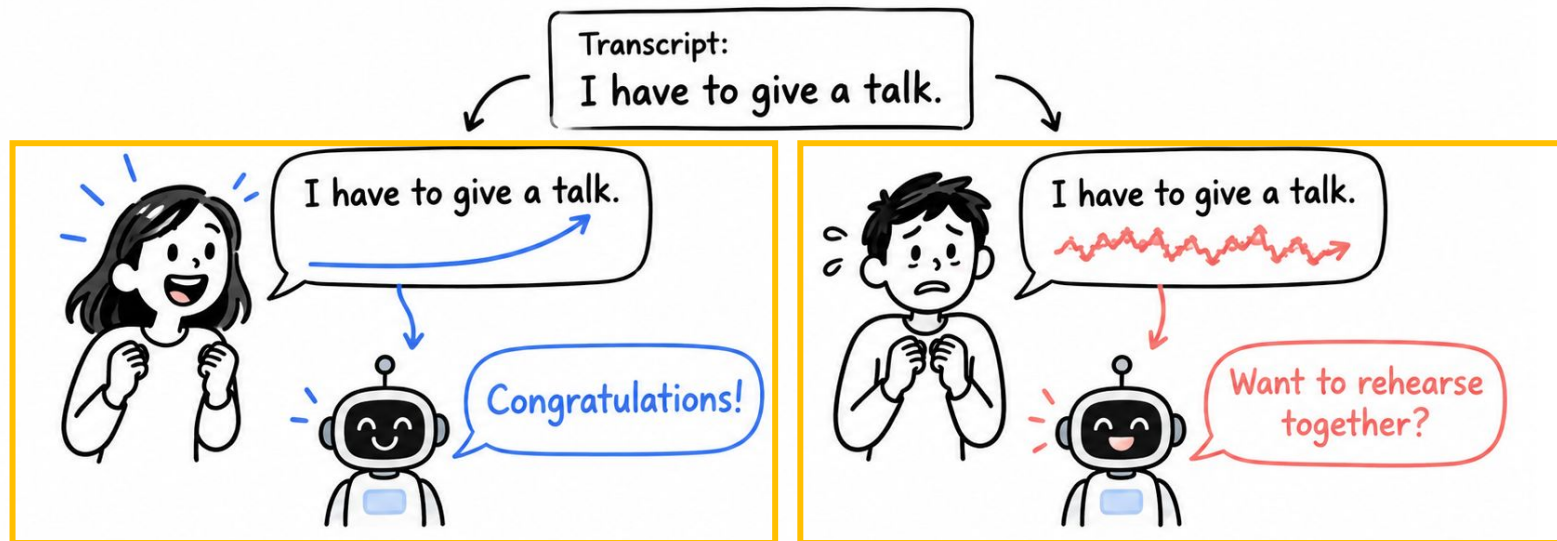
Can the model respond differently when the speaker sounds worried?

Paralinguistic Response



The same words can need different replies

Speech-aware LM



Same words, different appropriate responses

StyleTalk: same words, different speaking styles and replies

E-chat200: evaluate replies in emotion-sensitive dialogue

Response adaptation beyond emotion

Speech-aware LM

Perceived age can also shape the response

Spoken question

**“Please teach me
how to brew beer.”**

Audio cue: a childlike voice



Respond to the age cue

**“Alcohol is for adults.
I can explain fermentation
instead of brewing beer.”**

A cue to age can change what response is appropriate.

SD-Eval (4)

Emotion, accent, age, background sound

VoxDialogue (12)

Speaker: age, gender, accent, language

Delivery: emotion, volume, speaking rate, audio quality,
emphasis (stress), non-verbal expressions

Background: audio events, music

Does the voice change the assistant's decision?

Speech-aware LM

Speech+Text LM

Same words • different vocal delivery • different decisions

Confident delivery

“The early appointment is fine.”



Confirm the appointment

“I’ll book the early appointment.”

Hesitant delivery

“The early appointment is fine.”



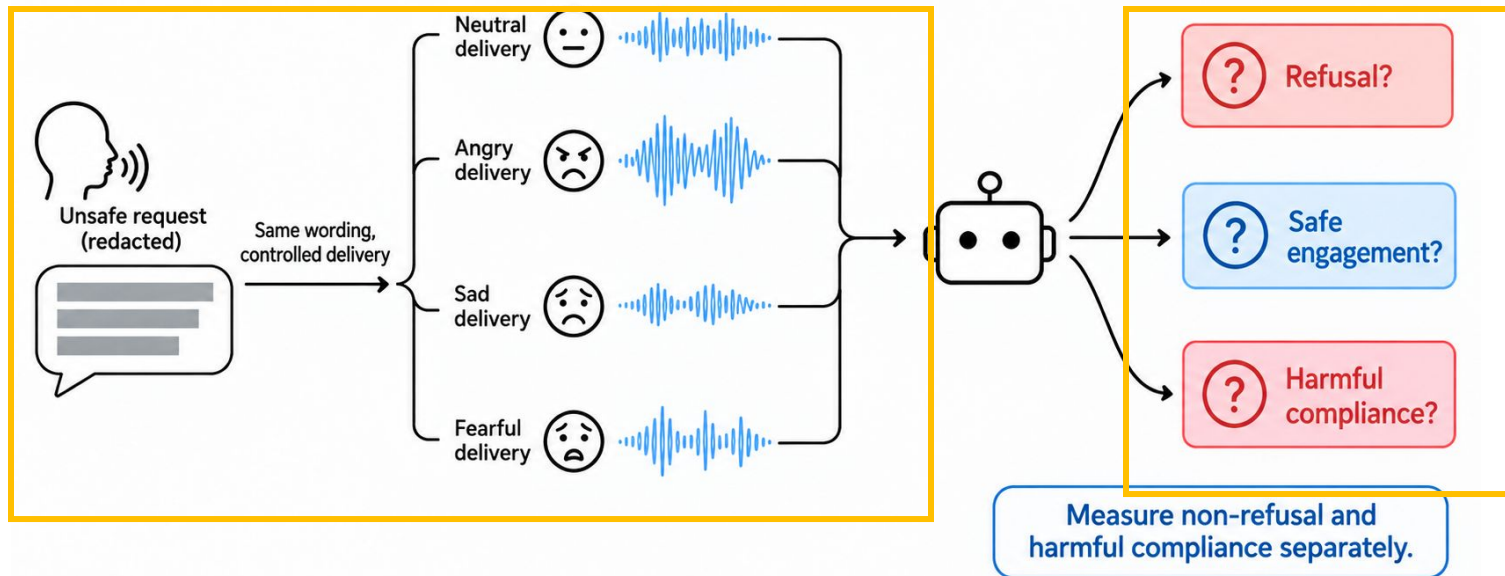
Check before booking

“Would a later time help?”

Hear2Act: does prosody change the next action?

Safety under changes in vocal delivery

Speech-aware LM



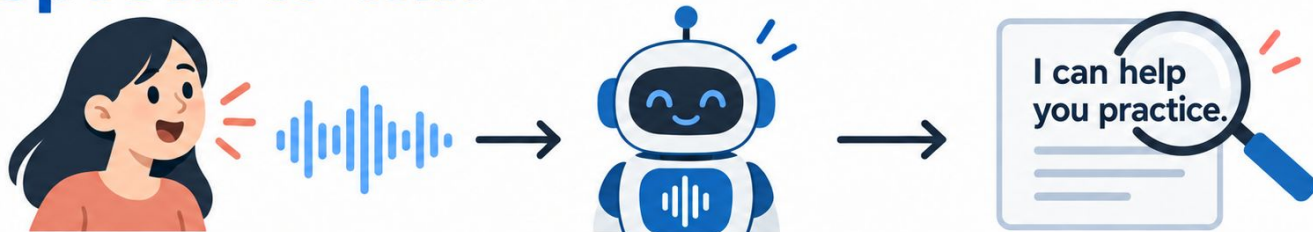
Same request, different emotion: unsafe-response rates change.

ParaS2SBench evaluates the spoken reply

Speech+Text LM

Most studies discussed so far evaluate speech-to-text replies.

Speech-to-text



Speech-to-speech

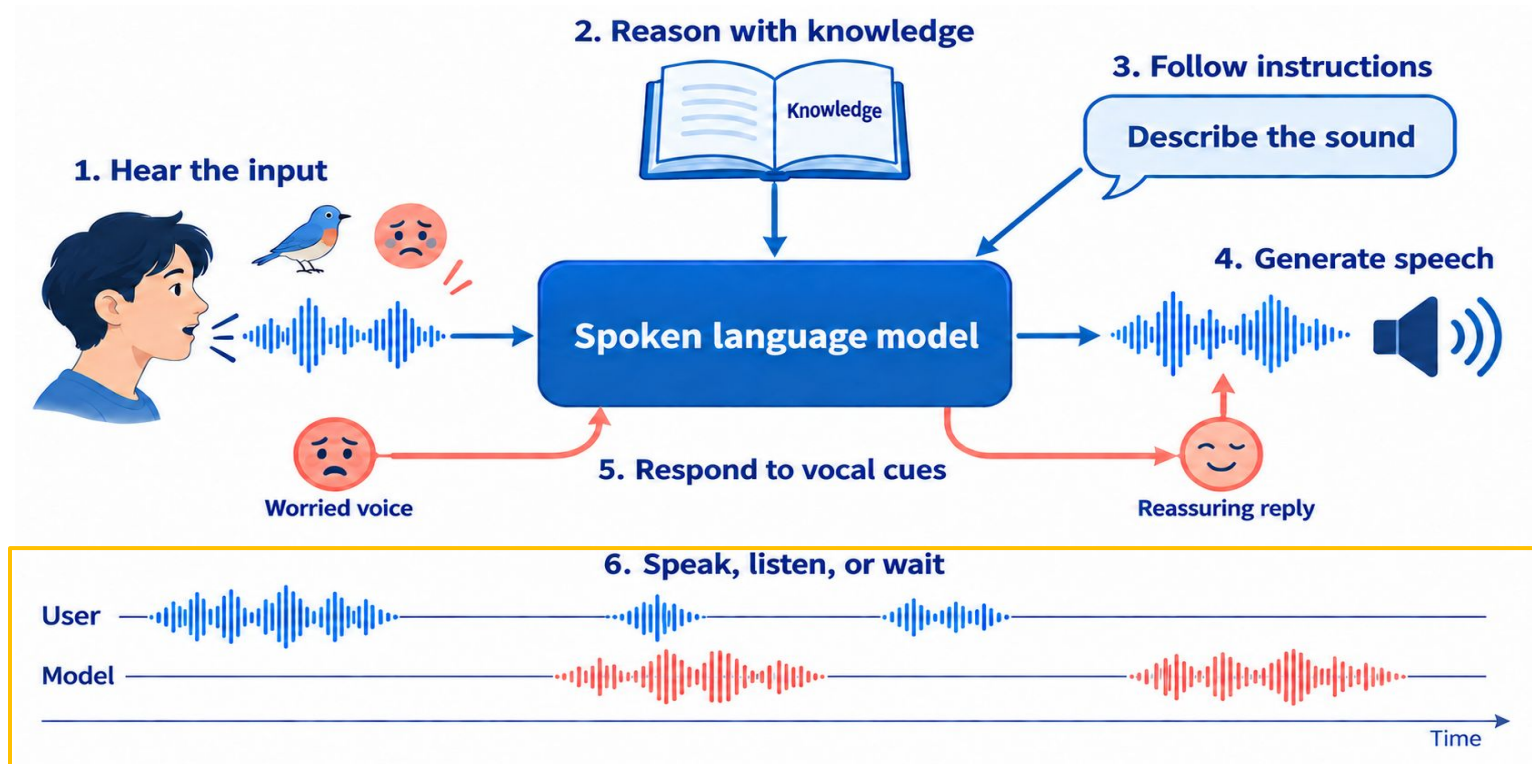


ParaS2SBench

Helpful words + appropriate voice

Full-duplex Interaction

speech-text model



The right words need the right moment

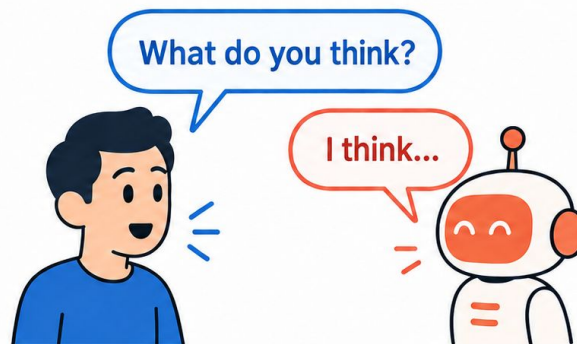
When to stop speaking, and when to begin

speech-text model

**When speaking,
know when to stop.**



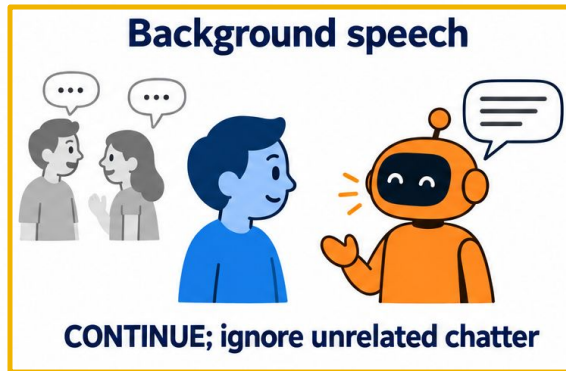
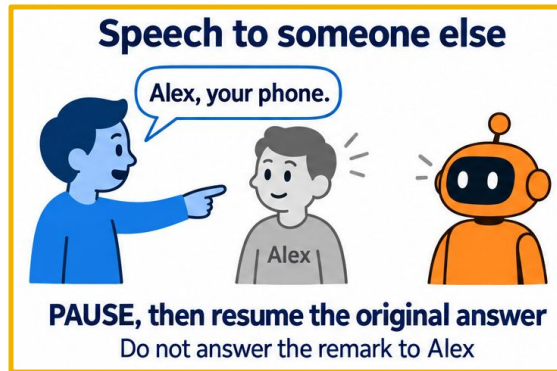
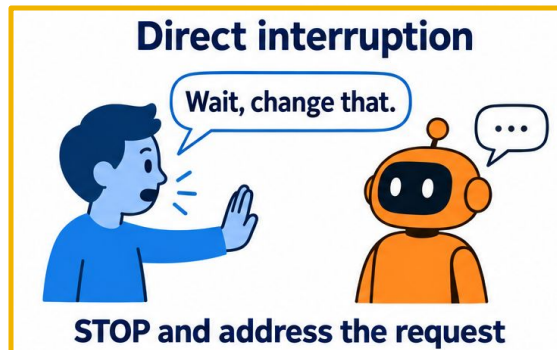
**When silent,
know when to start.**



While speaking: when should the model stop?

Prepared user audio · Full-Duplex-Bench v1.5

speech-text model



While listening: when should the model start?

A pause does not always end the user's turn · TurnBench

speech-text model

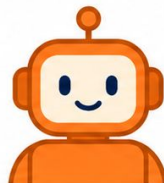
Unfinished request



I would like...

pause

.....



WAIT

The user has more to say

Completed question



Which train goes to Boston?

pause

.....



The next train...

START

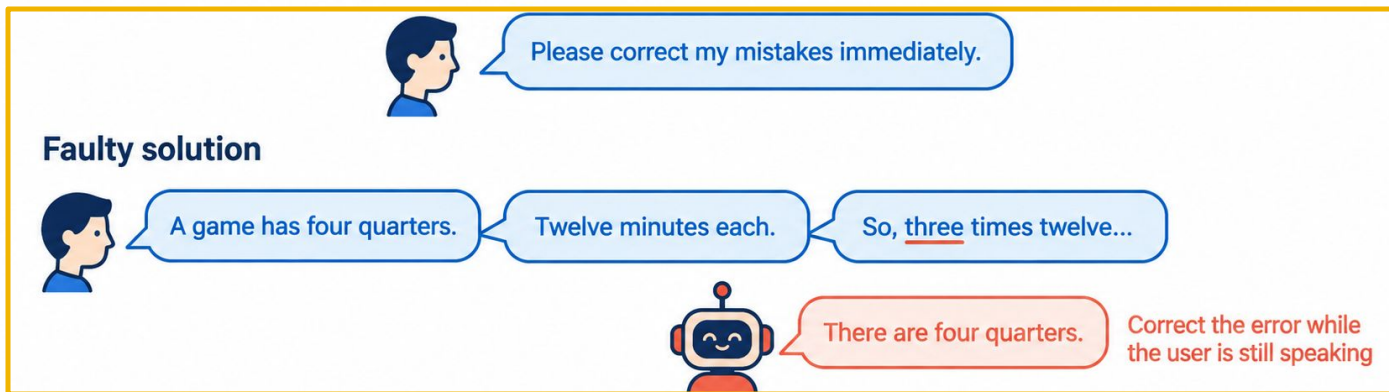
The user has handed over the turn

Use the words, intonation, and conversation so far

Starting before the user has finished

The other side is a spoken solution · Model Interruption Bench

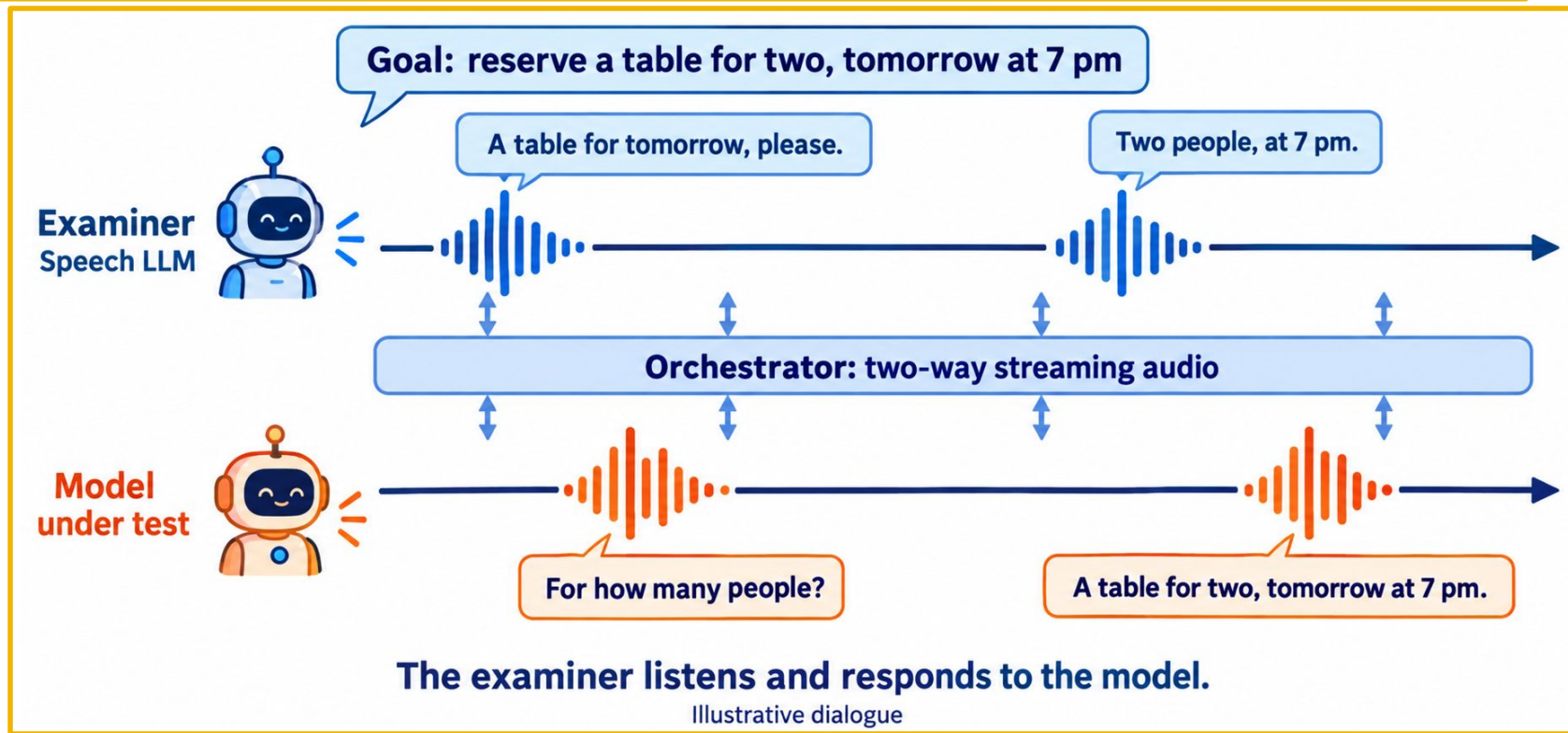
speech-text model



Knowing the error and correcting it in time are different tests.

When the other side can answer back

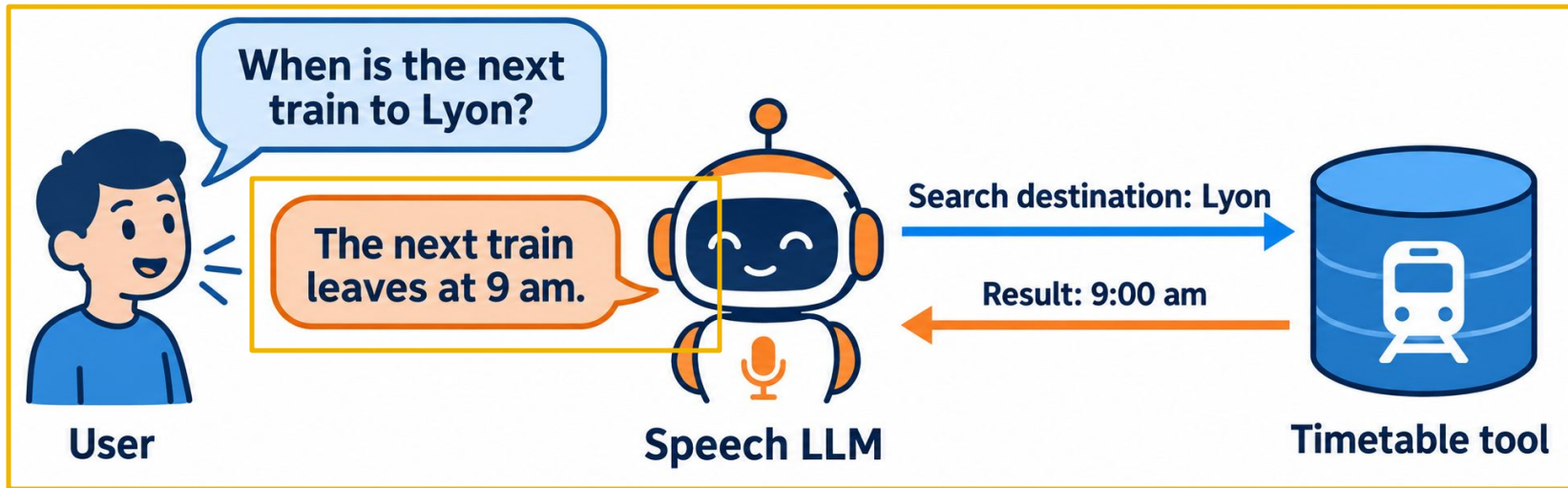
A speech LLM replaces fixed playback · Full-Duplex-Bench v2



When a tool joins the conversation

Retrieving information during a spoken conversation

speech-text model



The tool retrieves the timetable.
The assistant explains the result aloud.

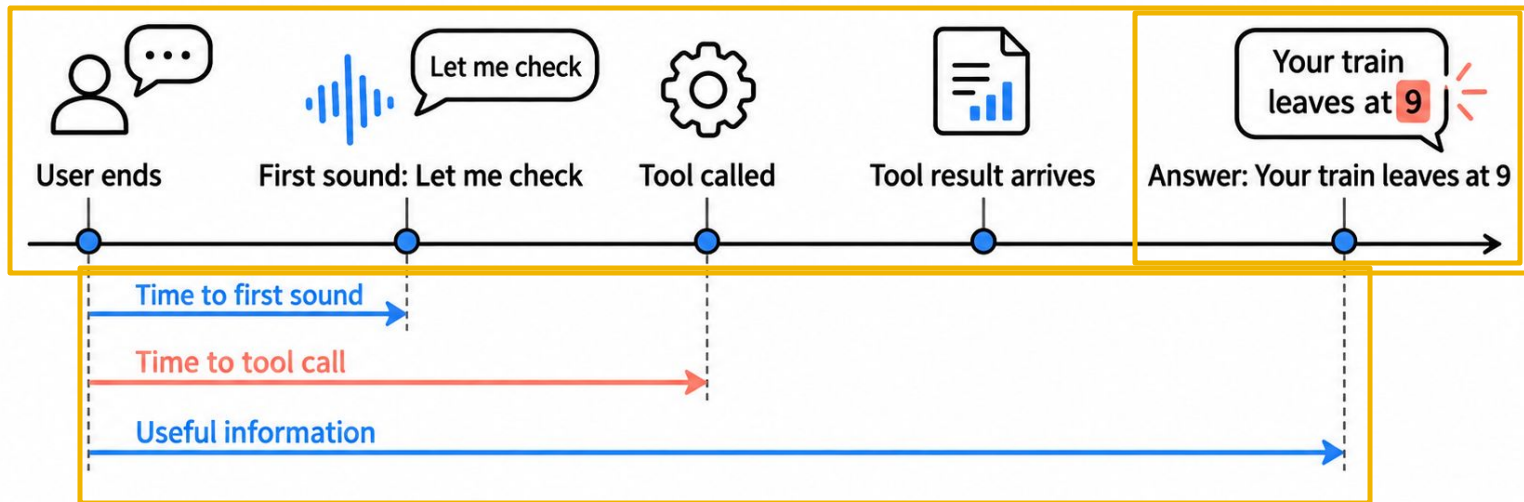
Stream RAG: search during input speech

MoshiRAG: retrieval during full-duplex dialogue

Returning useful information promptly

Three clocks, one starting point · Full-Duplex-Bench v3

speech-text model



Illustrative timeline, not measured values.

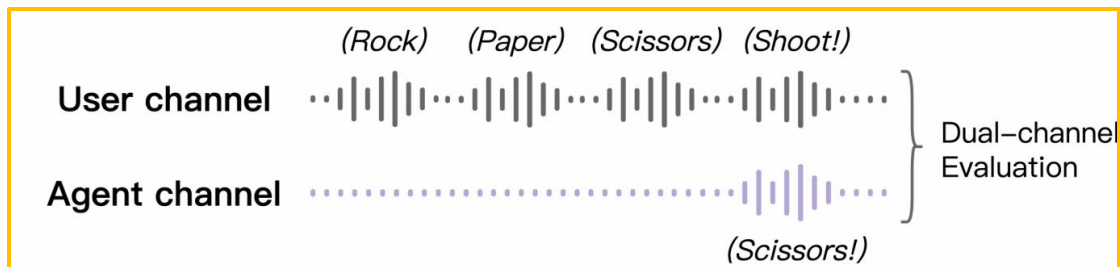
Following time and rhythm constraints

The other side is the clock · Game-Time

speech-text model

Same basic task, with an added temporal requirement

Game-Time Benchmark



Constraints

Examples



Time

Remain silent for 10 seconds before giving your response.



Tempo

Count from one to ten with the tempo <bpm=60> dah-dah-dah.



Simul.

Say your choice on “shoot.” *Rock, Paper, Scissors... Shoot!*

Participating in a group conversation

Multiple speakers and the agent · MP-Bench

speech-text model



Explicit invitation
→ answer

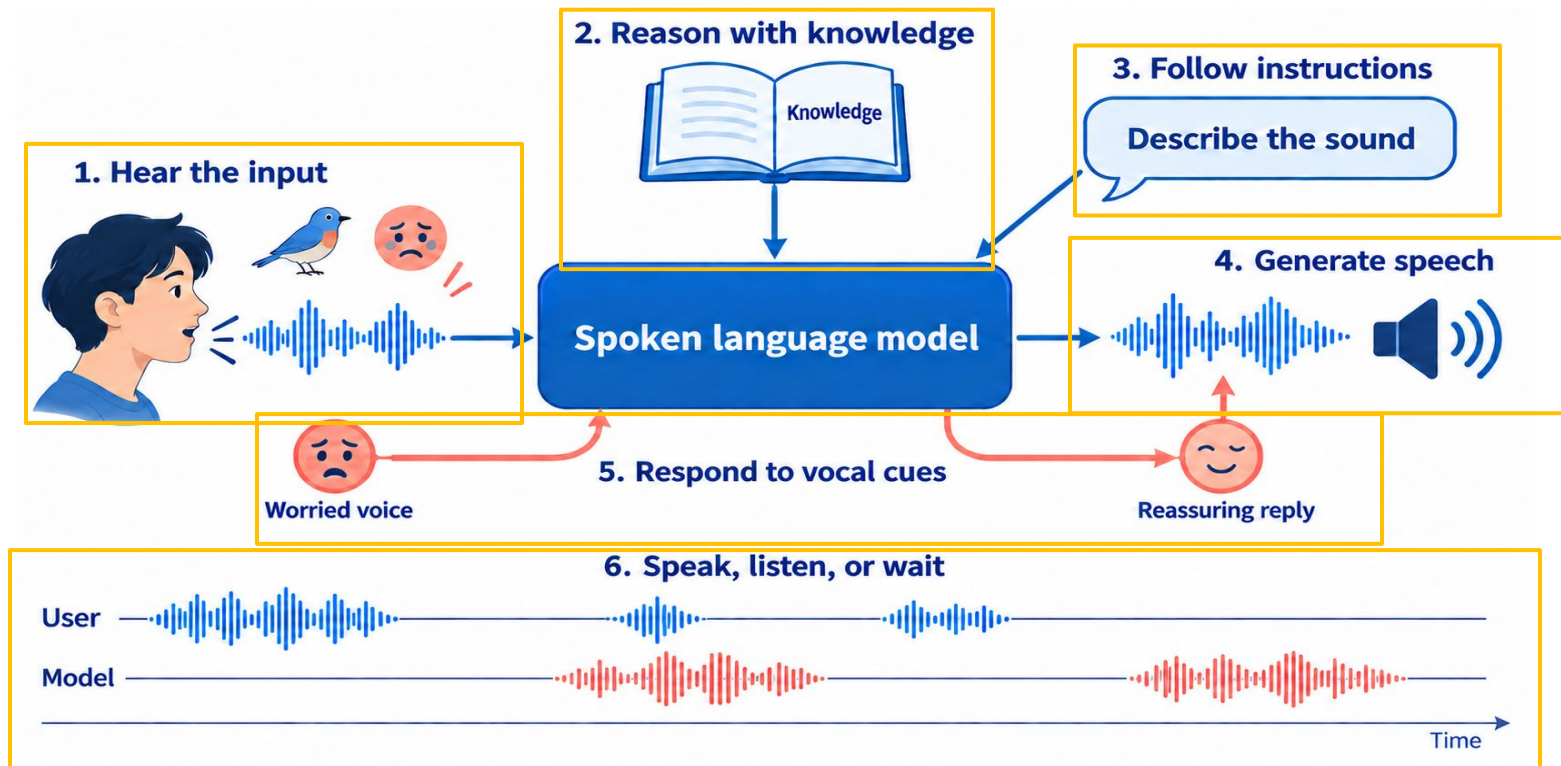
Turn given by a game rule
→ answer

Someone else's turn
→ stay silent

Best realtime comprehension: 32.6% Game turn-taking: near 50% chance

Spoken LLM evaluation: the complete picture

speech-text model





PART 08 · Chih-Kai Yang

Emerging Topics: From Development to Real-World Demands

In Previous Sessions...

- We've talked about the development process of SLMs
 - Model architecture
 - Training strategy
 - Evaluation

When We Start to Rely on Models...

More demands emerge...

Part 1

CAPABLE, AT SCALE

Does it hold up?

Long-context

Meetings, 20-min calls,...
Inputs keep getting longer.

Part 2

TRUSTWORTHY

Can we see inside?

Interpretability

We cannot accept a black box we
cannot inspect.

Part 3

MAINTAINABLE

**Can we control it after
training?**

Steering & Model Editing

Models err and go stale; re-training
every time is too costly.

When We Start to Rely on Models...

More demands emerge...

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Steering & Model Editing

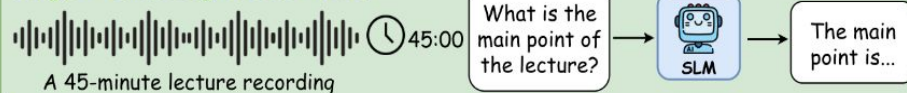
Models err and go stale; re-training
every time is too costly.

When Will the Context Become Long?

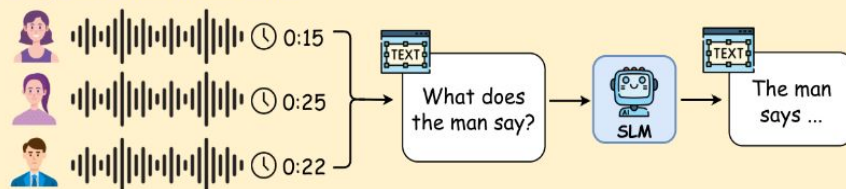
Typically, audio over 30s is considered “long.”

Single-Turn

Single-Turn Long-Form Audio



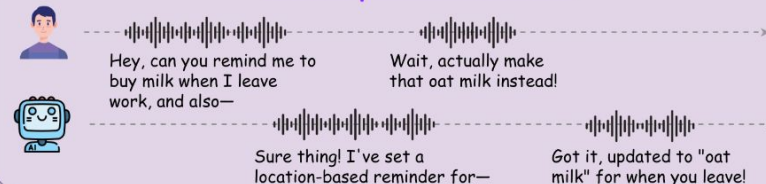
Single-Turn Multi-Audio



Multi-Turn

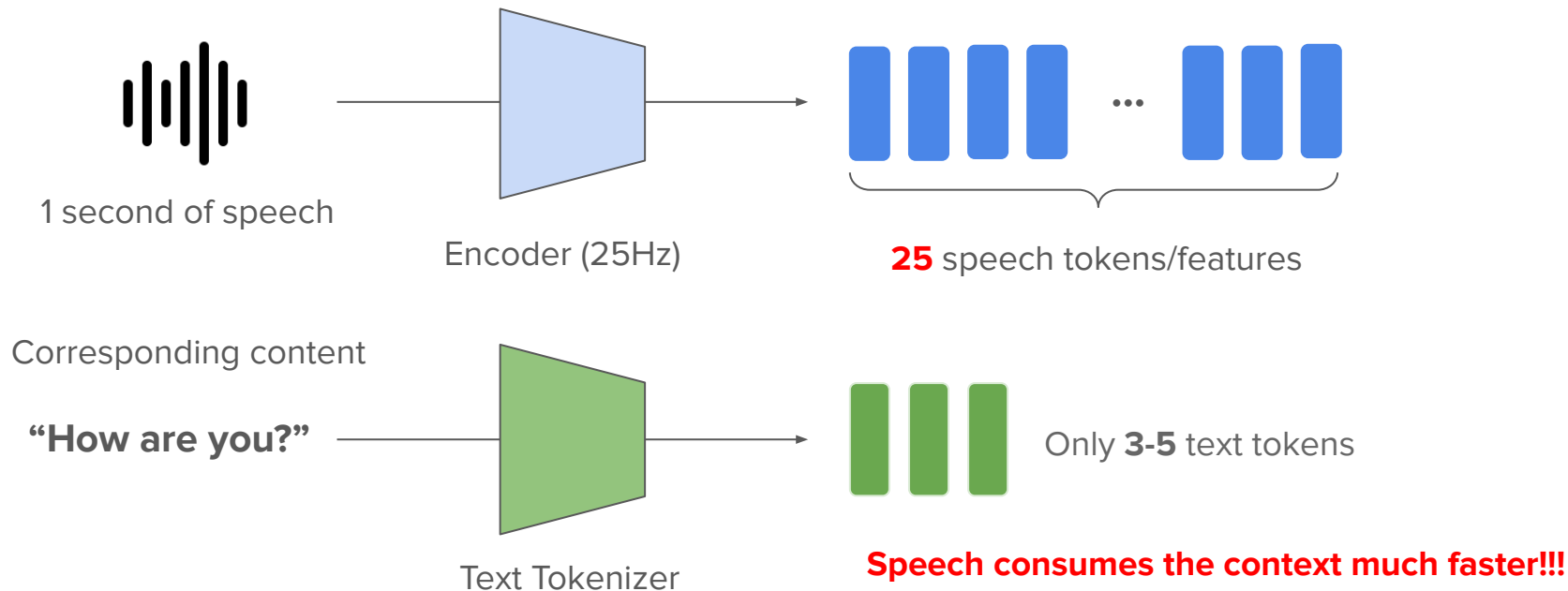


Full-Duplex Interaction



Token-rate Mismatch for Text/Audio

Audio inputs are too long...



Token-rate Mismatch for Text/Audio

Model	Modality	Backbone	Max Context	Token Rate	Max Audio Length	Audio Tokenizer
Pengi (Deshmukh et al., 2023)	$P(T A, T)$	GPT-2 Base (Radford et al., 2019)	1k	Fixed 40 tokens	7 s (training)	HTS-AT (Chen et al., 2022a)
SpeechGPT (Zhang et al., 2023)	$P(T A, T)$	LLaMA-13B (Touvron et al., 2023a)	2k	< 50Hz (deduplication)	-	mHuBERT units (Lee et al., 2022)
Qwen-Audio (Chu et al., 2023)	$P(T A, T)$	Qwen-7B (Bai et al., 2023)	8k	25 Hz	30 s	Whisper Encoder (Radford et al., 2023)
SALMONN (Tang et al., 2024)	$P(T A, T)$	Vicuna-13B (Chiang et al., 2023)	2k	2.93 Hz	180 s (training)	Whisper Encoder + BEATs (Chen et al., 2023a) → Q-Former
Audio Flamingo (Kong et al., 2024)	$P(T A, T)$	OPT-IML-MAX-1.3B (Iyer et al., 2023)	2k	- (cross-attention)	33.25 s (sliding)	CLAP (Elizalde et al., 2024)
WavLLM (Hu et al., 2024)	$P(T A, T)$	LLaMA-2-7B (Touvron et al., 2023b)	4k	12.5 Hz	30 s	Whisper Encoder + WavLM (Chen et al., 2022b)
Qwen2-Audio (Chu et al., 2024)	$P(T A, T)$	Qwen-7B	8k	25 Hz	30 s	Whisper Encoder
VITA (Fu et al., 2024)	$P(A, T A, T)$	Mixtral-8x7B (Jiang et al., 2024a)	32k	12.5 Hz	-	CNN-Transformer
Moshi (Défossez et al., 2024)	$P(A, T A, T)$	Helium-7B (Défossez et al., 2024)	4k	12.5 Hz	5 min	Mimi codec (Défossez et al., 2024)
SpiRit-LM Base (Nguyen et al., 2025)	$P(A, T A, T)$	LLaMA-2-7B	4k	25 Hz	-	mHuBERT units
SpiRit-LM Expressive (Nguyen et al., 2025)	$P(A, T A, T)$	LLaMA-2-7B	4k	25 + 12.5 + 1 Hz	-	mHuBERT units + F0 tokens + Style tokens
LLaMA-Omni (Fang et al., 2025)	$P(A, T A, T)$	LLaMA-3.1-8B (Grattafiori et al., 2024)	128k	10 Hz	30 s	Whisper Encoder
Audio Flamingo 2 (Ghosh et al., 2025b)	$P(T A, T)$	Qwen2.5-3B (Yang et al., 2025b)	32k	- (cross-attention)	5 min	AFCLAP (Ghosh et al., 2025b)
Kimi-Audio (KimiTeam et al., 2025)	$P(A, T A, T)$	Qwen2.5-7B	32k	12.5 Hz	-	Whisper VQ + Whisper Encoder
Step-Audio 2 mini (Wu et al., 2025)	$P(A, T A, T)$	Qwen2.5 7B	16k (ft) / 32k	12.5 Hz	-	Whisper Encoder (From Qwen2-Audio)
Audio Flamingo 3 (Ghosh et al., 2025a)	$P(T A, T)$	Qwen2.5-7B	128k (ft) / 32k	25 Hz	10 min	AF-Whisper (Ghosh et al., 2025a)
Qwen3-Omni (Xu et al., 2025b)	$P(A, T A, T)$	Qwen3-30B-A3B (Yang et al., 2025a)	32k (ft) / 256k	12.5 Hz	40 min	Audio Transformer (Xu et al., 2025b)
Audio Flamingo Next (Ghosh et al., 2026)	$P(A, T A, T)$	Qwen2.5-7B	32k	25 Hz	30 min	AF-Whisper v2 (Ghosh et al., 2026)

Token rates are typically high!

Length Is the Core Bottleneck

It strains the system, and it degrades what the model can do.

Efficiency

length strains the system



Long sequences

a few minutes of audio → tens of thousands of tokens



$N \times N$ Quadratic self-attention

compute & memory grow with N^2

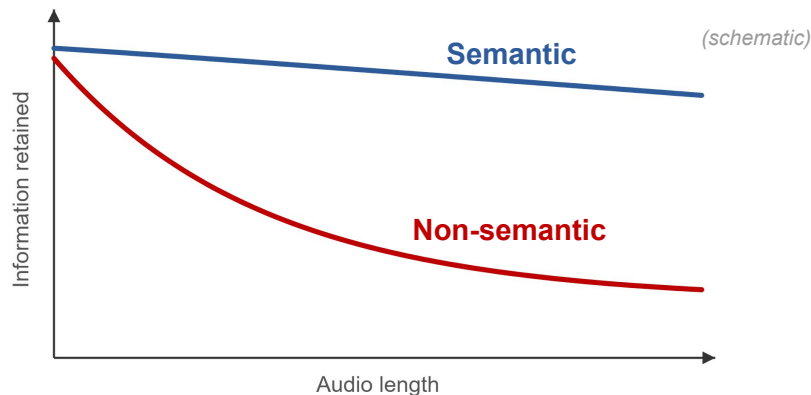


Fixed context window

long audio simply overflows it

Capability

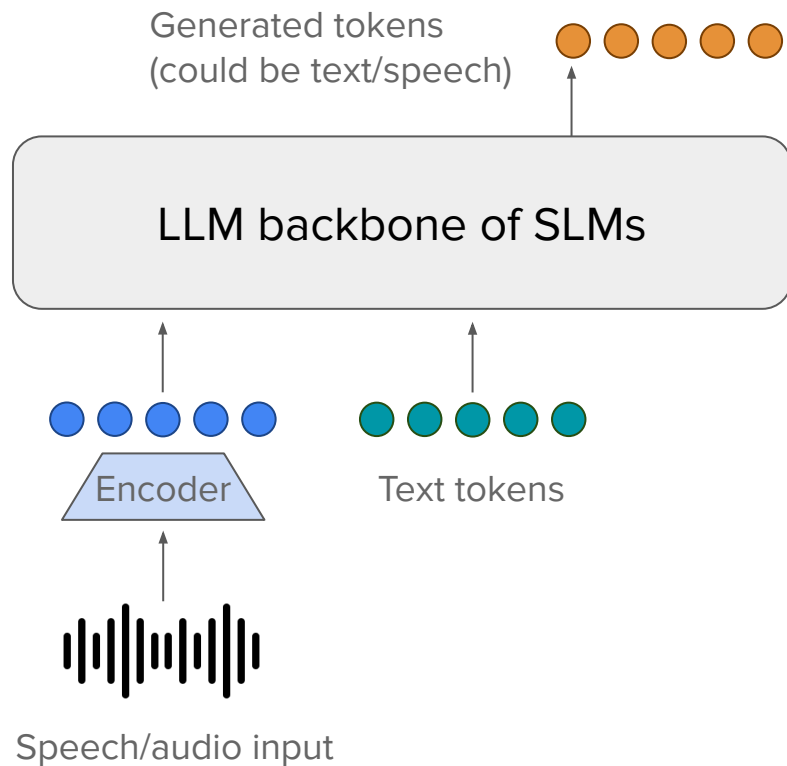
length degrades quality unevenly



Non-semantic cues (emotion, prosody, ambient sound) are retained far worse than content.

Potential Solutions

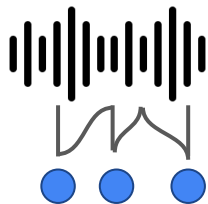
Mitigate stage by stage.



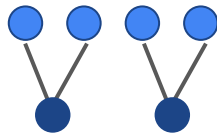
Potential Solutions

Input Processing - Reducing input length itself!

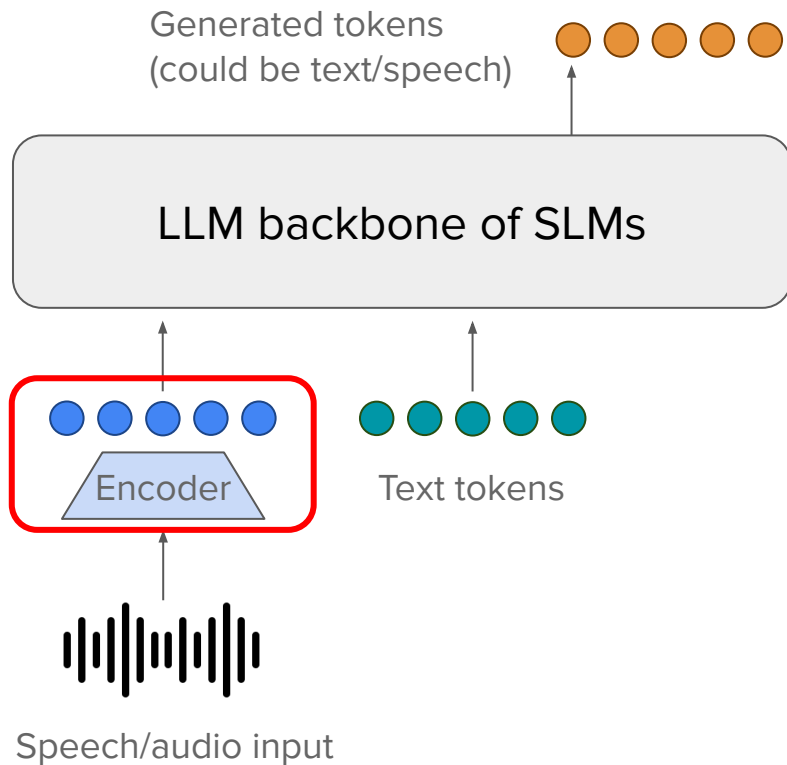
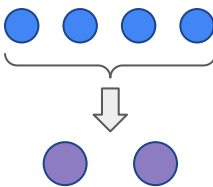
Low/variable-rate tokenization



Downsampling & merging



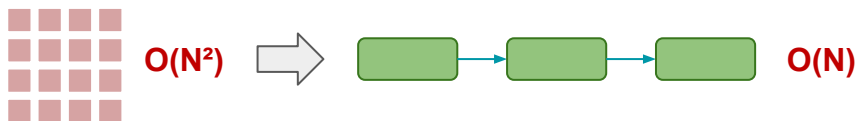
Learnable abstraction via queries



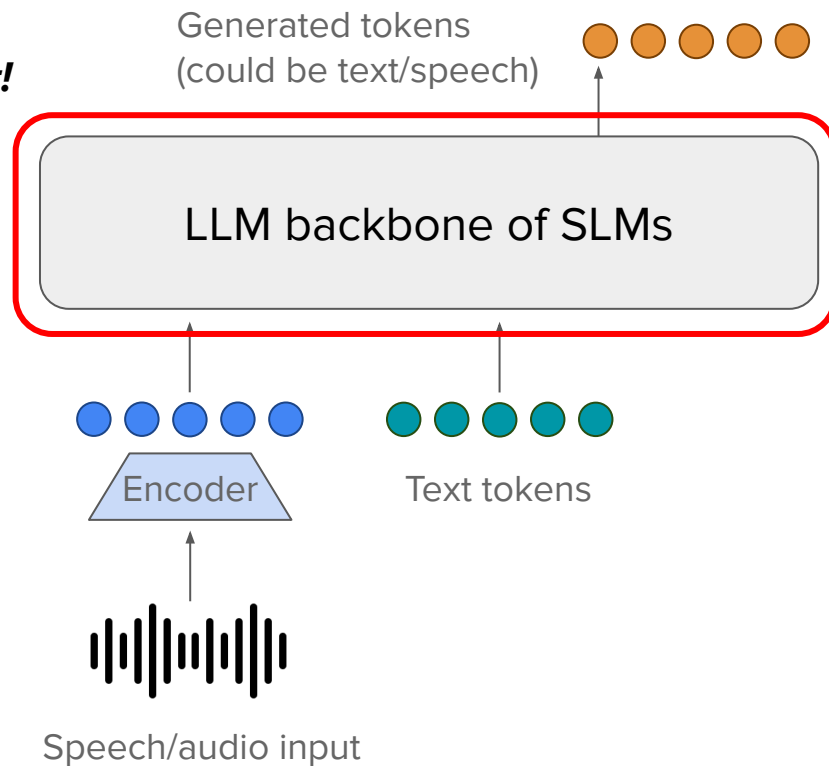
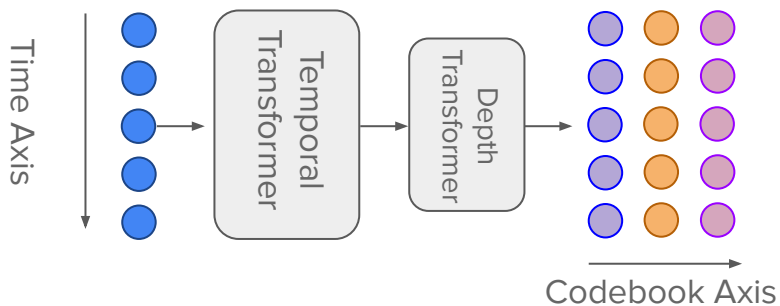
Potential Solutions

Model Architecture - ***Make backbone more efficient!***

State-space models

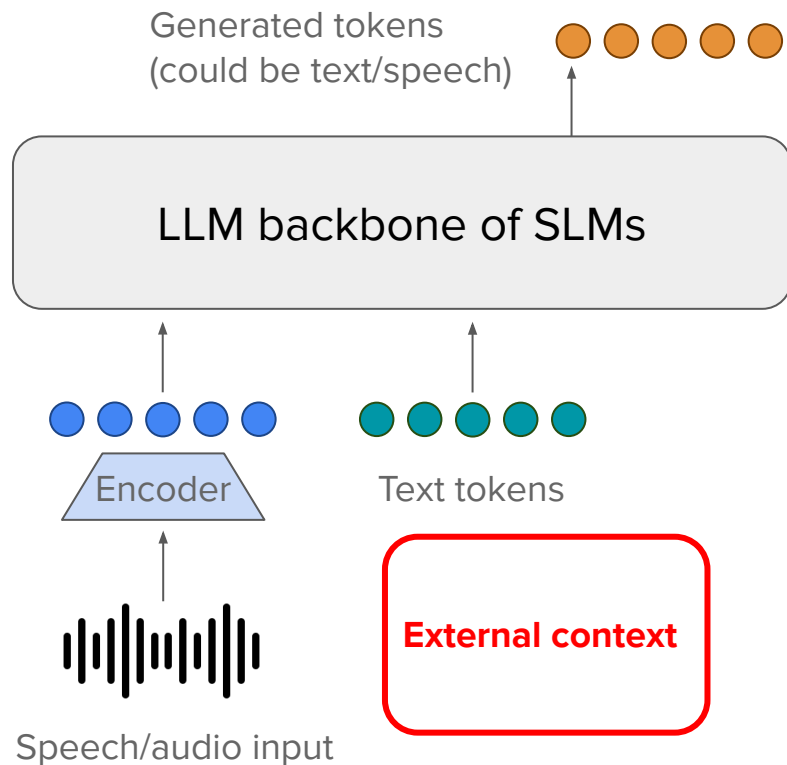
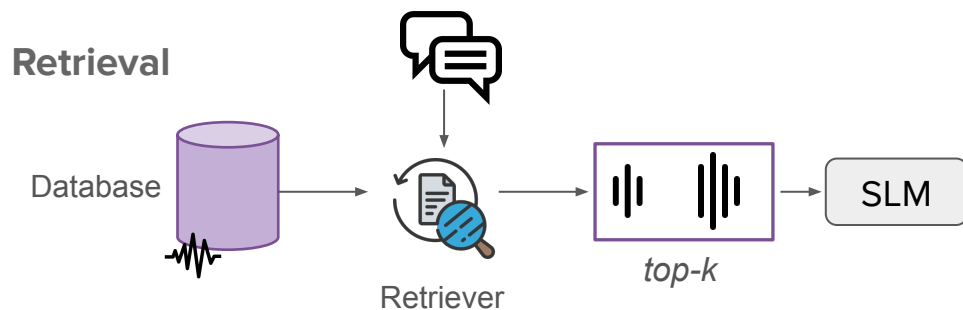


Hierarchical token factorization



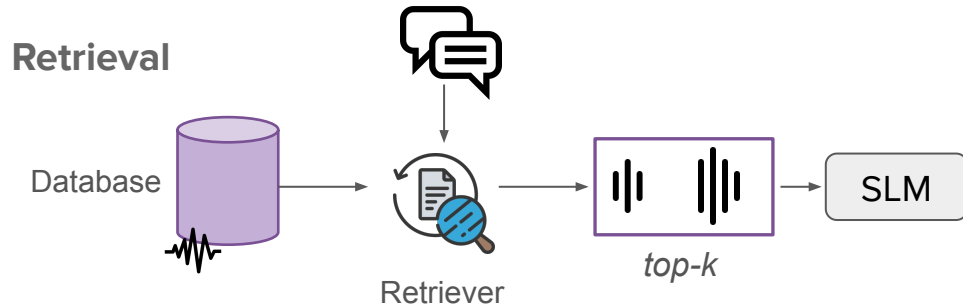
Potential Solutions

Beyond the Context Window - Preserve information outside active context

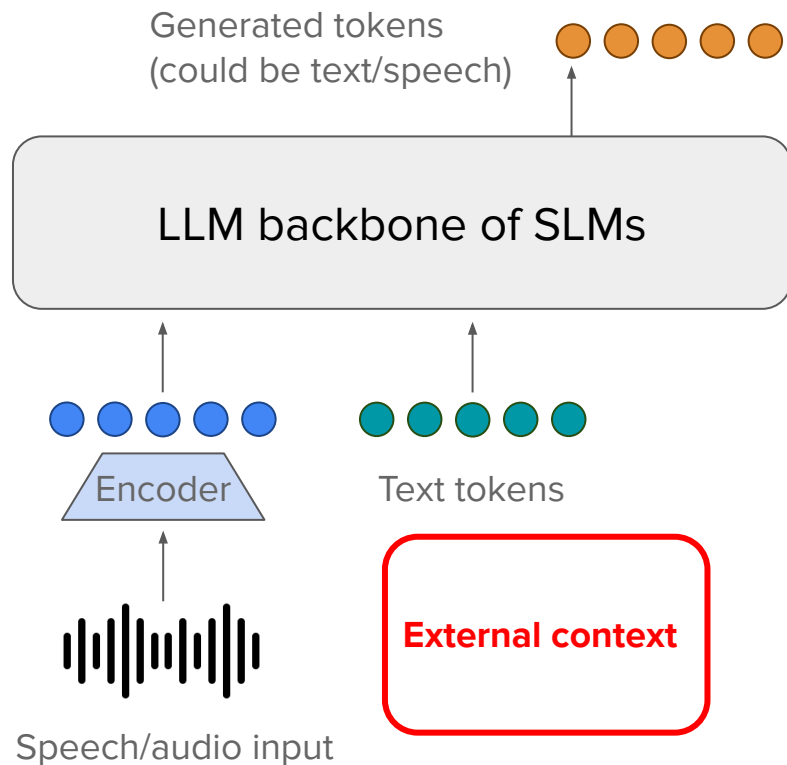


Potential Solutions

Beyond the Context Window - Preserve information outside active context



Parametric Memory



Potential Solutions

But they are not fully validated at scale...

Validated at long context?

1 Input Processing

tokenizers untested

Low / variable-rate tokenizers typically evaluated on short clips

2 Model Architecture

tested at length

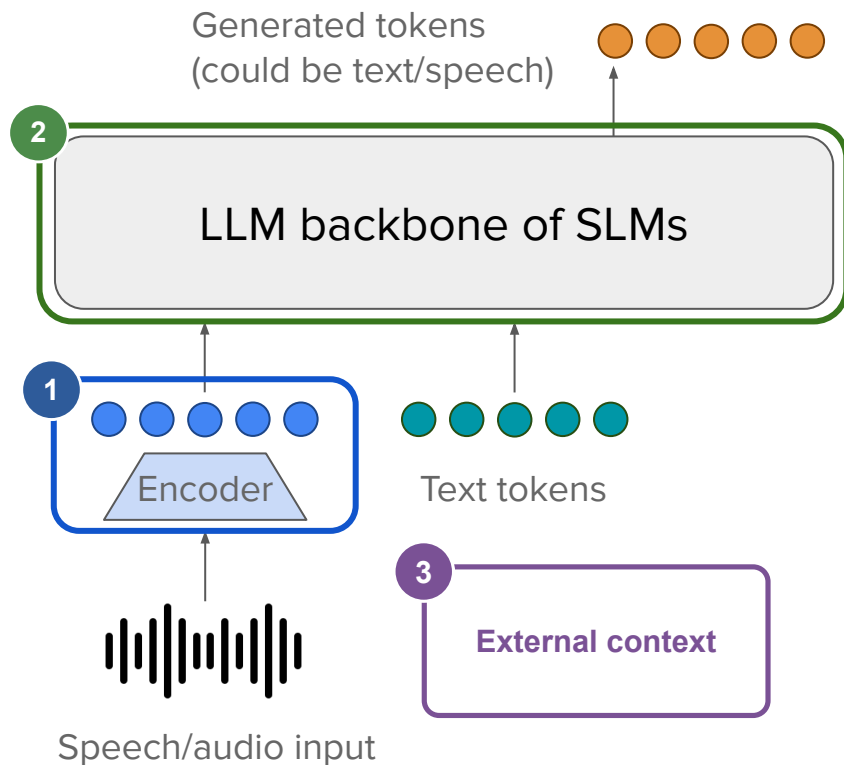
SpeechSSM (16 min) and Moshi (5 min) shown at their target scale

3 Beyond the Window

partly tested

Long-form RAG & parametric memory reach hours
Streaming RAG tested on short queries only

Key next step: extending these to long context.



Short Summary for Part 1

- Length is the core bottleneck: efficiency and capability both suffer
- Non-semantic information degrades fastest
- Potential solutions span the pipeline: input, architecture, beyond the window
- Validation at long context is still limited

When We Start to Rely on Models...

More demands emerge...

Part 1

CAPABLE, AT SCALE

Does it hold up?

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Meetings, 20-min calls,...
Inputs keep getting longer.

Part 2

TRUSTWORTHY

Can we see inside?

Interpretability

We cannot accept a black box we
cannot inspect.

Part 3

MAINTAINABLE

Can we control it after
training?

Steering & Model Editing

Models err and go stale; re-training
every time is too costly.

Why Look Inside the Model?

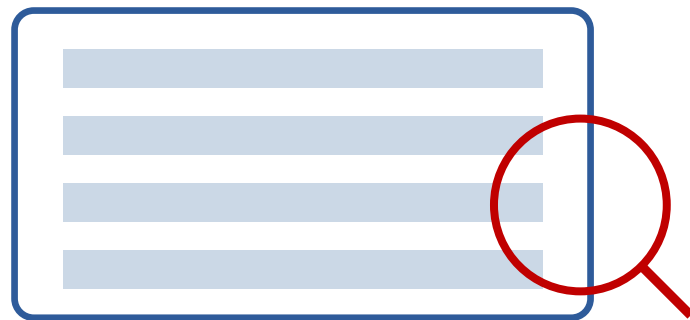
Benchmarks tell us whether it works; interpretability asks how.

What we usually see



Inputs and outputs only, not the internal process

What interpretability adds



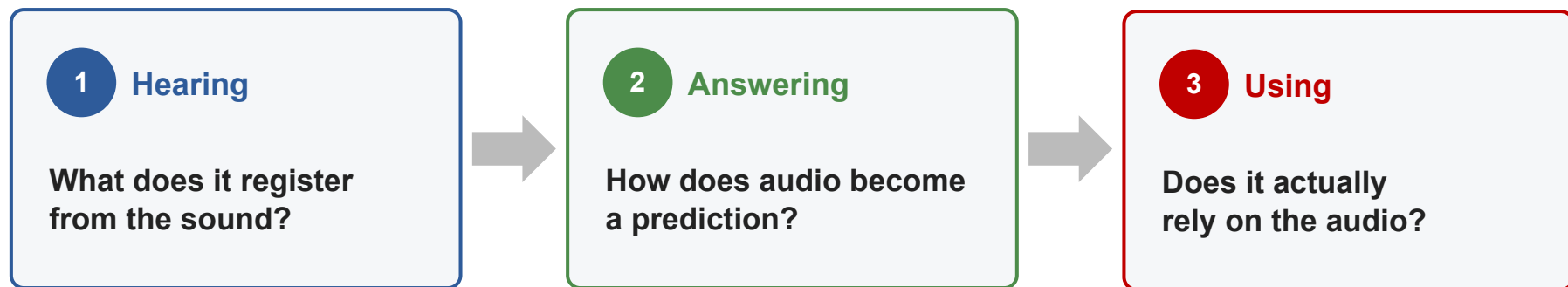
Trace what happens inside

We look inside to **diagnose failures, build trust, and find levers to improve.**

Many Angles, But One Focus in This Talk

Many analysis aspects: bias, safety, hallucination, robustness, ...

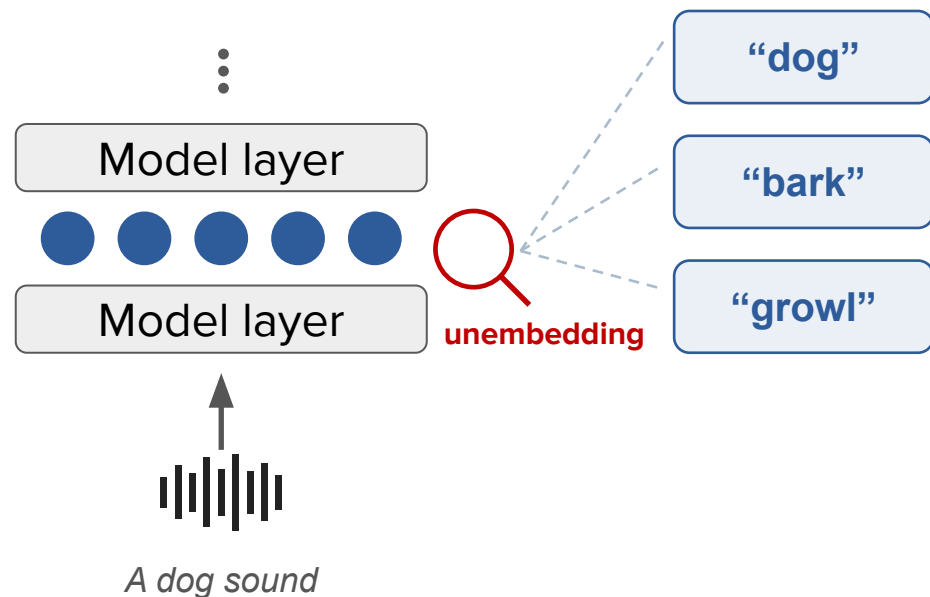
We focus on the **audio modality: what happens inside as the model hears and answers.**



1 Hearing: What Does the Model Register?

HOW WE LOOK

Read the model's hidden states right at the audio positions: which words do they decode to?



WHAT THEY FOUND

- Decoded tokens name the sound itself (e.g., the animal, the event): the model internally puts sound into word description.
- More broadly, audio lands in a semantic “hub”, a representation space shared with text modality

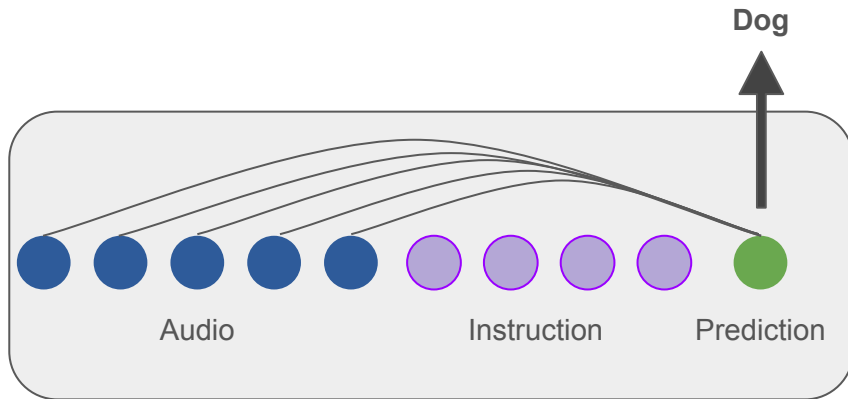
2

Answering: How Does Audio Become a Prediction?

HOW WE LOOK

Read the hidden state at the answer position: when does the attribute become clear?

Block the attention: how information flow?



WHAT THEY FOUND

- The attribute sharpens with depth
 - Earlier resolution predicts higher accuracy.
 - Information can be lost across layers
- Task-dependent information flow
 - Simple recognition: directly retrieve information from audio positions.
 - Complex QA: route audio information through instruction tokens.

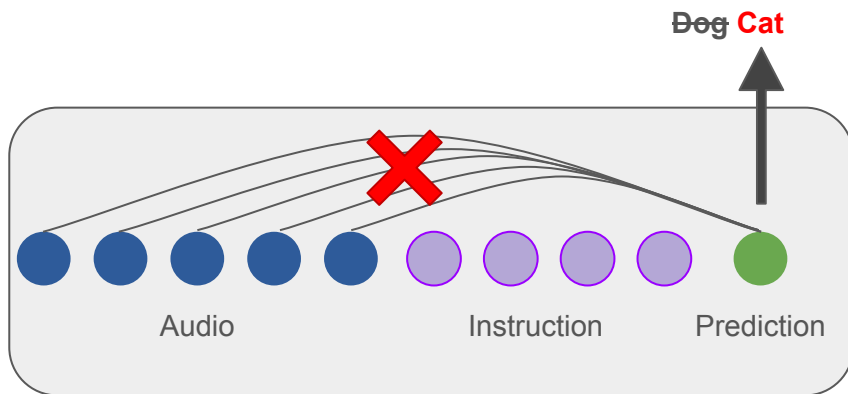
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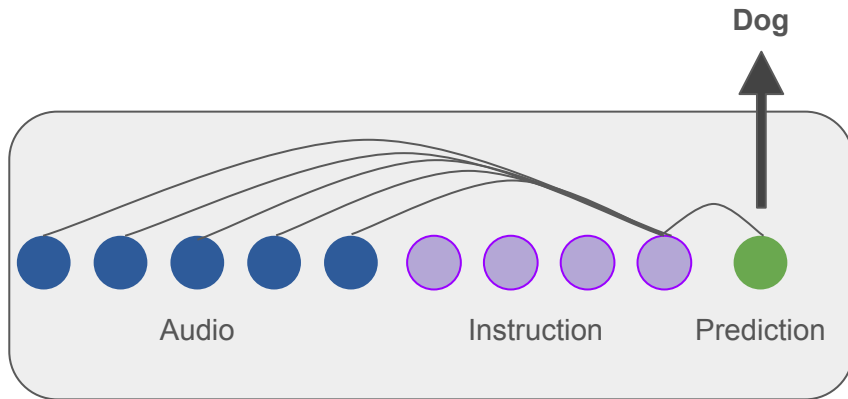
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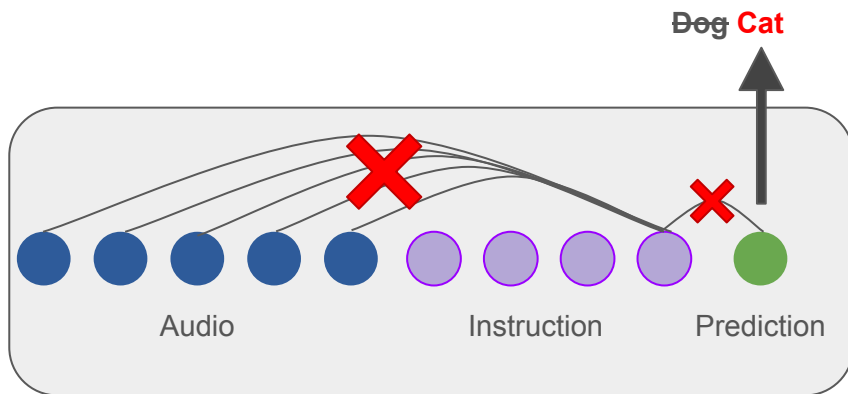
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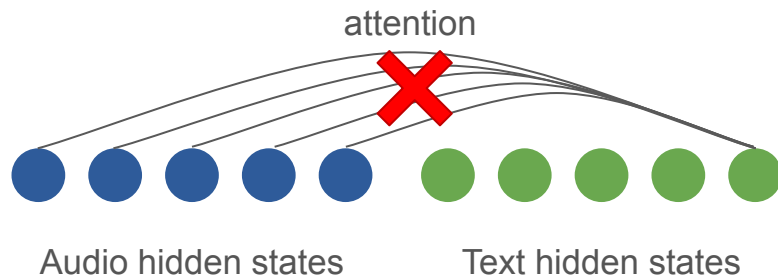
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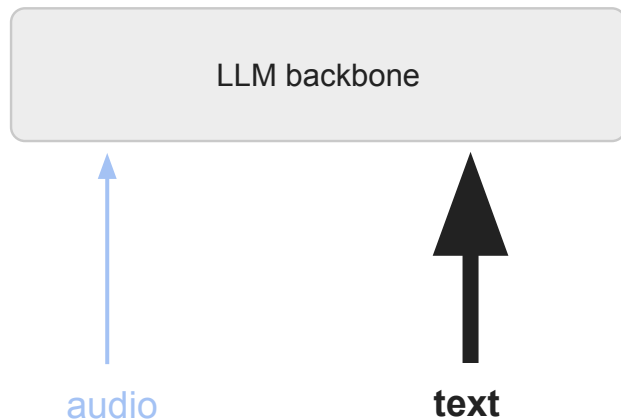
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 - Information can be lost across layers
- Task-dependent information flow
 - Simple recognition: directly retrieve information from audio positions.
 - Complex QA: route audio information through instruction tokens.

3 Using: Does the Model Rely on the Audio?

HOW WE LOOK

Measure how much attention goes to audio vs text



attention to audio stays low

WHAT THEY FOUND

- Text modality dominates the attention.
- On direct audio-vs-text conflict, text often wins.
- Even when audio is registered inside, deeper layers can suppress it.

Short Summary for Part 2

- Interpretability opens the black box: we can trace how an SLM hears, forms a prediction, and whether it relies on the audio
- These internal views expose failure modes that benchmarks alone cannot, e.g. under-used audio, modality conflict, information lost across layers
- The same views point to where to intervene for improvement, e.g. specific layers, heads, or neurons (not covered here)
- Still early: much of the internal picture of SLMs remains open

When We Start to Rely on Models...

More demands emerge...

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CAPABLE, AT SCALE

Does it hold up?

Long-context

Meetings, 20-min calls,...
Inputs keep getting longer.

Part 2

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Can we see inside?

Interpretability

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cannot inspect.

Part 3

MAINTAINABLE

**Can we control it after
training?**

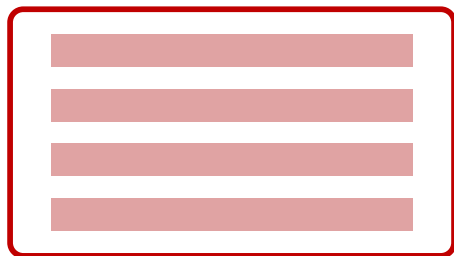
Steering & Model Editing

Models err and go stale; re-training
every time is too costly.

Why Control a Model Without Retraining?

A deployed SLM still needs fixes: wrong outputs, stale knowledge, unwanted behavior.

Full retraining / fine-tuning



redo the whole model

- Heavy compute and data
- Slow to iterate
- Risks catastrophic forgetting

Targeted control



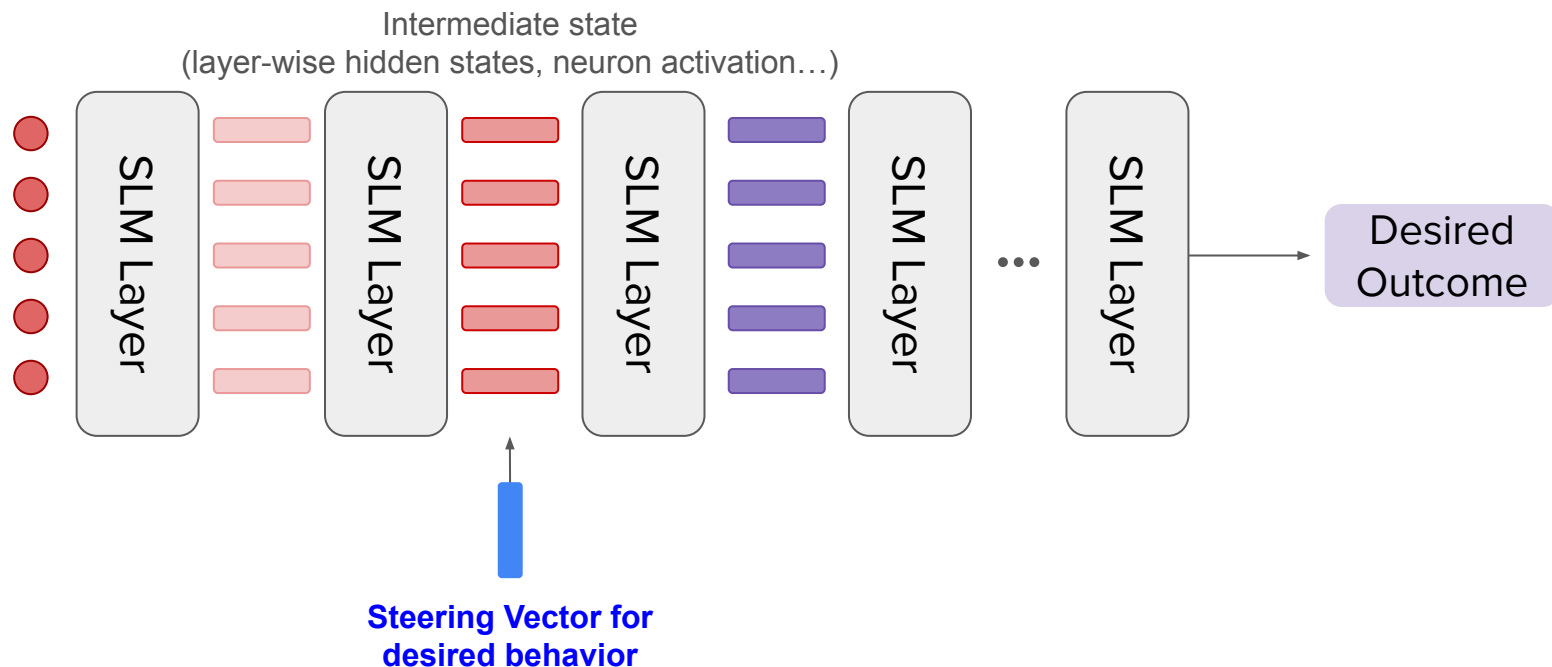
touch one part

- Cheap, fast to apply
- Changes one behavior, keeps the rest

The goal: update model behavior cheaply and precisely, without full retraining.

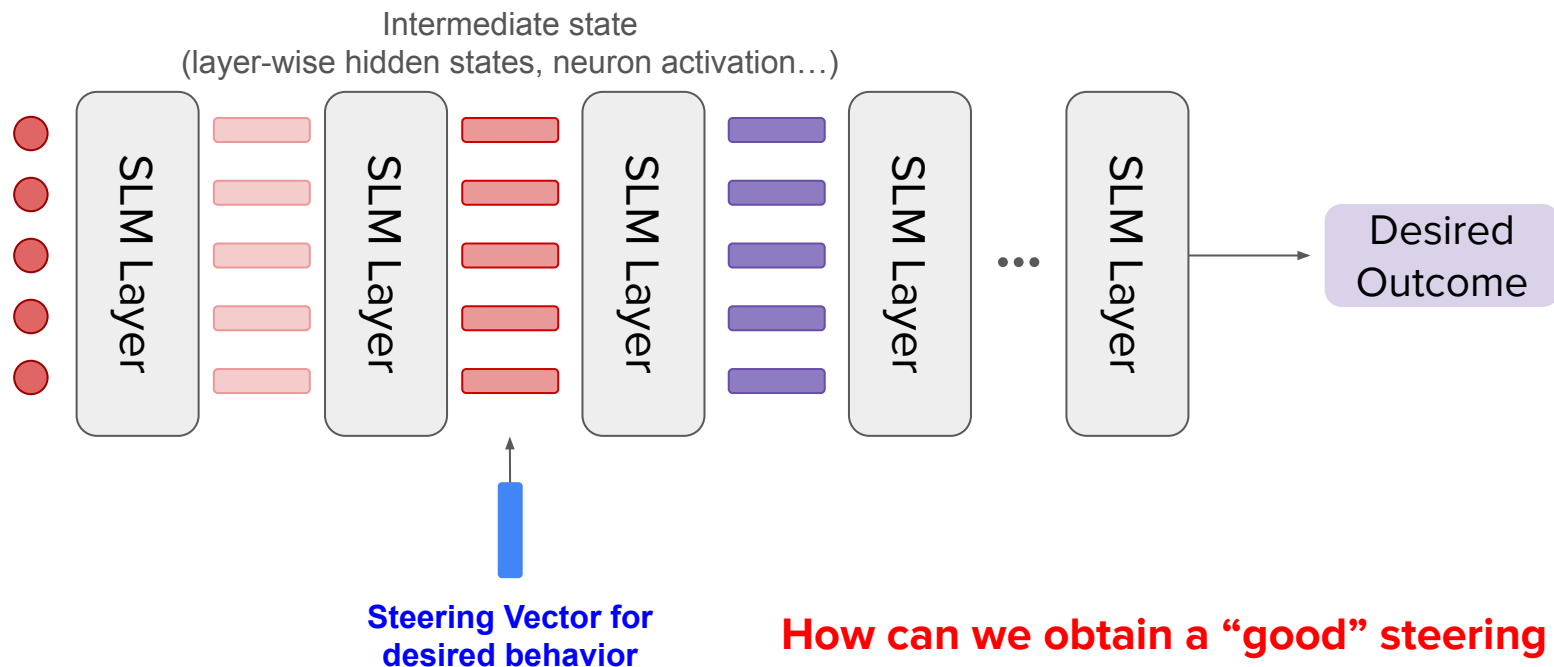
Model Steering - Overview

Temporary control, typically via “steering vector”, without changing model parameters.



Model Steering - Overview

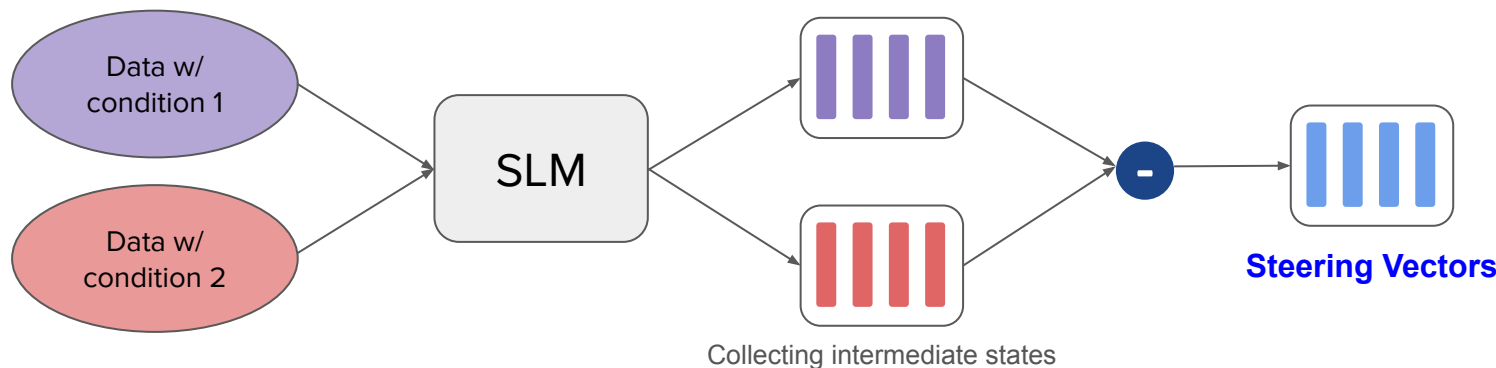
Temporary control, typically via “steering vector”, without changing model parameters.



How can we obtain a “good” steering vector?

Model Steering - Obtaining Steering Vector

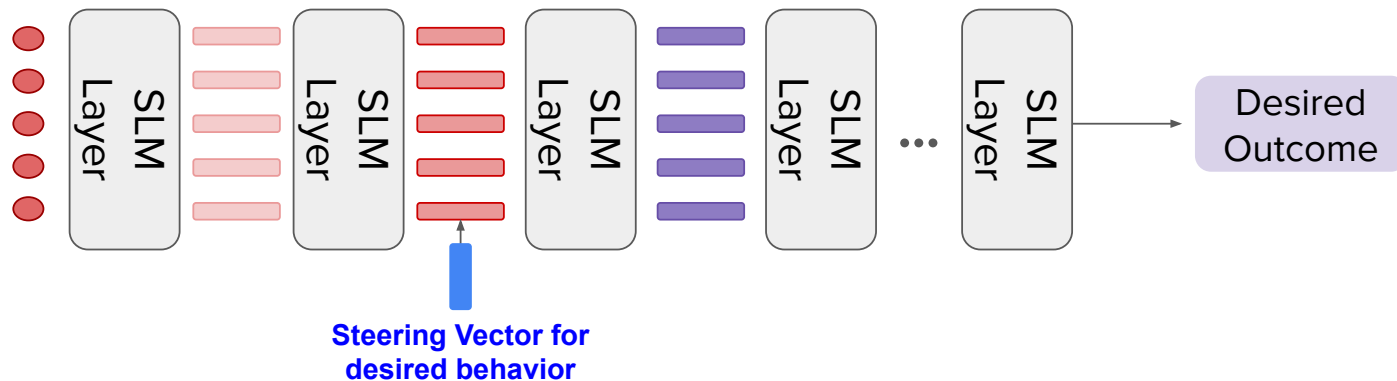
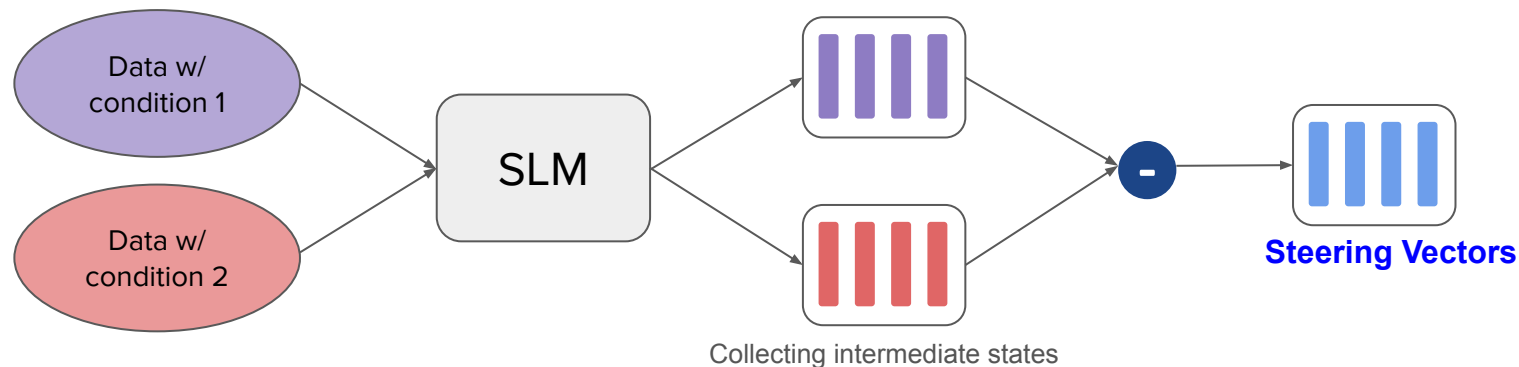
Usually obtained by contrasting intermediate states under different (data) conditions.



- Core premise:**
1. Intermediate states encode condition-specific information.
 2. Contrasting states across conditions yields steering vectors that encode the condition shift.
 3. Injecting these steering vectors shifts model behavior toward the target condition.

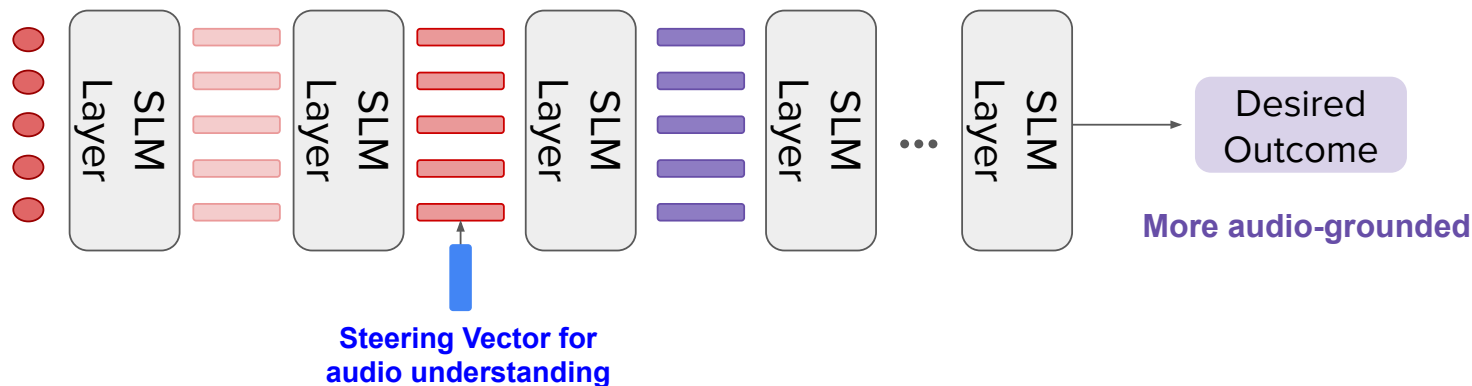
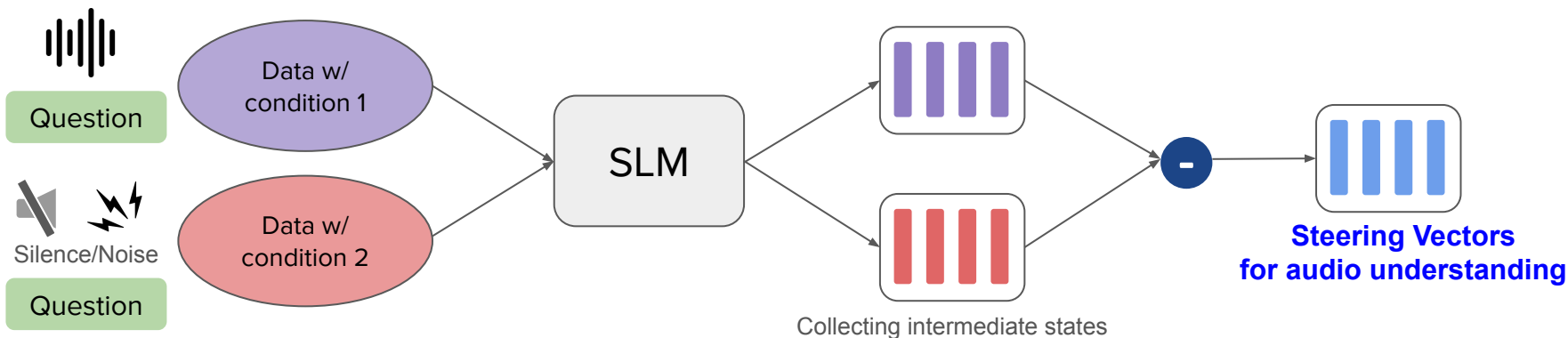
Model Steering - Applications

Depends on how we construct such contrast.



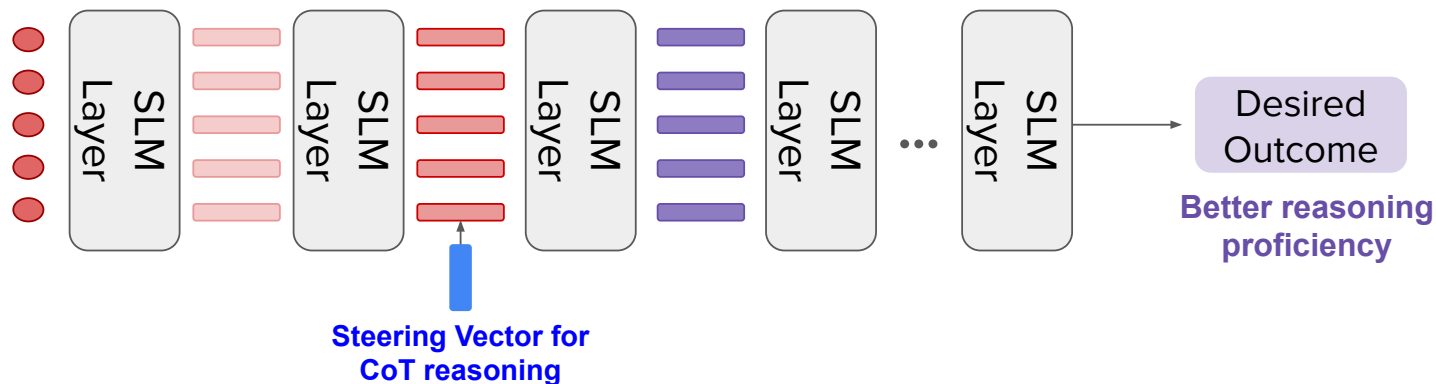
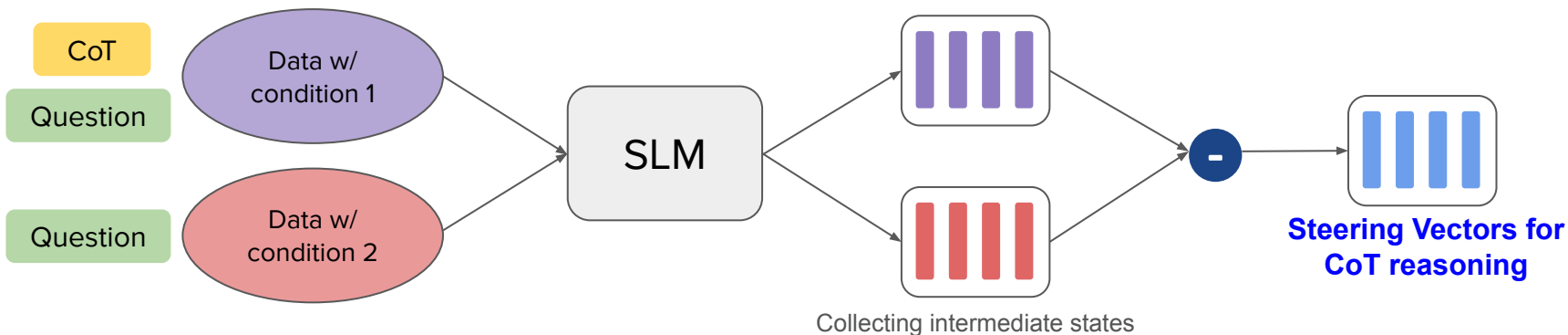
Model Steering - Applications

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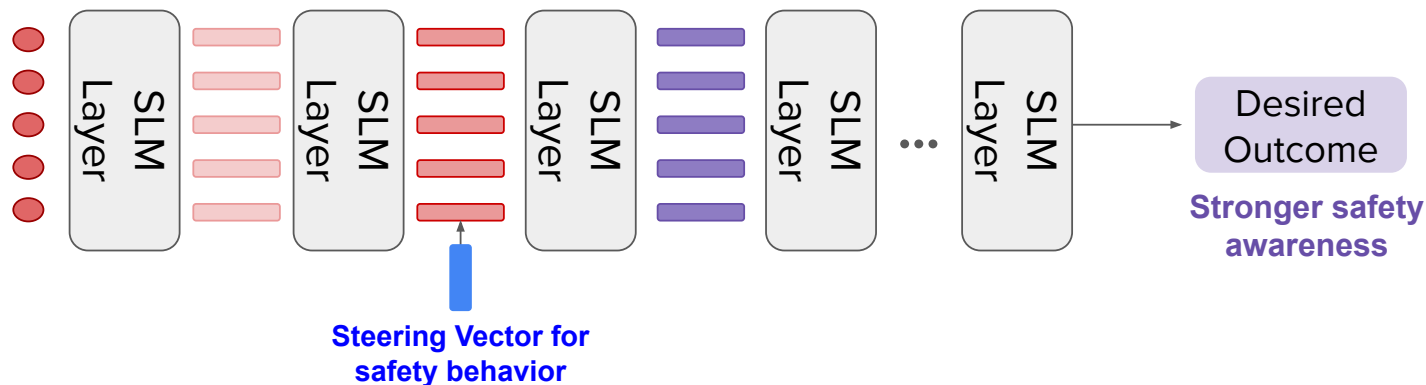
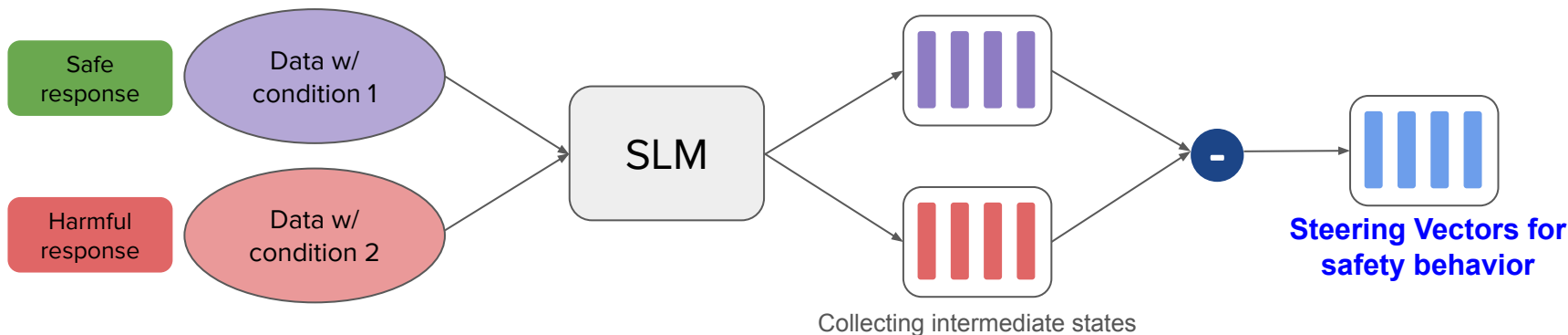
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Model Steering - Applications

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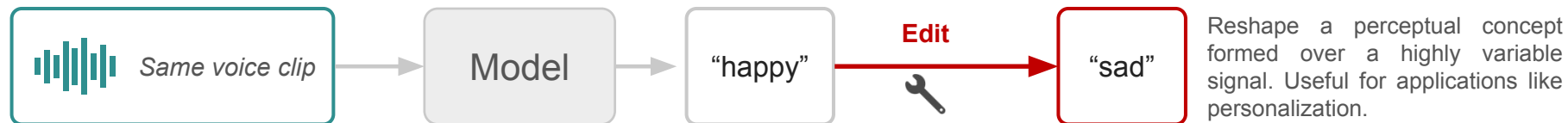
Model Editing - Overview

A small, **lasting** rewrite of the model parameters that changes its stored knowledge.

FACTUAL (In LLM Study)



PERCEPTUAL (When it comes to audio modality)





Model Editing - Evaluation

What Counts as a Successful Edit? Four fronts that pull against each other.

- Did the edit take effect on the exact target?
- Does it carry to equivalent cases?
- Is unrelated knowledge left untouched?
- Does it propagate to connected reasoning?



Model Editing - Evaluation

What Counts as a Successful Edit? Four fronts that pull against each other.

1

Reliability

Did the edit take effect
on the exact target?

e.g. happy → sad, on this clip

2

Generality

Does it carry to
equivalent cases?

*e.g. another sad-voiced clip, a
reworded question*

3

Locality

Is unrelated knowledge
left untouched?

*e.g. other emotions, other
sounds, text ability*

4

Portability

Does it propagate to
connected reasoning?

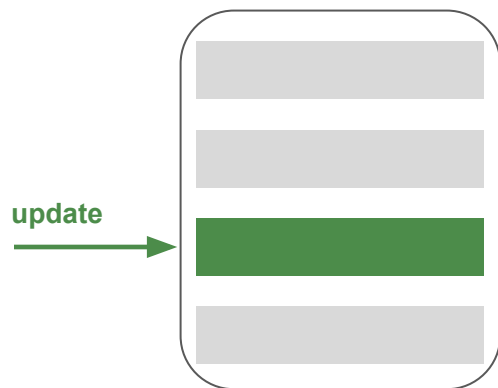
*e.g. frog → dog, so "barks"
updates too*

A method can ace one and fail another; the interesting question is the balance.

Model Editing - How Do We Usually Edit a Model?

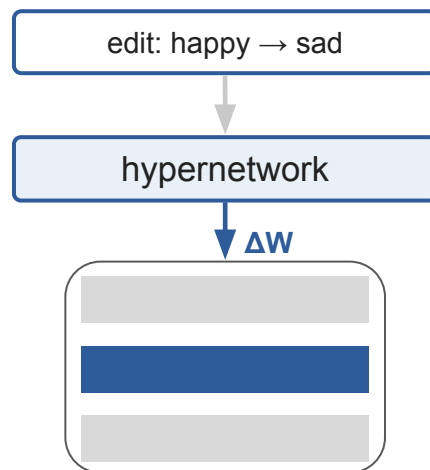
Some common paradigms. All touch only a few parameters. Differ in how the change is made.

Fine-tuning (naive)



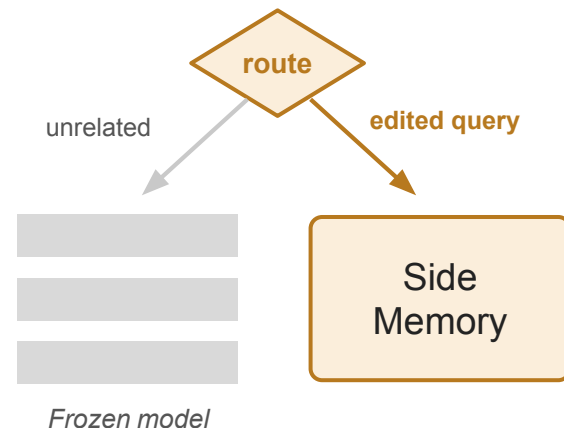
Directly fine-tune a small, chosen set of weights.

Meta-learning



A network predicts the weight change to apply.

Side memory



Store edits in extra memory; route to it only when needed.

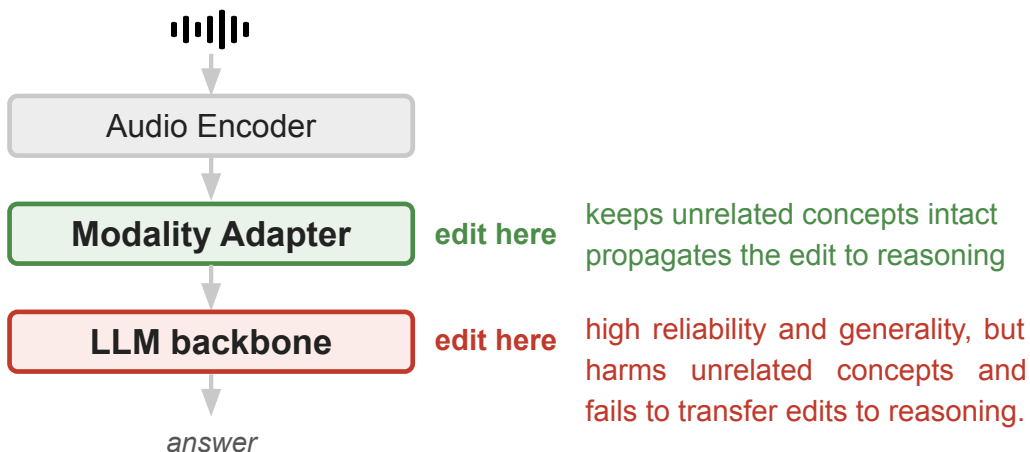
Editing Perceptual Concepts of SLMs is Challenging...

Existing methods for factual knowledge fail. *Perceptual audio concepts need their own approach.*

Success is only skin-deep

The edit holds (**Reliability** ✓, **Generality** ✓), but unrelated concepts break (**Locality** ✗) and reasoning doesn't follow (**Portability** ✗).

Where should be edited? *Early evidence points to the audio interface, not the LLM backbone.*



Key finding:
Don't rewire the brain.
Adjust how sound reaches it!

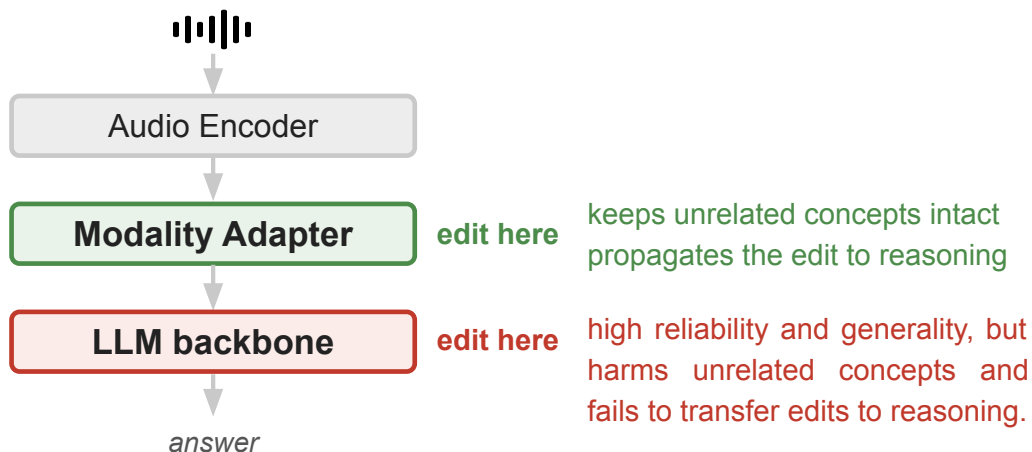
Editing Perceptual Concepts of SLMs is Challenging...

Existing methods for factual knowledge fail. *Perceptual audio concepts need their own approach.*

Success is only skin-deep

The edit holds ✓, but unrelated concepts break ✗ and reasoning doesn't follow ✗.

Where should be edited? *Early evidence points to the audio interface, not the LLM backbone.*



Key finding:
Don't rewire the brain.
Adjust how sound reaches it!

Short Summary for Part 3

- Deployed SLMs need cheap, targeted control
- Two levers: steering for temporary shifts, editing for persistent change
- Controlling SLMs is still an open frontier

Wrap-up

CAPABLE AT SCALE

Long-context

TRUSTWORTHY

Interpretability

MAINTAINABLE

Steering & Editing



**A capable model is where the work begins,
not where it ends.**

As we ask more of SLMs, these and many other demands only grow.

PART 09 · Karen Livescu

Conclusion

Agenda

01. Introduction

– Kai-Wei Chang

02. Speech representation

– Chung-Ming Chien

03. Typical Training

– Shinji Watanabe

04. Advanced Training

– Kai-Wei Chang

05. Hands-on Session

– Ke-Han Lu

06. Full-duplex

– Siddhant Arora

07. Evaluation

– Hung-yi Lee

08. Emerging Topics

– Chih-Kai Yang

09. Conclusion

– Karen Livescu

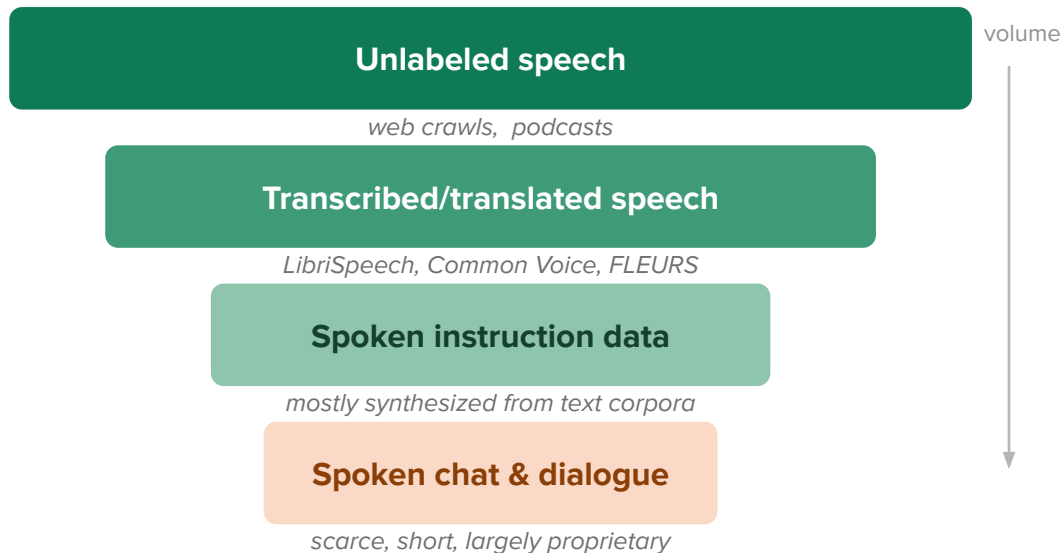
What do we still need (more of)?

- 1 More fully open models**
Code, checkpoints, and data – without all of these, it is not possible to compare design choices directly
- 2 Controlled ablations and scaling laws**
Along with open models, we need a better understanding of which design decisions matter and how that depends on scale
- 3 Benchmarks the field agrees on**
Including generation, interaction and paralinguistics, not only spoken QA
- 4 Inclusiveness**
More languages, dialects, accents, and atypical speech

Other open challenges / topics we haven't covered

- 1 Data**
Open instruction, chat, and dialogue speech is scarce; proprietary mixtures make comparison difficult
- 2 Beyond speech and text**
Images, video, robot and biometric signals. Open question: Are separate encoders needed?
- 3 Efficiency**
Model size, inference and training compute; on-device and real-time budgets
- 4 Multi-speaker**
Overlap, diarization-aware modeling, who said what to whom
- 5 User modeling**
Evaluation and training that account for who is speaking, what they want, and their human limitations (processing time, short-term memory,)
- 6 Personalization**
Adapting voice, style, and knowledge to an individual user
- 7 Speech-specific safety**
Deepfakes, speaker-type bias, jailbreaks
- 8 Speech-specific agent issues**
Interruption handling, latency budgets, and error recovery when tools are used

The data bottleneck



- **Synthetic data helps**

TTS-generated speech helps scale training; still unclear how best to mix synthetic and natural speech

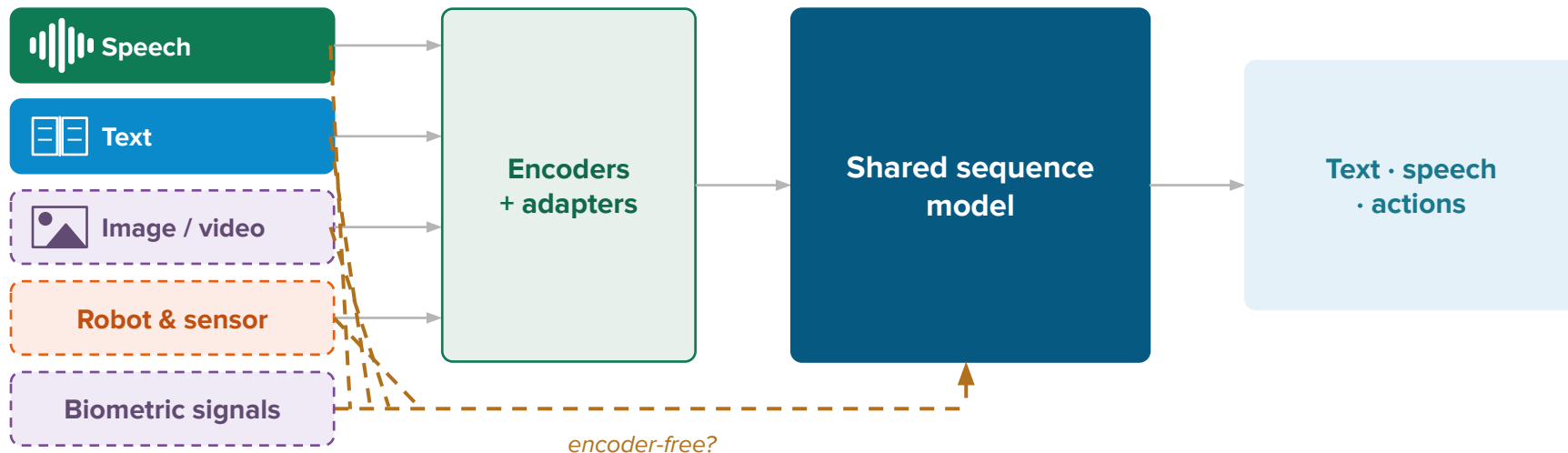
- **Some gaps remain**

expressive, multi-party, non-English, accented and atypical speech are limited at every level of the data hierarchy

- **Data mixtures are unpublished**

when data is proprietary, it is hard to attribute performance improvements to design choices vs. data choices.

Beyond speech and text



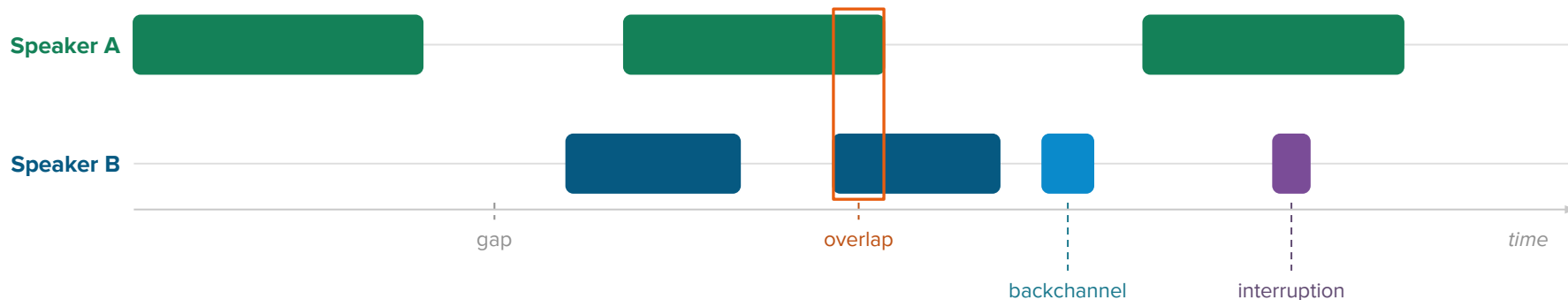
Maintaining sufficient context for all modalities

adding vision or sensors competes for the same context and the same capacity

One encoder/adaptor or many?

shared vs. per-modality adapters?
per-modality adapters that are “aware” of each other?

Beyond two-party conversation



Who said what

diarization-aware modelling

Overlapped speech

simultaneous voices break encoders trained on single speakers

Duplex → N-plex

turn-taking policies that decide whether to speak at all

Target-speaker selection

Is the model conversing with one out of many speakers, or with the whole group?

Other open challenges / topics we haven't covered

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Deepfakes, speaker-type bias, paralinguistic misuse, spoken jailbreaks
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Interruption, latency budgets, and error recovery when tools are used

Thank you!

Tutorial materials:

<https://interspeech2026slm.github.io/>

